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Improving Scansion with Syntax: an Investigation into the Effectiveness of a Syntactic Analysis of Poetry by Computer using Phonological Scansion Theory

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Improving Scansion with Syntax:

an Investigation into the Effectiveness of a Syntactic Analysis of Poetry by Computer using Phonological Scansion Theory

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by

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Preface

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Abstract

Two thousand years ago, Quintilian (1922), a Roman poetry expert, described his method of finding the rhythm in lines of poetry (technically called 'scansion'):

> "it is not so important for us to consider the actual feet (syllables paired up *according to the poem's meter and* with one syllable emphasised) but rather the general rhythmical effect of the phrase"

English poetry experts have instead focused on the stressed and unstressed syllables in feet, producing scansions that they themselves admit are often inaccurate and subjective (Wright, 1994). Their theories are used in the latest computer scansion programs, the best of which, like Hartman's Scandroid, scan barely as well as the undergraduates they are designed to help (Hartman, 2005). Figure 1 illustrates the inaccuracy and range of expert and computer scansions of a difficult line in Milton's Paradise Lost: the closer to the 'expected' point, the better the scansion.

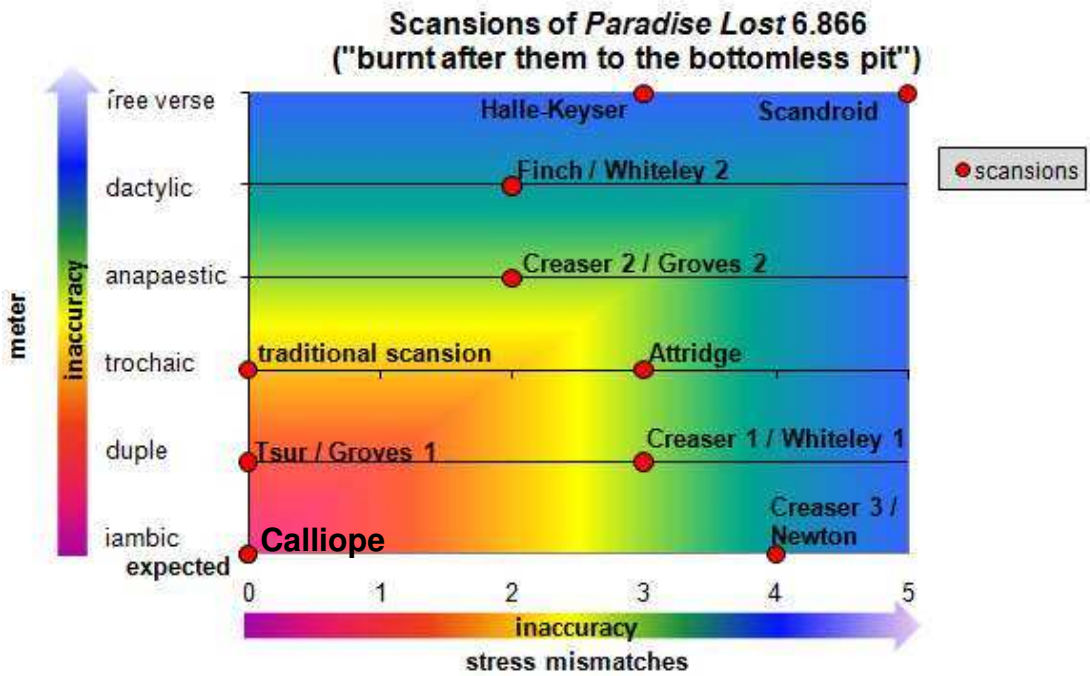


Figure 1: accuracy of scansion systems

This dissertation evaluates a new computer scansion application, Calliope, which follows Quintilian's phrase-based approach in two ways. Firstly, it addresses problems in identifying word stress by referencing syntactic data produced by the Antelope Natural Language Processing Parser (Proxem, 2007). Secondly, it implements a new scansion method based on research over the last twenty years into the influence of syntax on scansion (particularly Hayes and Kaun, 1996). This is the first time that syntax has been systematically integrated into a scansion program and the first time that some of these widely accepted research conclusions have been used to develop a scansion procedure.

Calliope is assessed for speed and accuracy in producing stress assignments and scansions in lines of poetry by comparing it to Hartman's program. Expert assessments serve as a benchmark, and non-expert assessments are used to identify acceptable alternatives. Using the same criteria, Calliope is also assessed against the most popular scansion methods (including systems developed by Fabb, Groves and Plamondon).

Compared to Scandroid, Calliope is far superior in assigning stresses. It is also much more effective in identifying meter, more accurate in predicting line scansion and identifies a wider range of meters. Compared to popular scansion methods, it equals or betters their performance in the same categories – see Figure 1. It seems that syntax makes a significant, but largely unexploited, contribution in determining both word stress and scansion in poetry. Suggestions are made for future research in this area.

Chapter 1 Introduction

In this chapter I will describe the main problems with existing computer scansion applications, and suggest a solution and ways to evaluate it against existing solutions.

Throughout the dissertation I will use the following symbols and abbreviations (the meanings are explained further in the text):

Symbol	Meaning
Stress	
w	Weakest stress
n	Low stress
m	Moderate stress/Secondary stress
s	Strongest stress/Primary stress
x	Metrically insignificant stress
Metrical Stress	
W	Weak stress
S	Strong stress
Musical Beat	
W	Weak beat
M	Moderate beat
S	Strong beat
Metrical Symbols	
|	Foot break
||	Caesura
Phonological Symbols	
\	Word break
<>	Intonational Unit (beginning and ending)
{}	Phonological Phrase (beginning and ending)
[]	Clitic Phrase/Group (beginning and ending)
/\	Polysyllabic Word (beginning and ending)
IU	Intonational Unit
PP	Phonological Phrase
CP	Clitic Phrase/Group
Scandroid Symbols (used by Hartman, 2005)	
/	Strong stress
|	Foot break
x	Weak stress

Table 1: symbols used in the dissertation

1.1 Description of the problem domain

1.1.1 Scansion difficulties

Most poetry in English from C15 onwards has been composed to conform to a rhythmical pattern of stressed and unstressed syllables. For example, iambic pentameter poems are composed of lines which ideally have an unstressed syllable followed by a stressed one repeated five times – its metrical scheme (Table 2c). Scansion is the process of identifying the pattern of stresses in any given line (Table 2b). Classical Scansion matches the stressed/unstressed pattern of syllables to the ideal pattern for a given meter. It allows, to a limited degree, unstressed syllables where stressed are expected and, to a lesser degree, the reverse – in Table 2, for example, the stresses in "black hairs grow on" (sssw) do not match the expected metrical scheme (wsws). This method is by far the most popular form of scansion, and has recently been implemented in numerous computer analyses of poetry.

"If hairs be wires, black hairs grow on her head" (Shakespeare, Sonnet 130.4)		
a	Line	If hairs be wires, black hairs grow on her head
b	Stress pattern	If(w) hairs(s) be(w) wires(s), black(s) hairs(s) grow(s) on(w) her(w) head(s) ws | ws | ss | sw | ws
c	Metrical scheme	WS|WS|WS|WS|WS (Iambic Pentameter)

Table 2: Classical Scansion

Computer-based scansion is, in fact, increasingly desirable: given the decline in the teaching of scansion in English courses, and the complexity involved in matching stress patterns to metrical schemes, very few people, even literary experts, are equipped to appreciate what is arguably an essential element of most English poetry (Liberman, 2004). Computer applications offer the possibility of providing accurate scansions, generated from objective principles, for whichever poem a user chooses, whether previously scanned or not. Expert scansions often disagree, and many are not explained

from first principles – a system which rigidly and transparently applies fixed principles would be a useful didactic tool, even if the end result was flawed. One which approximated reliable expert scansion would be even more useful. However, existing applications have significant deficiencies, especially when compared to improvements made to traditional scansion theory by linguists over the past forty years or more.

Existing computer scansion applications implement a range of scansion procedures. Traditional scansion is adopted by Raabe's program (1975), Bellin's LISP Scansion program (1997), Plamondon's AnalysePoems (2006) and one algorithm used in Hartman's Scandroid (2005). They all suffer from the shortcomings of the traditional method, for example, inadequate assessment of stress patterns and consequent inaccuracies in identifying meter. A more detailed discussion of these is provided below (1.1.2).

Dilligan and Lynn's program (1973) uses an early linguistic theory (Halle-Keyser) to identify whether lines are metrical or not. However, it does not produce the scansion of individual lines.

Hartman (2005) also uses a simplified version of this theory to derive scansion. However, it is limited to pentameter verse, biased towards iambic pentameter, and not very sophisticated in its application of the theory. Whilst the bias towards pentameter can be removed, Hartman himself acknowledges that his system is deficient in its assignment of stress, and subjective in its application of scansion.

Fabb's Bracketted Grid Theory (2003a, 2003b and 2006) imposes regular patterns on the line to determine which pattern is the closest fit, using the Generative Metrics concept of the stress maximum to anchor a line's stresses against the expected metrical stresses. It is unable to determine some meters correctly (for example, Catalectic Trochaic Trimeter).

Hayward's system (1996a and 1996b) references a number of linguistic concepts, but does not rigorously implement the theory behind them. It also requires user assessment of features of the line to work. Although the system uses a computer to process the interaction of stress-determining features of adjacent syllables, it does not use it to analyse the features themselves. The system is not a good candidate for full automation.

There is no program, nor any scansion procedure, which implements the phonological metrics theory of a core group of linguistic experts (Hayes, 2005). The theory has a relatively long history – it is derived from early observations by Magnuson and Ryder (1971) and Kiparsky (1977) that metrical stress matches word stress more strictly at the end of phrases than at the start, and is combined with an analysis of phrases based on phonology first formulated by Selkirk (1978). Its assumptions have been verified experimentally on numerous occasions: for example, Hayes (1984a), Hayes and Kaun (1996) and Tarlinskaja (1997). However, it has not been used by literature experts because it is perceived to be unwieldy compared to traditional scansion methods (Wallace, 1996 and Liberman, 2004).

1.1.2 Stress Assignment difficulties

In addition to scansion weaknesses, existing programs also suffer from inaccuracies in stress assignments. Since the stress pattern of a line forms the basis of scansion, errors at this level are extremely serious. Most programs assume only two levels of stress (for example, Scandroid). AnalysePoems assumes three. However, research over the past forty years has determined that even words which appear to have equal stress are differentiated under certain circumstances, some of which are relevant in poetry. In practice this means that at least four levels of stress may be necessary for determining rhythm accurately (Jespersen cited in Wallace, 1996).

Moreover, stress patterns in words interact with their syntactic function, producing different outputs in specific contexts. Some examples are given in the following discussion.

Since at least C16, it has been observed that the rules which determine when monosyllables receive stress are not very straightforward (Puttenham, 1569 and Brown, 1963). More recently, linguists have classified monosyllables into closed (stressed) or open (unstressed) categories (Hammond, 1991, Attridge, 1995 and Cooper, 1997). The closed class consists of content words – nouns, adjectives, verbs, adverbs and demonstrative pronouns. The open class consists of non-content words: determiners, prepositions, conjunctions, personal pronouns etc. (STELLA Project, n.d.). For example, 'of' is generally unstressed, whereas 'can' (noun or verb) is generally stressed.

Words with more than one syntactic function are stressed in some syntactic contexts, but unstressed in others. These fall into two categories. The first consists of synonyms which have different syntactic classes each with a distinct stress pattern. For example, Table 3a and b.

No	Word	First Stress Pattern	Second Stress Pattern
a	convict	the(w) **con(s)vict(w)** stays(s) [noun]	he(w) **con(w)victs(s)** me(w) [verb]
b	that	**that(s)** man(s) [demonstrative]	I(w) saw(s) **that(w)** it(w) was(s) [conjunction]
c	can	I(w) **can(w)** see(s) [modal verb]	I(w) **can(s)** [main verb]

Table 3: lexical stress dependent on syntax

The second consists of words which have different stress patterns depending on their precise syntactic functions. For example, in Table 3c, when 'can' functions as a modal verb it is generally unstressed, but when it is a main verb, it is stressed.

Linguistic theories like these have made stress assignment much more accurate. However, only one computer system references them – Hayward's; but it requires user input to do it, and does not make use of linguistic theories systematically.

1.2 Phonological Scansion

None of the existing computer systems combines accurate stress assignment with a sophisticated scansion theory backed up by linguistic research. However, having synthesised a scansion procedure from research in phonological metrics, I hope to demonstrate its effectiveness in a computer application. Below, I assess it against three preliminary tests for the procedure which are difficult for scansion procedures generally.

1.2.1 Comparison with expert scansion

In Table 3 expert human scansion of five lines of poetry are compared to the scansion produced by this phonological scansion procedure and three recent systems by Plamondon (AnalysePoems), Hartman (Scandroid) and Fabb (Bracketted Grid Theory), to demonstrate which is closest to the expert scansion: a process I will replicate experimentally below (3.5.3). Scansions which match the expert scansion are coloured green.

System	Scansion	Comment
“If hairs be wires, black wires grow on her head” (Shakespeare, Sonnet 130.4)		
Raffel (1992)	If(w) hairs(s) be(w) wires(s), black(w) wires(s) grow(s) on(w) her(w) head(s). ws|ws|ws|sw|ws	An expert scansion
Anstead et al. (1998)	ws|ws|ws|ws|ws	A graduate’s scansion. It violates the natural rhythm of grow(s) on(w) by forcing a iambic rhythm to match the meter
AnalysePoems, Plamondon (2006)	if(0) hairs(2) be(0) WIRES(2) black(2)WIRES(2) grow(2) on(0) her(0) head(2) ws|ws|ss|sw|ws	The scansion has an unmetrical series of 4 strong stresses. It does not take into account the break after “be wires”
Scandroid, Hartman (2005)	x / | / /|x / | x / | x / ws|ss|ws|ws|ws	The scansion has an unmetrical series of 3 strong stresses. It does not takes into account the break after “be wires”
Phonological Scansion	Xsw]s}n]sw]xws} = ws|ws|-ms|sw|ws	Matches the expert scansion. The m marks an intermediate stress which matches the |ws| pattern in the third foot
“Through Eden took their solitary way” (Milton, Paradise Lost 12.659)		
Milton (1674)	“English heroic verse without rhyme” i.e./ unrhymed iambic pentameter ws|ws|ws|ws|ws	Expert scansion, in fact the author’s own assessment
Scandroid, Hartman (2005)	/ / x / x | x /|x x| / sswsw|ws|ww|s	The Scandroid gave up trying to scan this line – it found no significant series of iambs based on inadequate stress assignments.
Bracketted Grid Theory, Fabb (2003b)	ws|ws|ws|ws|ws	Matches expert scansion
Phonological Scansion	Xsw}s}xswsw]s} = ws|ws|ws|ws|ws	Matches Fabb’s analysis and expert scansion

System	Scansion	Comment
“There they are my fifty men and women” (Browning, One More Word 1)		
Halpern (1962)	“Browning's trochaic pentameter poem One Word More” i.e./ sw|sw|sw|sw|sw	An expert opinion
Scandroid, Hartman (2005)	x / | / | x /| x / | x / x ws|s|ws|ws|wsw	The Scandroid tries to force a iambic reading on a trochaic verse
Phonological Scansion	xnm>xsw]s}x[sw]> = sn|mw|sw|sw|sw	produces a regular trochaic pentameter scansion
“What had I given to hear the soft sweep” (Charlotte Bronte, Diving 9)		
Bracketted Grid Theory, Fabb (2003b)	s|wws|wws|wws	I could not find an expert scansion for this verse, but it is probably dactylic trimeter as Fabb suggests
Scandroid, Hartman (2005)	x /| x /|x x | / x | / / ws|ws|ww|sw|ss	The Scandroid tries to scan the anapaests as iambs, even when forced to look for anapaests. Inaccurate stress assignment, and a failure to prioritise the end of the phrases produces the error
Phonological Scansion	s]xxsw]ws}xms} = s|wws|wws|wms	Matches Fabb’s scansion (and probably the intended meter)
“Gone were but the winter” (Christina Rossetti, Spring Quiet 1)		
Fussell (1979)	sw|sw|sw	Expert scansion cited in Fabb (2006)
Bracketted Grid Theory, Fabb (2006)	wws|wws	Fabb scans the entire poem with only two feet per line
Scandroid, Hartman (2005)	/ | x x | x /| x s|ww|ws|w	The Scandroid fails to find a significant series of iambs and gives up the scansion. Inadequate stress assignment is partly to blame
Phonological Scansion	Sw]/xxsw} = sw|nw|sw	Matches the expert scansion
Key Expert scansions Scansion matching expert scansion		

Table 4: test verses for computer scansion systems

The table shows that the Phonological Scansion process is the only one to consistently match the expert scansion for the chosen lines.

1.2.2 Identification of rhythm in ambiguous stress patterns

Phonological scansion is also able to reliably identify the poet's intended rhythm in lines whose stress pattern fits two or more metrical patterns. For example, the stress pattern swswsws can be scanned as iambic (-s|ws|ws|ws) or trochaic (sw|sw|sw|s-). Atkins (1942) called this "the great metrical riddle on which so much time and so much ink has been expended". Tsur (1998) and Tarlinskaja (1993) conclude that the only reliable way to identify which is the correct rhythm is by a comparison with previous verses, although Hascall (1971) and Wright (1988) note that the rhythms of individual phrases or words help sensitive critics identify the meter within verses (supported by Atkins, 1942, Attridge, 1982, Fabb, 2002 and Wheatley, 2006 following Cureton, 1996). Relying on this assumption, the phonological scansion procedure is able to identify meter without reference to other verses by identifying the rhythm of polysyllabic words – see

Table 5 which analyses key lines from the poem "Epitaph on the Countess Dowager of Pembroke".

Scansion type	Analysis	Comment
Willam Browne, "Epitaph on the Countess Dowager of Pembroke", lines 2 and 6		
Overall stress pattern	swswsws	Either catalectic trochaic or headless iambic tetrameter
Line 2	**Lies the subject of all verse**	Hascall's scansion is **catalectic trochaic**
Stress pattern	Lies(s) the(w) sub(s)ject(w) of(n) all(m) verse(s)	
Phonological scansion (meter)	xxsw}xms>	
Phonological scansion (rhythm)	(iambic=none, trochaic= 'sub(s)ject(w)')	One polysyllabic word is trochaic. There are no iambic rhythms. So, the verse rhythm is trochaic.
Overall Scansion	sw|sw|nm|s-	**catalectic trochaic**
Line 6	**Time shall throw a dart at thee**	Hascall's scansion is **headless iambic**
Stress pattern	Time(s) shall(m) throw(s) a(w) dart(s) at(n) thee(m)	
Phonological scansion (meter)	xms}ws}nm>	
Phonological scansion (rhythm)	(Iambic='shall(m) throw(s)', 'a(w) dart(s)', trochaic=none)	One phrasal verb, and one end of phrase are iambic. There are no trochaic rhythms. The verse rhythm is iambic.
Overall Scansion	-s|ms|ws|ns	**headless iambic**
Key Scansion matching expert scansion		

Table 5: phonological scansion of rhythm (after Hascall, 1971)

Hascall (1971) notes it took "an exceptionally sensitive and knowledgeable critic" to realise that the poem's first three lines are trochaic and the last three iambic. The phonological scansion procedure is able to replicate this sensitivity.

1.2.3 Assessment of Phonological Scansion procedure

In order to assess how effective the Phonological Scansion procedure is, I have classified verses according to difficulty. The procedure should be able to identify meter and scansion of the following categories of verses with decreasing accuracy, but still approximate expert scansion better than Scandroid and AnalysePoems:

No.	Category	Example
1	Verses of varying numbers of feet and types of feet (iambic, trochaic, dactylic and anapaestic) with no irregular feet	Iambic Pentameter: eg/ Shakespeare Sonnet 130.3 "If snow be white, why then her breasts are dun" Trochaic Trimeter: eg/C. Rossetti Spring Quiet 1 "Gone were but the winter"
2	Verses of varying number and type of feet with irregular scansion in phonologically insignificant syllables	Shakespeare Sonnet 130.2 "My mistress' eyes are nothing like the sun" (Iambic Pentameter)
3	Verses written in Sapphics, Alcaics (fixed syllable meter)	Sapphics: eg/ Swinburne Sapphics (3x Sw/sx/sw/ws/ws/s, 1x Sw/ws/s) Alcaics: eg/ Tennyson, Milton (2x s/sw/ss/sww/sx/s, 1x s/sw/ss/sw/sx, 1x sww/sww/sw/sx)
4	Verses of varying number and type of feet with irregular scansion in one phonologically significant syllable (including headless lines and feminine endings)	Shakespeare Sonnet 130.1 "Coral is far more red than her lips' red" (Iambic Pentameter)
5	Verses of varying number and type of feet with irregular scansion in more than one phonologically significant syllable.	C. Bronte Diving 1 "'What had I given to hear the soft sweep" (Anapaestic Tetrameter)
6	Verses of varying number and type of feet where, for phonologically significant syllables, the number showing irregular scansion is equal to or greater than the number showing regular scansion	Browning Lost Leader 2.10 "There would be doubt hesitation and pain" (Dactylic tetrameter)
7	Free verse showing some regular feet	Frost Out, Out
8	Free verse without regular feet	Whitman Out of the cradle rocking endlessly

Table 6: categories to assess computer implementation of phonological scansion

T. Criteria for success of the theory

I have defined the following objectives to measure the theory's success rate:

T1. For categories 1-6 the procedure should match the meter identified by experts, and produce a scansion which is close or identical to expert scansion in the majority of cases (95%).

T2. For categories 7-8, which are notoriously difficult to analyse, the procedure is less likely to succeed and should match 70% of scansions.

T3. It should also either match or improve the competing computer system scansions.

In addition, Hayes (2005) has implemented an algorithm which 'textsets' (matching lexical stresses in lyrics to beats in the music, similar to scansion): it replicates human text-setting 71% of the time.

T4. A procedure which replicates expert scansion better than 71% of the time would be a good result for the research.

1.3 Contribution to knowledge

The research tests whether a system employing linguistic theories of stress assignment and scansion makes an improvement in the identification of scansion by existing computer systems.

Over the past few years there has been an increase in computer scansion applications. This has been facilitated by the ease of implementing and publishing applications, but is probably motivated by a desire to make scansion more accessible, countering a perceived decline in the use and knowledge of scansion by students of English -compare Thomas (1913) with Walsh (1963), Woods et al. (1982) and Liberman (2004). However, the most popular applications use theories of scansion which do not fully

engage with recent developments in scansion theory made by linguists. According to Wallace (1996), these linguistic theories are ignored by English experts because they are thought to be too difficult and inaccurate. This research will assess whether a recent linguistic theory can be effectively implemented, and whether it produces results closer to expert scansions than the existing systems. If it is successful, it will demonstrate that linguistic scansion theory outperforms traditional theories, at least in computer applications, challenging its rejection by English experts.

From the perspective of computer science, this research is the first implementation of linguistic theories of stress assignment with phonological phrasing starting from syntactic data – speech processing implements both, but starts from phonetic data (Atterer, 2000). The research may be of use in demonstrating how successful an implementation from syntactic data is. The use of the computer could also open the application up to a wide audience over the Internet, and, using speech technology to recite the output, could make prosody more accessible and immediate to casual users who are baffled by scansion notations – a similar application has started to do this for Latin (Butterman, 2007).

The scansion procedure used in the research is a new synthesis of observations made by linguists working on the relationship between phonological phrases and metrics - this in itself is a small contribution, making the theory more practicable and more accessible to non-linguists. However, the present research also offers the opportunity of testing the theory's effectiveness by assessing its performance against traditional, and more recent, scansion procedures on a wide variety of data and on a large scale. Linguistic prosodic theories have rarely been tested against a large corpus of data, so the opportunity of automatically processing large corpora to test the theory should be beneficial. If it is unsuccessful, it will help to show where the theory fails to account for expert

assumptions in scansion. If it is successful, it will verify the theory, and support the conclusions of previous experiments. This may have implications for more recent theories by Fabb, Groves and others which do not reference phonological data to any significant extent. This project also implements, for the first time as far as I am aware, key recommendations from the latest attempt to automatically assign text to meter by one of the most respected scholars in metrical phonology – Hayes (2005) concludes a recent analysis of assigning meter:

> "I believe to obtain a better account of textsetting, we will need to draw on the resources of the theory of generative metrics (Halle and Keyser 1969 and much subsequent work). This research tradition, though seldom focused specifically on the textsetting problem, has yielded many results that (as the findings of Hayes and Kaun 1996 indicate) are directly applicable to the formulation of an adequate constraint set for textsetting."

1.4 Aims and objectives of the research project

The project has two aims: to assess whether the use of syntax improves stress assignment functions in computer applications, and whether it improves the identification of meter and scansion in verse.

To achieve the first aim, I have developed an application using a syntactic parser and coded stress assignment theories developed by Hayes and others to deal with compound and nuclear stress, and stress clashes. The function's output is assessed against expert opinion. It is then implemented in Scandroid to see if there is a noticeable difference.

To achieve the second aim, I have developed a scansion procedure based on research into the influence of syntax on scansion by Selkirk, Hayes and others. This is

implemented in an application (called Calliope) and the output tested for accuracy against the original procedure. The application's output, along with the scansion of Scandroid and three other competing theories, is then tested to see how accurately they replicate expert scansion. The scansion from the procedure is also assessed by non-experts. These two tests will determine whether Scandroid or the application produce a more natural scansion. These objectives are quantified further below (2.7).

1.5 Overview of the dissertation

In chapter 2, I outline the deficiencies in Classical Scansion identified by linguistics and literature experts, and the principal solutions they have devised. The chapter concludes with an assessment of how these theories are used in computer applications, and my own proposal for addressing the problems. Chapter 3 describes the experiments which I have used to evaluate the proposal, and Chapter 4 analyses the results. Chapter 5 summarises my conclusions. I have included an index and glossary (p. 145) as well as references and a bibliography (p. 131).

The Appendices contain a discussion of the Phonological Scansion procedure (p. 148) and more detailed comparisons of the procedure with expert scansion (p. 174), Scandroid (p. 179), Raabe's program (p. 186), AnalysePoems (p. 182), Bracketted Grid theory (p. 188) and Base and Template theory (p. 194). I have also included samples of the written tests given to test subjects (p. 200), the data (p. 206) from all the tests run (whether on computer or subjects), and sample output from Calliope (p. 221). This material supplements and supports the argument in the main part of the dissertation.

Chapter 2 Literature Review

2.1 Classical Scansion

2.1.1 Description

Most modern English poetry is based on meters identified from patterns of stress in words and phrases. These patterns are combined with a restriction on the number of syllables in poetic lines, and are termed 'accentual-syllabic' verse. Within this verse tradition, a wide variety of meters, derived originally from Classical Greek and Latin, are used. By far the most common is iambic pentameter, composed of five 'feet' of consecutive weak then strong stresses. In common with most meters, it also permits additional patterns as long as the basic meter of the poem can still be identified. Examples of iambic pentameter are given below:

Ref	Line and scansion	Scansion
(Shakespeare, Sonnet 18.1-2)		
a	"Shall I compare thee to a summer's day" Shall(s) I(w)| com(s)pare(w)| thee(w) to(s)| a(w) sum(s)|mer's(w) day(s)	sw|ws|ws|ws|ws
b	"Thou art more lovely, and more temperate" Thou(w) art(s)| more(w) love(s)|ly(w), and(s)| more(w) tem(s)|per(w)ate(s)	ws|ws|ws|ws|ws

Table 7: Classical scansion of Sonnet 18:1-2

In these two lines, there are five feet in each line. Each foot is separated from the others by |, and contains two syllables. Each syllable can be either stressed or unstressed. The pattern 'ws' is a 'iamb', and five in a line comprise a iambic pentameter (Table 7b). The pentameter also allows, among other patterns, 'sw', termed a 'trochee'. However these other patterns are only permitted where the iambs are still considered to be so dominant that the audience considers the line basically iambic. In practical terms, this means there are more iambs than any other pattern, as in Table 7a.

I will refer to this application of Classical meters to accentual-syllabic verse as 'Classical scansion'.

2.1.2 Use of Classical Scansion

Classical scansion is by far the most popular system used by English literature experts (STELLA Project, n.d.). Consequently, it is also the most popular system used in computer analyses of English poetry, including Bellin (1996), Hayward (1996a) Mamede (2004), Hartman (2005) and Plamondon (2005) – in fact, I have found no computer application which does not use it, although Hartman (2005) complements it with a theory loosely based in linguistics.

However, the system has been rejected by scholars as a good method for assessing the underlying rhythm of poetry since Jespersen in 1900 (and most recently by Attridge, 1982, Hayes, 1989, Cureton, 1996, Wesling, 1996 and Fitzgerald, 2007). The main problems are:

1. difficulties in matching actual stress to perceived metrical rhythms,
2. rhythmical effects which are ignored by Classical scansion,
3. the existence of valid poetry which does not use these patterns at all.

Each of these is discussed below. They have led to the system being side-lined in favour of more flexible analyses of poetry. Stauder (2000) comments:

> "The relevance of scansion to the reading and understanding of poetry has often seemed arbitrary or irrelevant, especially as formalist criticism was eclipsed in the 1980s and 90s by various forms of post-structuralist criticism…while literary critics, in small numbers, have continued to apply older forms of metrical analysis to texts, the most exciting work on rhythm has been done by scholars

interested in understanding rhythm in the broadest ways possible, including its temporal, cognitive and hermeneutical significance."

2.1.3 Subjectivity of Scansion

Scansion has one difficulty which applies regardless of the method used. Most literary scholars assume it is subjective, allowing multiple correct scansions. Some adopt this position for ideological reasons (Wright, 1994); for others it is based on ambiguities of stress assignments determined by the meaning of sentences, even though linguists have long since disambiguated this feature (Halliday, 1967 and Levinson, 1983) as shown in Table 8.

Stress Pattern		Context	Expert Support
"**To be or not to be that is the question**" **(Shakespeare, Hamlet 3.1)**			
a	To be or not to be that(n) **is**(s) the question	Follows an assertion that "to be or not to be" is not the question	Philip Burton (mentor of Richard Burton)
b	To be or not to be **that**(s) is(m) the question	Follows a discussion about what the question is	Jose Ferrer and David Garrick (famous actors)
Since this is the first line that Hamlet speaks in the scene, having entered alone and without any context, it is difficult to determine the pragmatic emphasis. However the most natural reading of the line is B. A would seem extremely forced, and is perhaps motivated by the desire to produce a completely regular iambic rhythm.			

Table 8: pragmatic variations of Hamlet, from data in Packard (1992) and Daalder (2006)

Others (Wright, 1994 and Tsur, 1998 and 2006) propose that rhythm is determined by the interaction of meter and stress with the "performance" of the line's dramatic features (although this type of scansion matches expert scansion less frequently than Phonological Scansion which ignores dramatic features of the line).

For whatever reason, some scholars believe all verse scansions are subjective (Nearing, 1962, Holder reviewed in Finch, 1998, Tsur, 1998 and accepted by Plamondon, 2006), although the majority believe that only scansions of some difficult lines are subjective (Woods, 1984, Adams, 1997, Finch, 1998 and accepted by Hartman, 1996). Even if the majority is correct, the identification of agreed criteria for assessing scansion becomes

problematic, albeit in a limited but nevertheless unpredictable set of lines. Figure 2 illustrates the range of expert scansions (detailed in Table 28) of a difficult line: the closer a scansion is plotted to the origin of the scatter graph, the more accurate it is – scansions are assessed for stress mismatches (as described in Stauder, 2000 and Equation 1) and correct meter assignments against the expectation of iambic pentameter and natural stress.

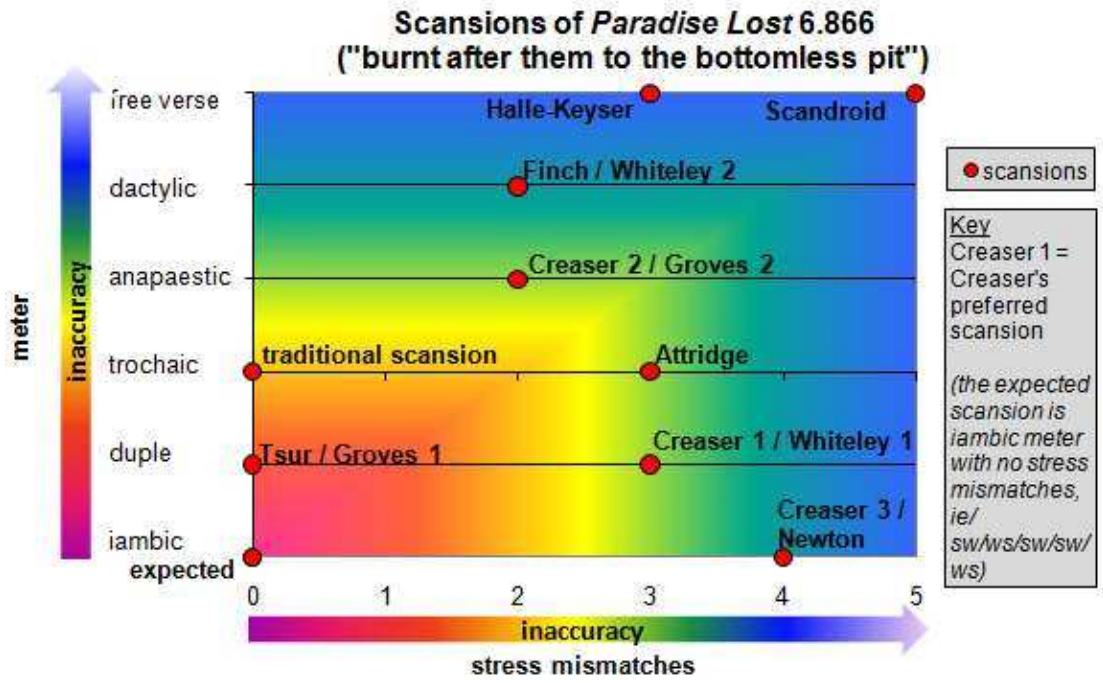


Figure 2: accuracy of expert scansions of Paradise Lost 6.866

Labelling Mismatches (stress mismatches) = 1 x (strong stress where weak stress expected) + 2 x (weak stress where strong stress expected)

Equation 1*: Stauder's Labelling Mismatch formula*

Linguistic analyses produce objective and potentially universal descriptions of the perceived rhythm in poetry. This current proposal is grounded in such approaches, but will also try to define some objective literary criteria where there is a general consensus.

2.2 Weaknesses of Classical Scansion

2.2.1 Rhythmical Prose and Classical Scansion

There are two main exceptions to the accentual-syllabic type of most English poetry. A few poets used different categories to identify metrical rhythm, for example purely syllabic, purely accentual (stress), or by syllabic quantity (based on the prosodic systems of the Greek and Classical Latin quantitative models of English meter). However, many of these experiments employed the classifications used for accentual-syllabic, and so could superficially be included within the system of Classical scansion. The second exception cannot – it is the recent trend to reject meter entirely and compose 'free verse' with extremely varied rhythmical patterns (for example, T.S. Eliot), or none at all (William Carlos Williams). Finally, some forms of prose resist metrical analysis but have a recognisable rhythm (Cummings and Herum, 1967): Cureton (1992) gives as an example the King James translation of Psalm 24 which illustrates the translation's "mid- and high- level ('oratorical') rhythms":

> "The Earth is the Lord's, and the fullness thereof;
> the world, and they that dwell therein.
> For he hath founded it upon the seas,
> and established it upon the floods…"

These types are not fully accommodated in the traditional scansion methods of English poetry, and yet have a definite rhythm.

2.2.2 Experience of the Reader

Theorists have long questioned the validity of applying the concept of the foot to verse. Ironically, the Classical Romans, whose scansion model introduced the foot to English,

played down the concept of metrical feet and, instead, emphasised natural speech rhythms (Quintilian, 95).

In English verse, even the validity of the concept of the foot has now been questioned. For example, Halle and Keyser (1966, 1971 and 1999) make no reference to the Classical concept of foot in their studies of verse stress, and Attridge (1982) and Tsur (1998) note that one variant of the iambic tetrameter verse can also be classified as trochaic tetrameter (see Table 9), but the rhythmic experience of the reader is identical in both cases (see also 1.2.2).

Scansion	**Classification**
"Here the anthem doth commence" (Shakespeare The Turtle and the Phoenix 21)	
Here(s) the(w) an(s)them(w) doth(s) com(w)mence(s)	Stresses
s| w s | w s | w s	Iambic Tetrameter
s w | s w | s w | s	Trochaic Tetrameter

Table 9: Iambic and Trochaic analysis of the same line

For such critics, and perhaps most readers, the strong stress pattern is more significant than the pattern of weak stresses between the strong (Hayes, 1995 and Halle, 1999).

2.2.3 Attridge's Beat-Offbeat Scansion Theory

This has led Attridge (1982, 1990 and 1995) to develop a system of scansion which sidelines feet and the number of unstressed syllables, marking only stressed syllables and patterns of unstressed intervening syllables. Variation in the number of weak stresses between strong undermines the principles of Classical scansion, and yet does not destroy the perceived rhythm; which may be one of the reasons why the system is the most popular modern alternative to Classical scansion, having been adopted by Willett (1997), Creaser (2001 and 2007) and Paterson (2002) among others, and used to teach scansion at Princeton (Showalter, 2003). Shapiro (1985) calls it "unquestionably...

the best account we now have of English versification". Examples of Attridge's basic theory are given in Table 10b ("o" is a weak stress which Attridge calls an "offbeat", "b" is a strong stress, termed a "beat", "ŏ" is a "double offbeat" consisting of two weak stresses sounded as one, and "ô" is an implied weak stress that does not appear in the lexical stresses of the line, but occurs as a significant temporal break between two strong stresses).

Scansion Type		**Scansion**
"He left me; I called after him aloud" (Wordsworth The Prelude 5.133)		
a	Classical Scansion	"He left me; I called after him aloud" w s| w w| s s|w s| w s
b	Attridge	o B ŏ B ô B o B o B
c	Extended Attridge	–s –s +s +s
"o" is a weak stress which Attridge calls an "offbeat", "b" is a strong stress, termed a "beat", "ŏ" is a "double offbeat" consisting of two weak stresses sounded as one, "ô" is an implied weak stress that does not appear in the lexical stresses of the line, but occurs as a significant temporal break between two strong stresses "+s" marks slightly increased stress "–s" mark slightly decreased stress		

Table 10: Attridge's scansion system from Attridge (1982)

The scansion easily produces a regular alternation of strong and weak "beats". However, it ignores the popular, and probably valid, assumption that most English poets worked in terms of stresses not beats – that is, they usually interpreted two weak stresses as two weak stresses, not a single "beat", and would not normally have inserted an additional beat between two strong stresses (as Attridge does). Attridge (1982) himself notes that the theory only deals with two levels of stress when some lines require more to reveal intended poetic effects. His solution is to use the +s/-s notation to mark modified stress levels (see Table 10c), however the double notation is a little cumbersome. Finally, in a linguistic assessment, Hayes (1984b) rejects Attridge's

theory because, among other deficiencies, it allows some lines which are avoided by poets but are excluded by more accurate Generative Metrical accounts.

2.2.4 Hayward's Connectionist Scansion Method

Hayward (1996a and 1996b) develops a computer scansion method based on Jakobson's theoretical distinction between the metrical pattern of a verse and how it is actually delivered (discussed in Bernhart, 1995). The user is required to assess a given verse for (among other things) prosodic features of alliteration, the basic syntactic function of words (whether they are verbs or nouns), and the user's own interpretation of the verse. An analysis is given below:

Category	**Syllable Scansion**
"The clouds that gather round the setting sun" (Wordsworth, Intimations of Immortality 195)	
Intonation	0000000100
Lexical	0001000100
Prosodic	0100010101
Syntactic	0201000001
Interpretive	0100000100
Final result	0.0/ 0.821/ 0.0/ 0.698/ 0.0/ 0.543/ 0.0/ 0.884/ 0.0/ 0.660
Hayward's description of the process:	
This coding represents a rise in intonation at 'setting', stress on the first syllables of 'gather' and 'setting', assonance between 'clouds' and 'round' and alliteration between 'setting' and 'sun', the function of 'clouds' as the subject of a sentence, with 'gather' and 'sun' acting as verb in a dependent clause and object of a preposition, and an interpretation of the line focussing on the 'clouds' and the fact that the sun is 'setting', in contrast to the bright days described in the preceding lines. Each position representing metrical stress is also connected to its neighboring positions, with a negative weight to decrease the stress on adjacent syllables. Finally, a bias for stress on even numbered syllables is built into the system. After inputs for all connected units are assigned, the system is sent through a series of 30 cycles in which inputs and activations from and to each of the sixty nodes are measured and averaged. What is finally achieved is a measurement of the potential activation of metrical stress for each of the ten positions for that particular line of poetry.	

Table 11: Hayward's scansion method

The need for user input makes this theory inappropriate for a fully automated scansion system. Whilst its use of subjectivity acknowledges a driving principle of recent literary

theories, the degree to which this operates in scansion is under fierce debate (Addison, 1994), and Hayward's system runs the risk of ignoring valid linguistic principles in favour of a subjective analysis without any firm objective basis.

2.2.5 Relative Levels of Stress

In its purest form Classical scansion deals with absolute levels of stress – see the scansion by Fussell (1979) of Paradise Lost 2.700 given in Table 12a. However, most metrists subordinate the actual stresses to the metrical pattern, recognising that weak stresses can be sounded as strong, and vice versa, to maintain an overall rhythm. In 1900 Jespersen proposed that this is possible because syllables have a wider range of stress than just weak or strong (Wimsatt and Beardsley, 1959). In Table 12c-e I have given an estimation of the relative stress strengths for each syllable - the last foot 'add wings' has two stressed syllables, but the final one 'wings' is relatively stronger than the first 'add' and so I have given it a higher value.

System		**Scansion**
"False fugitive, and to thy speed add wings" (Milton, Paradise Lost 2.700)		
a	Pure Classical Scansion	s s | w s, | w w | w s | s s
b	Modified Classical Scansion	w s | w s, | w s | w s | w s
c	Jespersen analysis (adopted by Steele)	3 4 | 1 3 | 2 1 | 1 2 | 3 4
d	Trager-Smith analysis	´ ´ | ˇ ^ | ` ˇ | ˇ ` | ^ ´
e	Notation used in this proposal	m s |w m,| n w| w n| m s

Table 12: stress values and Classical scansion

The patterns of stresses can also be significant across feet as well as within them – for example, the stress pattern 1234 in "thy speed add wings" is what Womack (2000) calls a 'crescendoing counterpoint'. The rise of stress levels across the four syllables, whilst retaining a relative iambic rhythm within its two feet, also undermines the iambic pattern because the listener expects stress in the second syllable to be stronger than the stress in the third. Womack notes that this is a widely used poetic technique, especially common in Shakespeare.

Classical Scansion notation was modified to display these sorts of patterns by Trager and Smith (including Trager, 1962 – see Table 12d). The idea has been most recently applied by Steele (1999), and allows him to identify patterns of stress 'modulation' which can produce 'subtleties of rhythm.' All these revised systems require at least four levels of stress.

2.2.6 Syntactic Patterns

However, these theories do not account for other syntactic patterns like caesurae and enjambement. For example, they do not distinguish between the lines in the following table:

Line		Scansion	Type
a	The other two, slight air, and purging fire, (Shakespeare Sonnet 45.1)	w s w s‖, w s‖, w s w s	Two caesurae
b	For when these quicker elements are gone (Shakespeare Sonnet 45.5)	w s w s w s w s w s	No caesura

Table 13: the effect of caesurae on scansion

The syntactic breaks in Table 13a produce a very different rhythmical effect on the reader than the flowing example in Table 13b. Breaks within the line are called "caesurae", phrases which run across line breaks produce "enjambement"; but these phenomena are not systematically incorporated into Classical scansion, even though they are clearly relevant for rhythm (Cummings and Herum, 1967 and Hartman, 2006).

2.3 Temporal Prosody

Alternative theories have been developed to deal with these problems, including the traditionally strongest competitor to Classical scansion - temporal prosody. This analyses rhythm by the time taken to pronounce syllables, rather than their relative stress levels (Copeland, 2002). It was first proposed by Joshua Steele in 1775, partly

because Classical poetic models work on this pattern and partly because, before tapes, musical notation was the best method of recording poetry's rhythmic beat (Halle, 2005a) – Table 14 has an example.

Type	Notation
Line	As Peter was going to the hall, he met John
Musical Feet	As/ Peter was/ going to the/ hall/, he met/ John
Scansion	As(w)/ Pe(s)ter(w) was(w)/ go(s)ing(w) to(w) the(w)/ hall(s)/, he(w) met(s)/ John(s)
Steele (original notation)	(Δ)As(∴)/ Pe(Δ)ter(..) was(∴)/ go(Δ)ing(..) to the(∴)/ hall(Δ.. ∴)/, (Δ)he(..) met(∴)/ John(S.. ∴)
Steele (using musical stresses)	(SW)As(M)/ Pe(S)ter(W) was(M)/ go(S)ing(W) to the(M)/ hall(SWM)/, (S)he(W) met(M)/ John(SWM)

Table 14: Steele's musical notation (from Halle, 2005a and Steele, 1779)

It has been used most recently by Brown (1965) and Abercrombie (1965, cited in Daalder, 2006), and reassessed, though not entirely favourably, by Monelle (1989). It fails principally by ignoring items which appear significant to the reader – stress and syntax - in favour of something that is intuitively not significant- time (Holcombe, 2007): so, the literary critic Wallace (1996) dismisses it out of hand, as does the linguist Tarlinskaja, quoted by Halle (2005a).

2.4 Generative Metrics

2.4.1 Descriptive Theories

There is one group of researchers who have investigated the relationship between syntax and verse systematically - Generative Metrists analyse verse to determine the conditions under which a given verse is perceived as metrical. Fabb (2006) describes the main distinction between this and the traditional scansion discussed so far, using the analysis in Table 15 as an illustration:

Scansion type	Scansion	Comments
“Does the road wind uphill all the way?” (Christina Rossetti, Up-Hill 1)		
Classical scansion	wws/ss/ws/ws	2 substitutions for iambs: an anapaest (wws) and a spondee (ss). Four recognisable feet imply tetrameter, probably iambic.
Generative Metrics	wwss[sw]sws	The strong stress in the polysyllabic word ‘uphill’ is generally matched to a strong metrical stress. This, and the 9 syllable count, implies iambic tetrameter.
“May not the darkness hide it from my face?” (Christina Rossetti, Up-Hill 2)		
Classical scansion	ws/ws/ws/ws/ws	No substitutions – a regular iambic pentameter line
Generative Metrics	wsw[sw]swwws	The strong stress in the polysyllabic word ‘darkness’, marked by brackets, is preceded and followed by weak stresses: a pattern which always matches the strong stress to a strong metrical stress. This, and the 10 syllable count, implies iambic pentameter.

Table 15: Generative Metrics and traditional scansions compared (Fabb, 2006)

> “In metrical analyses such as Fussell (1979) as well as newer theories (e.g., Attridge 1982) the metrical form is molded to the rhythmic form, rather than being distinct from it...In the approach to metricality which sees rhythm as the expression of meter (Classical Scansion), these lines are well formed because the specific rhythms are permitted variations of iambic pentameter. In the approach to rhythm espoused in this paper (Generative Metrics), many aspects of these rhythms are simply irrelevant to the metricality of the lines; as we show below the lines are metrical because their numbers of syllables (nine and ten) can be counted by the metrical rules for iambic pentameter, and the polysyllables ‘uphill’ and ‘darkness’ are placed as they are in the line.”

One strand of Generative Metrics, which is led by the Slavic Metrists including most recently Tarlinskaja (1992, 1993, 1995 and 1997), and followed by investigators like Raabe (1975), Attridge (1982) and Duffell (2008), involves statistical analysis of stress patterns to determine which are more acceptable than others. For example, Tarlinskaja

(1993), marking weakest stress as 1, greatest stress as 5 and a caesura as ||, concludes that Shakespeare's earlier rhythm is typically [1 4 || 2 3 5], but his later rhythm is [1 4 3 || 2 5].

A second strand is primarily linguistic, and started with Halle and Keyser's proposal (1966 and 1971), revised by Levin (1973), that a fully stressed syllable between two unstressed syllables in the same word (termed a 'stress maximum') occurs only in positions where strong stress is expected in the meter. Magnuson and Ryder (1970 and 1971) and Kiparsky (1975) modified the theory to account for the fact that stressed monosyllabic words occur anywhere in the line not just in stressed positions, but polysyllabic words must place their stressed syllables in strong positions – that is, word boundaries are also a factor, not just lexical stress.

Table 16 shows the original Halle-Keyser predictions and some modifications made later – lines which are declared metrical should be frequently found, those which are unmetrical should be extremely rare.

Line/Stress/Meter		Halle-Keyser theory	Modifications
a	Line: Introduced grandfather to amuse friends Lexical: s w w \ s w w \ w \ w s \s Metrical: W S| W S| W S | W S| W S (declared metrical although never found, but reclassifed as unmetrical by the polysyllabic modification)	<u>metrical, but never found:</u> stress maximum ('-duced gra|ndfath-') in WSW position	<u>unmetrical, therefore never found</u>: Strong stress in polysyllabic words not in S position ('in|troduced|'= WSW, 'a|muse' = SW)
b	Line: Ode to the West Wind by Percy Bysshe Shelley Lexical: s \ w\ w \ s \ s \ w\ s w\ s \ s w Metrical: W S | W S |W S| W S| W S W (declared unmetrical although found, and reclassified as metrical by Kiparsky)	<u>unmetrical, but found</u>: stress maximum in 'by Pe|rcy' in WSW position	<u>metrical, and found</u>: monosyllables allowed to be stressed anywhere (stress maximum negated by stressed 'by' 'by Per|cy' = ssw)
c	Line: Though death doth consume, yet virtue preserves Lexical: w\ s\ s \ w s\ s \ s w \ w s Metrical:W S | W S| W S | W S | W S (declared metrical, but which is unmetrical according to Kiparsky and others)	<u>metrical, but declared unmetrical by experts</u> (Groves, 2007b)	<u>unmetrical and found, but opposed: 2</u> polysyllabic stresses in unstressed position (con|sume =ws, |virtue| =sw), violating the polysyllabic modification

Table 16: Halle-Keyser theory and changes by Kiparsky, and Magnuson and Ryder

2.4.2 Stress Rules

This analysis was developed further by the discovery that stress in a word can be modified by the stress of closely associated contiguous words in the same phrase. This linguistic stress theory is of crucial importance to scansion, which requires a realistic assessment of the stress patterns in a verse. The idea was first proposed by Newman in 1946, according to Krifka (2001), but Chomsky and Halle (1968), followed by Liberman and Prince (discussed in Hayes, 1984) identified two processes for it – the Nuclear Stress Rule and the Compound Stress Rule (the role of an additional stress assignment rule, Stress Clash, is still being debated – see the Appendices, p194). The first rule proposes that words receive increasingly greater stress from the beginning to the end of a sentence in a recursive assignment of stress. This additional stress is

assigned to the principal stress within each stressed word – an illustration is given in the table below:

Stress Analysis	Stress Rule Applied
"Mary ate sweet ice cream"	
[Mary [ate [sweet [ice cream]]]]	Phrases
[Mary(1) [ate(1) [sweet(1) [ice(1) cream(1)]]]]	lexical stress
[Mary [ate [sweet [ice(1) cream(2)]]]]	compound stress
[Mary [ate [sweet(2) [ice(1) cream(3)]]]]	nuclear stress, first cycle
[Mary [ate(2) [sweet(3) [ice(1) cream(4)]]]]	nuclear stress, second cycle
[Mary(2) [ate(3) [sweet(4) [ice(1) cream(5)]]]]	nuclear stress, third cycle

Table 17: Stress Rules (Krifka, 2001)

In the sentence "Mary ate sweet ice cream" Nuclear Stress is assigned recursively from the last main stressed word back, so "Mary" receives a stress level of 2 and "cream" 5. "Ice" receives a stress level of 1 because it is the subordinate element of the compound "ice cream"; "cream" has the principal stress. According to Chomsky and Halle, the Compound Stress Rule assigns additional stress to the rightmost element in compounds of two or more nouns – for example, in "ice cream" principal stress is assigned to "cream". Bolinger (1972), Sproat and Liberman (1987) and Giegerich (2006) modified it to accommodate some compounds, defined by their meaning, which are stressed on the leftmost element. For example, the first element of "metal bridge" is stressed because it expresses the substance of the second element, but "metal fatigue" is stressed on the second element because the relationship is one of association not substance: "fatigue in metal" not "fatigue made of metal".

Beaver (1971) summarises the impact of these stress assignment rules on verse using stress-maxima to determine metricality:

> "(1) The Compound Rule is apparently always operative (equivalently, never disregarded) in English verse.

(2) The Nuclear Stress Rule in its Noun phase apparently operates in… English, contrary to what some linguists have believed."

2.4.3 Phonological Analysis

Another key component in understanding the nature of stress assignment and metrical patterns involved a new type of analysis partly dependent on syntax: phonological and clitic phrases. First applied to meter by Kiparsky (1975), the theory was defined by Selkirk in 1972 (see Selkirk, 1981, 1984 and 1986). Although researchers differ about how to divide some syntactic structures into phonological phrases, especially intonational units, (von Heusinger, 1999 and Atterer, 2000), the general principles are agreed. Table 18 gives phonological analyses by four researchers. However, I will use Selkrik's analysis because it is more accurate, easier to implement on computer, and is the only one used by experts in Phonological Metrics (Hayes, 1989 and Cureton, 1992). According to Selkirk, clitic phrases consist of a content word (a verb, adverb, adjective or noun) and any non-content words to its right and dependent upon it. Phonological phrases consist of clitic phrases which are syntactically related at the next level up. Intonational units consist of phonological phrases within breaks marked in the text such as commas, semi-colons or full stops (Hayes, 1989). Although Wheeldon (2000) notes that non-syntactic information can also influence the formation of phonological phrases (a phonological phrase will end immediately after an emphasised word), since this type of information cannot be referenced accurately by a computer I have ignored it.

Ref	Phonological Analysis
clitic phrases are contained within square brackets [] phonological phrases are contained in curly brackets {} intonational units are contained in angle brackets <>	
Abney	<{[Given] [the chance]}, {[rabbits] [reproduce] [quickly]}>
	<{[More than fifteen carpenters] [are working]}{[in the house]}>
Bachenko and Fitzpatrick	<{[Given] [the chance]}>, <{[rabbits]} {[reproduce][quickly]}>
	<{[More than fifteen carpenters]}{[are working] [in the house]}>
Gee and Grosjean	<{[Given] [the chance]}>, <{[rabbits] [reproduce] [quickly]}>
	<{[More than fifteen carpenters]}{[are working]}{[in the house]}>
Selkirk/Cureton	<{[Given][the chance]}>, <{[rabbits]} {[reproduce] [quickly]}>
	<{[More than fifteen] [carpenters]} {[are working][in the house]}>

Table 18: phonological analyses of two sentences from Atterer (2000)

Selkirk's phonological theory accounts for changes in the way certain phrases are pronounced. For example "of" is never reduced to "o'" when it occurs at the end of a clitic phrase, and stress clash is resolved within phonological units, but unresolved outside them (Hayes, 1989 and 1995, Kim, 1999 and Zuraw, 2006).

This analysis also predicts prosodic patterns. In 1971 the literary critic Hascall noted his impression that phonological and metrical boundaries coincide in iambic but not trochaic verse. Since then, more scientific analyses have been made: for example, Hayes (1989) has shown that the interaction of syntax and clitic phrases predict the metrical stress assignments in the phrases in Longfellow's Hiawatha (Table 19), and may have a more general application (Fitzgerald, 2007). Where the clitic phrase is separated by one syntactic break the clitic scansion is preferred, where there are three or more syntactic breaks, the non-clitic scansion is preferred, otherwise there is apparently free variation:

Clitic Phrases	Syntactic Breaks	Scansion preferred
[the great][lakes]	1: the $_{NP}$(great lakes)	- x – (clitic)
[in great][flocks]	2: In $_{PP}$($_{NP}$ (great flocks))	- x – or x – x (either)
[that] [old feuds]	3: That $_{S}$($_{SUBJ}$($_{NP}$ (old feuds))	x - x (non clitic)
NP= noun phrase; PP = prepositional phrase, S = sentence, SUBJ = subject		

Table 19: Haye*s's* application of clitic phrases to scansion

This research has a direct impact on the identification of rhythm in poetry. According to Kiparsky (1975) phonological phrasing determines the location of caesurae in verse. More recent work on rhythm has accepted and generalised this conclusion: Barsch (1995) summarises:

> "The view that the principles of versification are related to the prosodic features of a language is shared by so many metrists of totally different schools that it is a banality".

These principles can identify rhythms in free verse - Gates and Laforgue (1990) have successfully used them to identify T. S. Eliot's "music of common speech". They have also been used to determine the idiosyncratic rules of individual poets: Youmans (1983) and Harvey (1996) have developed these for Milton, Shakespeare and Donne.

2.5 Phonological Metrics and Recent Linguistic Theories

The principles also make useful observations about the relationship between metrical and lexical stress patterns. Literary scholars such as Andersen (1909) had noticed that stress patterns in words affect the rhythm of the line even when it is counter to the overall metrical rhythm, and Steele (1779) recorded the match between strong stresses in polysyllabic words and musical stresses. However these patterns were only defined scientifically relatively recently, by Magnuson and Ryder (1971) and Kiparsky (1975):

lexical stress matched metrical stress more frequently within polysyllabic words than monosyllabic ones. Kiparsky (1977) also found that stresses at the end of phrases were much more strictly matched than at the beginning of phrases. These observations support a basic principle of Generative Metrics formulated by Kiparsky in 1968 ("beginnings free, endings strict"), and have been confirmed by Hayes (1984a, 1996, 1998 and 2000), Tarlinskaja (1993 and 1997), Fabb (2001) and Li (2004).

To demonstrate the effectiveness of this principle, the table below (Table 20) gives the scansion of a line using Plamondon and Hartman's systems, as well as a phonological analysis which results much more quickly in a firm identification of iambic pentameter.

Scansion		**System**
"My mistress, when she walks, treads on the ground" (Shakespeare Sonnet 130.12)		
a	My(1) mis(2)|tress(0), when(2)| she(1) walks(2)|, treads(2) on(0)| the(0) ground(2)|	AnalysePoems
b	My(w) mis(s)|tress(w), when(s)| she(w) walks(s), |treads(s) on(w) |the(w) ground(s)|	Scandroid
c	"[My(w) mis(s)|tress(w)], [when(w)| she(w) walks(s)], |[treads(s)] [on(w) |the(w) ground(s)|]"	Cureton's phonological analysis

Table 20: scansion of a line by AnalysePoems, Scandroid and Cureton

Whereas Scandroid and AnalysePoems need to reference the stress patterns in each of the five feet to determine the overall metrical pattern, the phonological scansion procedure only references stresses at the ends of the phonological units. The closing syllables of three of the four phonological units fit with the expected metrical stress patterns of iambic pentameter: "mis(s)|tress(w)", "|she(w) walks(s)|" and "|the(w) ground(s)|". Phonological theory claims that these are the key areas the listener targets to identify metrical patterns (Hanson, 1991, Hayes, 1989 and followed by Dahlgren, 2005). Hayes and Kaun (1996) conducted experiments on song lyrics leading to two conclusions which help in identifying poetic meter.

1. metrical and lexical stresses almost always coincide at the end of phonological phrases. In fact, the higher up the hierarchy, the more likely the match: clitic phrases 90% of the time, phonological phrases 97% and intonational units 99%. See Table 21b for an example. These conclusions are also backed up by recent research into matching lyrics to music, termed "textsetting": Halle (2004) notes that words grouped together in musical phrases sound most natural to audiences when the words also form phonological units – musical groupings rarely cross phonological boundaries.

Ref	Scansion	System
"Or how haps it I seek not to advance" (Shakespeare Henry VI Part 1 3.1.31)		
a	**Or(w) how(s)/ haps(s) it(w)/ I(w) seek(s)/ not(w) to(s)/ ad(w)vance(s)**	Word stress
comment	(1 foot is a trochee, 4 feet are iambs: ws/sw/ws/ws/ws)	
b	**<{[Or(w) how(s)]$_{CP}$ / [haps(s) it(w)]$_{CP}$ }$_{PP}$/ {[I(w) seek(s)/]$_{CP}$ [not(w) to(s)/ ad(w)vance(s)]$_{CP}$ }$_{PP}$ >$_{IU}$**	Prosodic stress
comment	Using the last two stresses in each phonological unit gives ws]/sw]}/ws]/xx/ws]}> (where x is a stress of irrelevant value). 3 clitic phrases match metrical stresses (ws/), as well as one clitic phrase's associated phonological phrase and intonational unit. Only the second clitic phrase (and its associated phonological phrase) does not match the iambic metrical stress: [haps(s) it(w)]}	
clitic phrases are labelled 'CP' and contained in square brackets []; phonological phrases are labelled 'PP' and contained in curly brackets {}; intonational units are labelled 'IU' and contained in angle brackets <>		

Table 21: word and prosodic stress (from Hayes and Kaun, 1996)

2. Hayes and Kaun (1996) also note (following Kiparsky, 1975) that metrical and lexical stresses coincide most often within polysyllabic words (98% of the time there is a match): for example in Table 21a, whereas the lexical stress in disyllabic words matches a iambic metrical scheme ('ad(w)vance(s)'), in monosyllabic words it does not necessarily match ('haps(s) it(w)').

2.5.1 Groves's Base and Template Theory

Two other scholars (Groves and Fabb) use Generative Metrics theories to prioritise stresses in the line. Groves (1998, 2001, 2007a and 2007b) has developed a scansion theory, based on the principle that the distribution of stress maxima determines the meter of a verse, which he calls Base and Template theory (2007b). He analyses each verse for linguistic stresses using the stress maxima to fix the strong stresses in a metrical template which should fit the metrical pattern. Table 22 gives a sample analysis, partly from Groves (2007a). The innovation in this theory is the additional prohibition against "an unstressed syllable adjacent to a fully stressed syllable in the same syntactic constituent from functioning as a metrical strong". For example, in Table 22d 'consume' breaks the prohibition by forcing the unstressed first syllable into a metrically strong position when it is adjacent to a stressed in the final syllable of the word, rendering the line unmetrical:

a	**Thomas Wyatt, *translation of Petrarch's Rime 140*.1-2**	
1	**Line**	**The longe love that in my thought doeth harbar**
	Linguistic stresses	Ŏ----a------A---- Ŏ O o------A o----A---o
	Metrical template	w---S S-----w S---w S---------w S---w
	Metrical pattern	w---S w-----S w---S w--------S w---S (Iambic Pentameter)
2	**Line**	**and in myn hert doeth kepe his residence**
	Linguistic stresses	O O o---A o-----A o---A-o O
	Metrical template	[w—S] w—S w----S w---S w-S
	Metrical pattern	w—S w—S w----S w---S w-S (Iambic Pentameter)
b	**(Milton, Paradise Lost 6.866) Burnt after them to the bottomless pit**	
	Linguistic stresses	A---a---o---A O o---A---o---a A
	Metrical template	S---w w--S S--w S--w w--S
	Metrical pattern	S---w S--w S--w S--w S--w (Trochaic Pentameter)
c	**(Chaucer) Though death doth consume, yet virtue preserves**	
	Linguistic stresses	Ŏ ---A a---o--A | <Ŏ—A>---o o---A
	Metrical template	w---S S---w S---w S---w w---S
	Metrical pattern	w---S w---S w---S w---S w---S (Iambic Pentameter)

Key:
O is a weak stress;
A is a strong stress;
a is a strong stress subordinated to a strong stress in the same syntactic unit;
o is a weak stress subordinated to a strong stress in the same syntactic unit;
Ŏ is a partially subordinated weak stress;
--- indicates the connected items are part of the same unit (metrical or syntactic);
| indicates an intonation break, used to calculate the possibility of a metrical stress not associated with a syllable (Groves, 2007b)
[] signifies an optionally reversible foot, <> marks out unmetrical patterns
Stress maximum, unstressed syllable next to a linked stressed syllable

Table 22: scansion system of Groves, partly from Groves (2007a and 2007b)

Groves (2007b) claims that the theory is better at identifying non-metrical lines than Halle-Keyser theory, Kiparsky's analysis or even Shakespeare's own editors over the years. The theory is designed to evaluate whether a given line is perceived by readers as irreconcilably at variance with its metrical pattern, however it cannot determine the meter of a line, except in extreme cases.

One weakness of the theory itself is that it assumes that stress maxima are always matched to strong stresses in the line: Magnuson and Ryder (1971), Beaver (1971) and Tsur (1998) have discovered a small number of lines where this is not the case.

Moreover, since stress maxima are rare in lines – Beaver (1968) finds only 1/3 of stressed syllables in Shakespeare's Sonnet 89 are stress maxima, and only three are found in Table 22 - Groves also uses strong stresses in word groups, whether in stress maxima or not. Consequently, his analysis has no way of distinguishing which stresses might be more important than others in determining meter– for example the iambic line Paradise Lost 6.866 has three trochaic and two iambic feet which Groves's system would incorrectly analyse as a trochaic pentameter, because more stresses match the trochaic scheme.

He also limits the effectiveness of the theory by acknowledging only two levels of stress. This leads him to see a clash of strong metrical stresses in the line 'Un(w)ru(s)ly(w) Mur(s)murs(w), or(s) ill(s)-tim'd(w) App(w)lause(s)' – 'or' requires strong stress after the weak '-murs', and 'ill' has primary stress in the compound 'ill-tim'd'. There is no stress clash with multiple stress levels: 'Un(w)ru(s)ly(w) Mur(s)murs(w), or(n) ill(s)-tim'd(m) App(w)lause(s)'.

Finally, this theory can only identify a match to one metrical pattern at a time, sometimes forcing a false reading. In the case of Wyatt's lines (Table 22a), there has been a long debate whether the poem is iambic pentameter or accentual (Schwartz, 1963). However, this ambiguity, and even the possibility of an accentual analysis, is obscured by the theory.

2.5.2 Fabb's Bracketted Grid Theory

Spurred on by the success of using grids to resolve stress clashes (pioneered by Hayes, 1984a and others) and to textset lyrics to music (Lerdahl and Halle, 1993), Halle and Keyser (1999) have recently proposed a Generative Metrics grid-based method of scansion relying on the assumptions that the distribution of stress maxima determine the

meter of a verse, and that verses are generally divided into feet of either two or three syllables (Fabb, 2002). The theory, called Bracketted Grid Theory, has been applied successfully to verse by Frost (Halle and Keyser, 1999), Arnold (Fabb, 2003a) and Rossetti (Fabb, 2006).

However the restriction that only stress maxima are diagnostic of meter makes it limited: as well as the concerns discussed above for Groves's theory, reliance on stress maxima causes the theory to identify meters incorrectly because Fabb assumes that stress maxima prefer to occupy one foot not straddle many. For example, he classifies Christina Rossetti's trochaic trimeter poem "Spring Quiet" with two uneven feet, not three regular ones. This undermines his claim that the theory reveals the regularising metrical pattern in an otherwise complex poem – see Table 23.

Scansion Method	Scansion	Comment
"Gone were but the Winter" (Christina Rossetti, Spring Quiet 1)		
Bracketted Grid Theory (Fabb, 2006)	x x x (x x (x x) x Gone were but the Winter wws|wsw	two strong stresses and two uneven feet assigned: syllables in the stress maximum "the Winter" are kept together in the final foot.
Fussell (1979), cited by Fabb (2006)	sw|sw|sw|	three regular trochaic feet

Table 23: Bracketted Grid Theory (Fabb, 2006)

2.5.3 Optimality Theory

Up to the mid 1990s Generative Metrics theory explained variations in metrical patterns by looking for fixed universal rules to produce scansion. However, Golston and Riad (1995) proposed that whilst many rules applied universally, others should be ignored or prioritised in particular contexts: this is an application of the Optimality Theory of Prince and Smolensky (1993). The theory does scan successfully, and has become

researchers' most popular method – for example, Hayes and MacEachern (1998), Friedberg (1999), Hammond (2004), Fitzgerald (2006) and Kiparsky (2006). However, since it has to be fine-tuned for each individual poet (Hammond, 2004) it is not ideal for scanning a wide range of poetry. Moreover, it cannot determine patterns from text, only evaluate the application of a given pattern to a text.

2.5.4 Hayes's Optimality Algorithm

Nevertheless, Hayes (2005) has implemented an Optimality algorithm which textsets words to music. Table 24 compares Hayes's first attempt at a scansion algorithm (the Syllabic Distribution Algorithm derived from Halle, 1999) to the more successful Optimality algorithm, which replicates the output of untrained human text-setters 71% of the time. Textsetting along similar lines has also been successfully implemented by Kim (1996), Dell and Halle (2005) and Halle (2005b and 2007).

Description	Results
Lexical Stresses	To(w) court(s) young(m) mai(s)dens(w) I(m) was(n) bent(s)
Binary Lexical Stresses (used by Hayes)	To(w) court(s) young(s) mai(s)dens(w) I(w) was(w) bent(s)
Musical Stresses	*MWSW MWSW MWSW…*
Human textsetters	To(M)(W) court(S)(W) young(M)(W) mai(S)(W)dens(M)(W) I(S)(W) was(M)(W) bent(S)(W)
Comments	Human textsetters assign strong lexical stresses (s and m) to both strong and medium musical stresses, preferring strong musical stresses, and weak lexical stresses (w and n) to weak and medium musical stresses.
Syllabic Distribution Algorithm	To(M)(W) court(S)(W)(M)(W) young(S)(W)(M)(W) maid(S)ens(W) I(M) was(W) bent(S)(W)
Comments	The Syllabic Distribution algorithm only assigns strong lexical stresses (s) to strong musical stresses. Weak lexical stresses (m, n and w) are assigned to medium and weak musical stresses, in order to fit the strong stress pattern.
Optimality Algorithm	To(M)(W) court(S)(W) young(M)(W) mai(S)(W)dens(M)(W) I(S)(W) was(M)(W) bent(S)(W)
Comments	The Optimality algorithm allows m and n lexical stresses in M and S positions to avoid breaks in the line like (W)(M)(W) seen in the first algorithm's output 'To(M)(W) court(S)(W)(M)(W) young(S)(W)(M)(W) maid(S)ens(W)'. This adjustment enables the second algorithm to approximate human textsetting preferences very well.
'*S'* is a strong musical stress, '*M'* a medium stress and '*W'* a weak stress	

Table 24: the Syllabic Distribution and Optimality Algorithms used by Hayes (2005)

However, Hayes himself proposes three improvements– using multiple levels of stress (Hayes uses only weak and strong stress), referencing word and phrase boundaries to identify metrical stresses along the lines of Hayes and Kaun (1996) as described above, and matching the number of musical stresses assigned to a syllable to the syllable's natural phonetic length (for example, short vowels should be assigned fewer musical stresses than long vowels). Only the last is adopted in Keshet's (2006) analysis of textsetting folk songs. However, given that Keshet's theory is based on the assumption that the temporal length of syllables is key to understanding metricality, when it is applied to poetry it encounters the same difficulties as the temporal prosodic theories described above. The first two suggestions, which are supported but not implemented by Gerber (2001), form the basis of the scansion procedure used in this current project.

2.6 Computer Applications

Classical Scansion, then, has significant weaknesses, some of which are addressed by the stress assignment and phonological theories of Generative Metrics.

In the next sections I will discuss how computer applications address the issues discussed above, focussing on the two applications in particular which are used as benchmarks to assess this project, Hartman's Scandroid and Plamondon's AnalysePoems.

2.6.1 Stress Assignment and Syllable Division

The most important pre-requisite for the effective scansion of a line is the identification of its natural stresses. According to linguistic theories, this necessitates at least four levels of stress. However, none of the applications uses four levels – Hayward uses only two; Hayes, AnalysePoems and Scandroid three. Although a middle level of stress can be resolved into either weak or strong stress depending on the (usually metrical) requirements, this leads, among other errors, to an unwarranted number of feet with only weak or strong stresses which are avoided by scansion experts (Baker, 1996).

Three approaches are taken to determine stress patterns: Scandroid uses two algorithms to allocate syllables (developed by Holzer) and stress (developed by Bernstein and Nessly). Since English is very inconsistent, he also uses an exception dictionary to make the results more accurate. Although this system minimises computer resources and is the most consistent at identifying unknown syllable and stress values, it is also, in fact, the least accurate. Hayward relies on user input. However, this is time-consuming, and seems counter-productive in an automated system. Bellin and Plamondon use a dictionary. This is the most reliable approach, but it consumes the largest memory space, and relies on the scansion to identify unknown words rather than deriving

scansion from them. However, none of these systems accounts for stress assignments dependent on phrases, not words. For example, the stress in compound phrases discussed above, which is determined by the meaning of the elements, is entirely overlooked – probably because this analysis requires both syntactic and semantic data which are considered to be beyond the scope of any system without user input.

2.6.2 Syntactic Data

Syntactic data alone is required to resolve stress assignment problems in a more frequent set of cases: synonyms with stress values differentiated by syntactic function (such as the noun-verb pair con(s)vict(w) and con(w)vict(s)) and words which are stressed as phrases (such as phrasal verbs like "have(w) on(s)" or modal verbs like "have(w) done(s)", as opposed to verbs like "have") – more details are given above (1.1.2). It is also used by native speakers (and linguists) to assign relatively stronger stresses to the governing words within phrases – for example, whereas all the applications scan "white cliff" as a spondee (ss), it is usually perceived as a iamb (ws) because "white" is dependent on "cliff".

Even when application designers openly acknowledge the importance of syntax (Hartman, 2005 and Hayes, 2005) they do not use it. One reason for this omission is that parsers needed to identify syntactic relationships are perceived as too difficult to implement, or too complicated for the benefits accrued. For example, Robey (1993) sides with Hayward's user input approach, but only as a counsel of desperation because

> "in a great many cases a syllable which is accented if a word is uttered on its own loses part or all of the accent when the word is combined with others."

However, syntactic analyses are performed by poetry composing programs such as those of Mamede (2004), Gervas (2005) and Hartman's AutoPoet (1996), but very few

use it to identify rhythm. Dilligan and Lynn (1973) describe an application of the Halle-Keyser stress maxima theory which is 99% effective in identifying metrical lines in a range of poets, but the theory does not identify line rhythm. Although Hayward (1996a) develops a scansion program which takes user inputs to identify syntactic units in subordinate and main clauses, it does not use a more sophisticated syntactic analysis. Automated applications which do not reference syntax are forced to rely on scansion to determine the probable stress – this is both circular and, at times, highly inaccurate.

2.6.3 Linguistic Theories

There are no applications which fully implement the scansion systems of the Generative Metrists, despite their usefulness and critical acclaim. Scandroid uses a version of Generative Metrics when traditional scansion fails to scan a verse. However, it is extremely simplistic, and is discussed in greater detail below. Hayward's system makes even less use of linguistic theories. Though he employs categories which he labels 'lexical' and 'syntactic' to help assign stress, these refer to extremely basic classifications. Hayes's system uses a theory that is still in the process of development, and which ignores potentially fundamental linguistic insights into poetic scansion, some pioneered by Hayes himself.

2.6.4 Scandroid (Hartman, 2005)

In addition to the generic problems detailed above, Scandroid has particular weaknesses. Since I use it and Plamondon's AnalysePoems to assess the Phonological Scansion application, Calliope, I will now describe both in more detail.

Scandroid uses two competing algorithms to determine scansion (see Table 25), neither of which has the backing of linguists. "Corral the Weird" identifies irregular line-

endings and beginnings, and assumes that the remainder of the line has a regular disyllabic foot structure. If the number of remaining syllables does not fit a disyllabic structure, it attempts to find one or more trisyllabic feet. “Maximise the Normal” locates the longest run of iambic patterns in the line (it includes sequences of weak stresses as iambic). It then divides the remaining syllables into disyllabic feet. Hartman identifies the second with Generative Metrics, although it is a very simplified application of the basic idea, and corresponds to no linguistic Generative Metrics theory.

The choice of which algorithm to use is also, as Hartman himself acknowledges, arbitrary at times: after assigning stresses to the line, the program produces variants with every combination of resolved ambiguous (middle level) stresses. Both algorithms are run on each variant, and the program usually chooses the scansion that is least “complex” (meaning most regular), although Hartman does not explain the precise mechanism.

The major weakness of the program is its bias towards iambs (and iambic pentameter in particular). It will also attempt to scan for anapaests, although in practice it cannot identify anapaestic meters very well.

These difficulties, coupled with an inadequate stress assignment system, produce a scansion which is as accurate as an unskilled undergraduate.

Phase	**Scansion**	**Comment**	**Example**
Line	**Than in the breath that from my mistress reeks (Shakespeare, Sonnet 130.12)**		
Dictionary	than(w) in(w) the(w) BREATH(?) that(w) from(w)my(w) MISTRESS(?) REEKS(?)	Stresses and syllables assigned from the dictionary.	“breath”, “mistress” and “reeks” are not found in the dictionary.
Stress Assignments	breath(s), mis(s)/tress(w), reeks(s)		Breath and reeks are calculated as single strong stresses, mistress as sw
Corral the Weird	The line has a regular iambic start and end – so divide the line into iambic feet. The line becomes: w%/ws/w%/ws/ws	Test the line by removing odd endings and beginnings and dividing the remainder into disyllabic feet	There are no non-iambic endings or beginnings. The line is divided into 5 feet. 2 stressless syllables are promoted to “ambiguous” (%) to match the rhythm of the iambic feet
Maximise the Normal	ww/ws/ww/ws/ws. Becomes w%/ws/w%/ws/ws	Test the line by finding the longest run of iambs within the line,	The longest run of iambs extends across the entire line.

Table 25: Scandroid process

2.6.5 AnalysePoems (Plamondon, 2006)

AnalysePoems also has particular difficulties. It does not assign stress consistently: since Plamondon uses the medial level of stress to determine the probability of strong or weak metrical stresses, he assigns weak stresses to syllables with lexical primary stress (for example, “was”, “asks” and “I”) if, in his opinion, they are significantly likely to avoid the expected strong metrical stresses in poetry. This forces the program to avoid promoting these words appropriately (see “than in” in Table 26).

Unlike Scandroid, the program scans for most types of duple and triple meter. However, when producing variant scansions of a line, it is unable to use alternate pronunciations in combination, limiting the accuracy of the final scansion.

Finally, contrary to Plamondon’s aim of not imposing a metrical structure on lines, the system chooses to make linguistic stresses fit metrical stresses (instead of the other way round) at key points in the process – in fact, it ignores or fails to identify the natural linguistic rhythm. For example, the system ignores until late on in the process the high probability that secondary stresses attract strong metrical stresses: only secondary

stresses which fit the proposed meter are sure to be stressed, other stresses which intentionally violate this meter may be ignored. So, the system is biased towards identifying regular metrical patterns, even where these are not probable from the linguistic stress patterns. This means that free verse poems may be classified as metrical because some of the feet in a given line are recognisably metrical.

<table>
<tr><th>Phase</th><th>Scansion</th><th>Comment</th><th>Examples</th></tr>
<tr><td>Line</td><td colspan="3">Than in the breath that from my mistress reeks (Shakespeare, Sonnet 130.12)</td></tr>
<tr><td>Dictionary</td><td>than(0) in(0) the(0) breath(2) that(1) from(1)my(1) MISTRESS(2) –(0) reeks(2)</td><td>Stresses and syllables are assigned from the dictionary.</td><td>“mistress” is not found in the dictionary.</td></tr>
<tr><td>Test 1: Meter determination</td><td><table><tr><td>No of Stress-less syllable types</td><td>Intial (s, ws, wws, wwws)</td><td>Medial (sws, swws, swwws)</td><td>Final (s, sw, sww)</td></tr><tr><td>line</td><td>3w=1</td><td>3w=1
1w=1</td><td>0w=1</td></tr><tr><td>poem</td><td></td><td></td><td></td></tr></table></td><td>The meter is determined by compiling 2 sets of 3 statistics: the most common and 2nd most common number of stressless syllables before the 1st and after the last stressed syllable, and between stressed syllables</td><td>the meter of the line is identified as iambic pentameter because it matches the iambic pattern of 1 initial, 0 final and 4 medial stressless syllables</td></tr>
<tr><td>Test 1b: Confidence Level</td><td>line: 50%, 50%
poem:</td><td>The confidence level is the ratio of the number of occurrences of the value over the total number of occurrences.</td><td>1 occurrence of 3 medial stressless syllables over 2 medial stressless syllables = 50%</td></tr>
<tr><td>Test 2: weak stress promotion</td><td>than(0) in(0) the(0) breath(2) that(1) from(2)my(1) mistress(2) –(0) reeks(2)</td><td>Secondary stresses which, if they were strong, would make the line closer to the expected medial stressless value are promoted.</td><td>“from” promoted to a metrical stress makes the % of 1 medial stressless syllables 100%. Stressless “in” is not promoted.</td></tr>
<tr><td>Test 3: Polysyllabic promotion</td><td>There are no polysyllabic words (of three or more syllables) in the line</td><td>Any 2ry stresses in polysyllables are promoted to strong metrical stresses</td><td></td></tr>
<tr><td>Test 4: Foot matching</td><td>There are no secondary stresses to promote</td><td>The stress pattern in individual feet for the proposed meter is used to promote secondary stresses to strong metrical stresses.</td><td>if “in” had a secondary stress, it would have been promoted because it would form a iambic foot (ws)</td></tr>
<tr><td>Test 5: Syllabic length</td><td>Poem: 10 syllables
Line: 9 syllables (therefore mistress = 2 syllables)</td><td>The average syllabic length is calculated, and the syllabic length of unidentified words deduced from it.</td><td>The average syllabic length of the poem is 10. “mistress” must be 2 syllables.</td></tr>
<tr><td>Test 6: Alternate pronunciations</td><td>There are no alternate pronunciations recorded for this line.</td><td>Alternate pronunciations recorded in the dictionary are tried out in the line, one by one.</td><td>If “mistress” (sw) had an alternate of “m’stress”(s) it would be assessed</td></tr>
<tr><td>Test 7: Elision</td><td>There are no elisions possible</td><td>2 vowels across a word boundary are elided</td><td>eg/ “my idol” would become “m’ idol”</td></tr>
<tr><td>Test 8: Trochaic promotion</td><td>than(0) in(0) the(0) breath(2) that(1) from(2)my(1) mistress(2) –(0) reeks(2)</td><td>A 2ry stress in the initial foot of a iambic poem are promoted to a metrical stress</td><td>“than” is not promoted because it is originally stressless.</td></tr>
<tr><td>Final Scansion</td><td>Wwwswswsws</td><td></td><td></td></tr>
</table>

Table 26: AnalysePoems process

As with Scandroid, AnalysePoems relies on inaccurate stress assignments and a scansion procedure which is not validated by experts. However, it is able to scan more

meters and as accurately as a skilled undergraduate. Its most serious defect is that it subordinates natural stress to meter, contrary to Plamondon's aim for it.

2.6.6 Summary Comparison of Key Applications

Table 27 gives a comparison of Scandroid, AnalysePoems and Phonological Scansion as implemented in Calliope. The best result is coloured green, an acceptable result yellow.

Category	Scandroid	AnalysePoems	Calliope
Stress Assignment Method	Algorithm	Dictionary	Dictionary
Syllable Division Method	Algorithm	Dictionary	Dictionary
Stresses Phrasal Verbs ("found out")	No	No	Yes
Consistent Stress Allocation	Yes	no	Yes
Differentiates synonyms ("convict", noun and verb)	No	No	Yes
Stresses Compounds	By meter	By meter	By meaning
Levels of stress	3	3	4
Meter determines stresses	Yes	Yes	No
Promotes weak stress to strong metrical positions	Yes	Yes	Yes
Demotes strong stresses to weak metrical positions	No	No	Yes
Elision	No	By spelling	By phonetics
Compares variant readings	No	One variant at a time	All variant combinations
Variants derived from combinations of words ("it was" -> "'twas")	No	No	Yes
Meter repertoire	Iambic, anapaestic	Iambic, trochaic, dactylic, anapaestic, free verse	Iambic, trochaic, dactylic, anapaestic, Amphibrachic, Fixed meters (Sapphic, Alcaic), Accentual, free verse
Scansion methods	Traditional, simplified Generative Metrics	Statistical	Phonological metrics
Methods supported by experts	No	No	Yes
Calculates complexity	No	Yes, method not approved by experts	Yes, method not approved by experts
User input required	No	No	No
Generates pyrrhics	Very rarely	Sometimes	No
Generates spondees	Frequently	Rarely	Very Rarely
Level of scansion	Bad undergraduate	undergraduate	Post-graduate

Table 27: comparison of Scandroid, AnalysePoems and Calliope processes

As noted by Hayes (2005) no system currently exists which integrates recent linguistic developments in stress assignment and metrical theory.

2.6.7 Phonological Scansion's Accuracy

The table below (Table 28) gives an analysis of a line from Paradise Lost, called a "strange Monster" by Bentley, an eminent and early editor (Norbrook, 1999), to show the potential of a phonological scansion procedure. An illustration of the data, already shown earlier in Figure 2, now includes an assessment of Phonological Scansion (Figure 3) – the closer to the 'expected' mark the better the scansion, as detailed in 2.1.3.

System	Scansion	Meter	Comment
“Burnt after them to the bottomless pit.” (Milton, Paradise Lost 6.866)			
Stresses	smwnnwswms	Iambic penta-meter	Found in a poem of iambic pentameter lines, the verse is probably intended as iambic pentameter.
Trad. Scansion	sw|ws|sw|sw|ws	Trochaic penta-meter	Three trochees and two iambs produce a net trochaic rhythm.
Attridge	B ô b o O ŏ B ŏ B	Trochaic penta-meter	An offbeat is inserted between “burnt” and “after”, and “them” is promoted to a beat.
Newton (1824)/ Creaser (2007)	ss|ws|ws|sw|ws	Iambic Penta-meter	Creaser comments: “the only way to give the actual line five beats in a pattern compatible with the norms would be through an absurd emphasis on ‘the’”.
Creaser (2007)/ Whiteley (1958)	ss|ws|ww|sw|ws	Irregular Penta-meter	This is Creaser’s first choice: “the line is most likely to be read by insisting on five beats in abnormal sequence”. Whiteley decides on a pentameter not tetrameter reading because: “it is...strange to give it [a meter] in which the poet was not writing”
Groves/ Tsur (inserted break after ‘them’)	Burnt after them, to the bottomless pit A-o-o-A o-o-A-o-a s sw|ws |ww|sw|w s	Irregular Penta-meter	According to Groves, Tsur claims “experienced readers…fudge a metrical reading by forcing a break where none exists in the syntax” otherwise an anapaestic reading is heard.
Groves/ Creaser (2007)	A-o-o-A-o-o-A-o-a s s|wws |wws |wws	Ana-paestic tetra-meter	Groves avoids this reading by changing the syntax of the line. This is Creaser’s second choice: “once the absence of the third beat is felt, the line makes an aptly vertiginous climax to a passage evoking a fall”
Finch (1993)/ Whiteley (1958)	sww|sww|sww|s	Dactylic Tetra-meter	Finch comments: “the line has three dactyls and is hence ‘alien’ to the blank verse context”. Whiteley notes the dactylic reading is possible.
Scandroid	ssw|ww|ws|ww|s	No pattern	Scandroid fails to find either iambic or anapaestic meter.
Halle-Keyser	sswsw[wsw]ws	unmetrica l	The stress maximum has wsw where SWS is expected in the meter: the line is unmetrical
Phonologic al Scansion	x|mw\n}xx|swm]s> =(iambic=fIU, 1PP; trochaic=1PW3; anapaestic=fIU, 1PW3; dactylic=1PP) 5|7 units are iambic =sm|wn|nw|sw|ms (sw|ws|sw|sw|ws)	Iambic Penta-meter	1/3 units are trochaic, but the units with most weight are iambic. This gives a relatively regular iambic rhythm (71%). There is no consistent triple rhythm – two units are dactylic, and one is anapaestic, so the line is neither anapaestic nor dactylic.

Table 28: scansions of Paradise Lost 6.886, illustrating Phonological Scansion

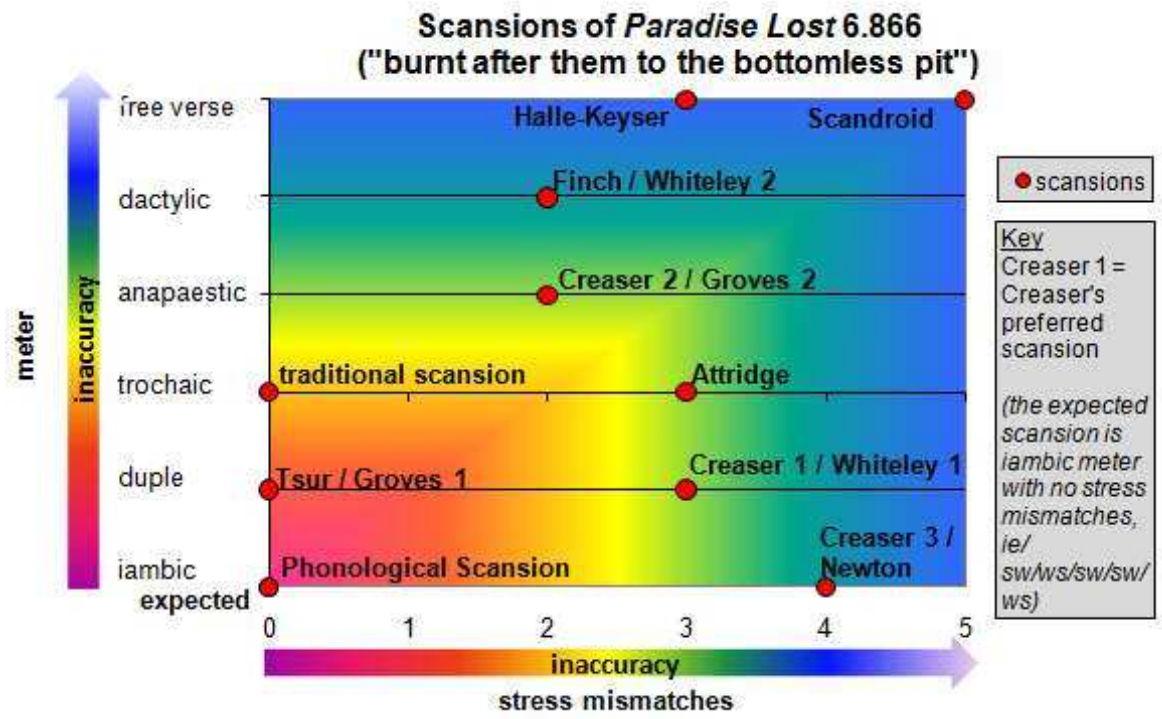


Figure 3: representation of the accuracy of scansions of Paradise Lost 6.866

Most experts expect a iambic pentameter line, because the surrounding lines are blank verse. However, since they do not distinguish more influential stresses, they are unable to identify how the natural stress pattern produces this rhythm. Phonological Scansion, relying on a prioritisation of stresses, is the only system which identifies the expected rhythm. Groves's theory and Tsur come close, but require the text to be amended to secure a pentameter rhythm which is neither iambic nor trochaic, although its feet are disyllabic (duple).

2.7 Research question

The question I hope to answer can be redefined in more specific terms: whether the following two uses of syntax improve computer scansion, initially using Scandroid as the benchmark:

1. the use of syntax in determining stress assignments

2. the use of syntax to identify meter (based on Hayes and Kaun, 1996) in the Phonological Scansion procedure

Syntactically determined stress assignments will be applied to Scandroid's program to identify improvements (Revised Scandroid). Phonological Scansion will be implemented in the Calliope application (which will also reference syntax to determine stress) and its results can be compared directly to the results of the previous application to determine whether syntactically-determined scansion makes any improvements.

A. Criteria for success of the applications

These computer applications would be successful if they:

1. identified acceptable lines which Scandroid rejects as unmetrical or impossible to scan
2. identified lines more quickly than Scandroid
3. corrected errors in Scandroid's scansion of lines (matching expert scansion where Scandroid failed)

B. Criteria for significant success of the applications

They would be significantly successful if they:

1. correctly identified more meters than Scandroid
2. correctly identified expert scansion more frequently than Scandroid
3. correctly identified more meters than all other competing scansion systems (Plamondon's AnalysePoems, Fabb's Bracketted Gird Theory)
4. correctly identified expert scansion more frequently than all other competing scansion systems

C. Criteria for overall success of Calliope

The hypothesis (that syntax improves scansion by both stress assignment and by identifying key elements to scan) would be proven if

1. both applications are at least successful (in the terms defined above) and
2. Calliope outperforms Revised Scandroid.

2.8 Summary

Most experts are agreed that Classical Scansion is not an adequate tool for scansion. The method needs to be modified to account for multiple stresses, and for syntactic patterns. Recent linguistic research has determined new methods of identifying metrical rhythm in poetry. In the 1990s, theories rejecting metrical scansion in favour of linguistic analyses of natural rhythms were popular. In this decade, the use of a rule-based scansion procedure has become the focus of scholarly activity. However, the phonological aspects of the initial research, and theories based on them, have been largely overlooked. I intend to use these insights, as well as linguistic models of stress assignment, to produce a more accurate scansion procedure than those currently used in computer scansion systems.

Chapter 3 Research Methods

3.1 Overview

The research depends on implementing three theories on the computer – first, Chomsky and Halle's stress assignment theory (Halle, 1998), with modifications for compound stress (Sproat and Liberman, 1987) and stress clash (Liberman and Prince, 1977); second, Selkirk's phonological phrase assignment theory (1981); and third, the phonological scansion theory derived from Hayes and Kaun (1996).

The first two theories are used to produce stress assignments derived from syntax, and the third theory is used to develop a scansion procedure based in syntax. Assessments of the effects of both modifications are used to assess whether syntax has a positive impact on scansion.

The computer implementation of all three theories was tested for accuracy against outputs detailed in the literature. The application was then compared with Scandroid to see which produces scansion closest to an expert's – measured by precise matches, and by matches with key features of expert scansion. Additionally, to determine the effect of the stress assignment theories, their output was processed by Scandroid to see if the resultant scansion was closer to expert scansion than the original Scandroid's. In each of these tests, non-expert opinions were used to assess the acceptable range of results. An assessment of the severity of any deviations was then made.

3.2 Stress Assignment tests

The first two theories were used to produce accurate stress patterns for verse which were analysed using Scandroid algorithms and the phonological scansion theory. The objective was to determine whether the phonological scansion theory improves Scandroid's algorithms when both have the same stress inputs. Each of the first two theories was tested as part of the overall assessment of the research.

One experiment was run to assess the most difficult and relevant stress assignment procedure – compound stress (nuclear stress is relatively straightforward and it is still debated whether stress clash operates in poetry)

I assembled a corpus of phrases with stress assignments identified and agreed by experts in the field: Bolinger (1972), Sproat and Liberman (1987) and Giegerich (2006). The experiment had the system process the corpus of data, and the results compared to the output predicted by the experts to assess how accurate the stress assignment function is. The results were analysed for statistically significant patterns underlying any discrepancies. Since the entire corpus of data can be tested in this way, it should be possible to identify conditions under which differences occur. The results are comparable to similar tests run on compound stress assignment algorithms by Sproat and Liberman (1987) and Lappe and Plag (2007) for two categories of compound stress – Sproat and Liberman recorded 80% success, and Lappe and Plag 95% success. So, the target for the experiment was an 80% success rate.

3.2.1 Computer Processing

(the results are given in 4.1.3)

Category	Description
Corpus	compound stress corpus
Test Subjects	Computer
Success Rate	The computer matches the expert assessment of the corpus in 80% of cases.
Method	the computer processes the corpus. Results are compared to expert assessments.
Assessment	The results for the computer are compared with expert assessments, and the distributions of errors are assessed by compound stress type.
References	Sproat & Liberman (1987) and Lappe & Plag (2007) Sproat & Liberman have 80% success, Lappe & Plag 95% success for their compound stress assignment algorithms

3.3 Phonological Phrase tests

To test phonological phrase assignments, a corpus of data was compiled from verses which have been assigned phonological classifications by experts, derived mainly from Selkirk (1981, 1984 and 1986), Hayes (1984a and 1989), Hayes and Kaun (1995 and 1996), Atterer (2000) and Cureton (1992, 1993 and 1997). The system processed the corpus and the output was compared to the expert assessments. Atterer (2000) has developed a method of statistically assessing the effectiveness of automatic prosodic phrase assignment against expert human assignments given in Equation 2. The data produced here can be easily modified to fit the parameters of Atterer's analysis.

Breaks-correct = ((Breaks-Deletion Errors-Substitution Errors)/Breaks) x 100%
Junctures-correct = ((Number of Junctures -Deletion Errors-Substitution Errors-Insertion Errors)/Number of Junctures) * 100%
Juncture-insertions = (Insertion Errors/Number of Junctures) * 100%

Equation 2*: Atterer's formulae for assessing phonological phrases*

3.3.1 Computer Processing

(the results are given in 4.1.1)

Category	Description
Corpus	entire phonological analysis corpus
Test Subjects	Computer
Success Rate	90% correct assignments
Method	The computer produces a phonological analysis which is compared to expert analyses
References	Hayes and Kaun (1995, 1996), Atterer (2000)
Assessment	the phonological data is assessed against expert scansions produced by Selkirk, Hayes and Cureton using the tests designed by Atterer (2000) and detailed in the footnote above. The results are assessed for statistical confidence.

3.4 Phonological Scansion theory

3.4.1 Phonological Scansion Procedure

I have synthesised the phonological scansion procedure from the theory outlined by Hayes and Kaun (1996), based on the idea that only the ends of phonological units are significant for meter assignment, and that larger phonological units have greater weight in this assignment – Table 29 gives an analysis of one line (further details are given in the Appendices on p. 148).

Type	**Analysis**
"Shall I compare thee to a summer's day?" (Shakespeare, Sonnet 130.1)	
Stresses	Shall(m) I(n) com(w)pare(s) thee(n) to(m) a(w) sum(s)mer's(w) day(s)?
Phonological Analysis	<{[Shall I /compare\ thee]} {[to a /summer's\] [day?]}>
Key Stresses	<{[xx/ws\n]}{[xx/sw\]s]}>
Phonological Scansion	10 syllables, 4 main stresses Duple units: Iambic (ws) = final IU ('mer's day'), 1PW2 ('a sum'), 1PW2 ('compare') [total=5] Trochaic (nx) =1PP ('thee to') [total=2] Triple units: Amphibrachic (wsw) = 1PW2 ('a summer's') [total=1] Dactylic (sww) =1PW4 ('Shall I com'), 1PW4 ('pare thee to') [total=4]
Analysis	The line matches 4/5 weighted units with a dactylic rhythm, but one unit is not matched (triple rhythms require 100% match): the line is not dactylic. The line is iambic with 5/7 weighted units matched (71%)
Result	**The analysis indicates that the line is iambic pentameter (mn|ws|nm|ws|ws). The iambic rhythm is strong (71% consistent).**

Analysis Symbol	**Scansion Symbol**	**Meaning**	**Scansion Weighting**
/ \	PW	Polysyllabic Word (incl. modal verb phrases)	1
[ ]	CP	Clitic Phrase	1
{ }	PP	Phonological Phrase	2
< >	IU	Intonational Unit	3

Table 29: Phonological Scansion of Shakespeare, Sonnet 130.1

3.4.2 Theory: Accuracy against Experts

(the results are given in 4.2.1)

In order to assess the effectiveness of the theory, I have collated a test corpus which includes the examples matching phonological phrases to meter given by Hayes and Kaun (1996) and Hayes (1984a), as well as examples of expert scansions of 17 individual verses and 32 poems – 372 verses in all. I can therefore determine whether the theory's output approximates expert scansion better than the scansions produced by the systems of Plamondon, Hartman and Fabb. The expert scansion used has been regularised by principles agreed in Baker (1996) to screen out idiosyncrasies. The sample size and the method are largely the same as those used by Hayes (2005) to test his textsetting algorithm allowing a comparison with his results.

Category	**Description**
Corpus	Poetry corpus of 372 verses, including 313 iambic poems (to assess Scandroid) and Shakespeare's Sonnet 130 (to assess AnalysePoems)
Test Subjects	1 test subject
Success Rate	Categories 1-6, 100%, categories 7-8 80%. Improves on competing theories' success rates or analyses, eg/ Hayes (71%), Fabb, Hartman, Plamondon, Groves
Tests Objectives	T1, T2, T3, T4
Method	The Phonological Scansion procedure is processed manually, by a human being. Results are compared to expert assessments and to competing procedures' results for verses in all eight categories of difficulty described above. The scansions will also be compared to non-expert scansions where these are available to determine if any approximate non-expert assumptions.
References	Hayes(1984), Hayes and Kaun (1996), Hayes (2005)
Assessment	The results are analysed for statistical confidence, and compared to each other. The iambic poems are used to assess Scandroid's performance, Sonnet 130 assesses AnalysePoems. Each assessment is presented in a separate results table, including the assessment of the entire corpus.

3.4.3 Phonological Scansion Application

The Phonological Scansion Theory is implemented in the Calliope application (the program flow is given in Figure 4).

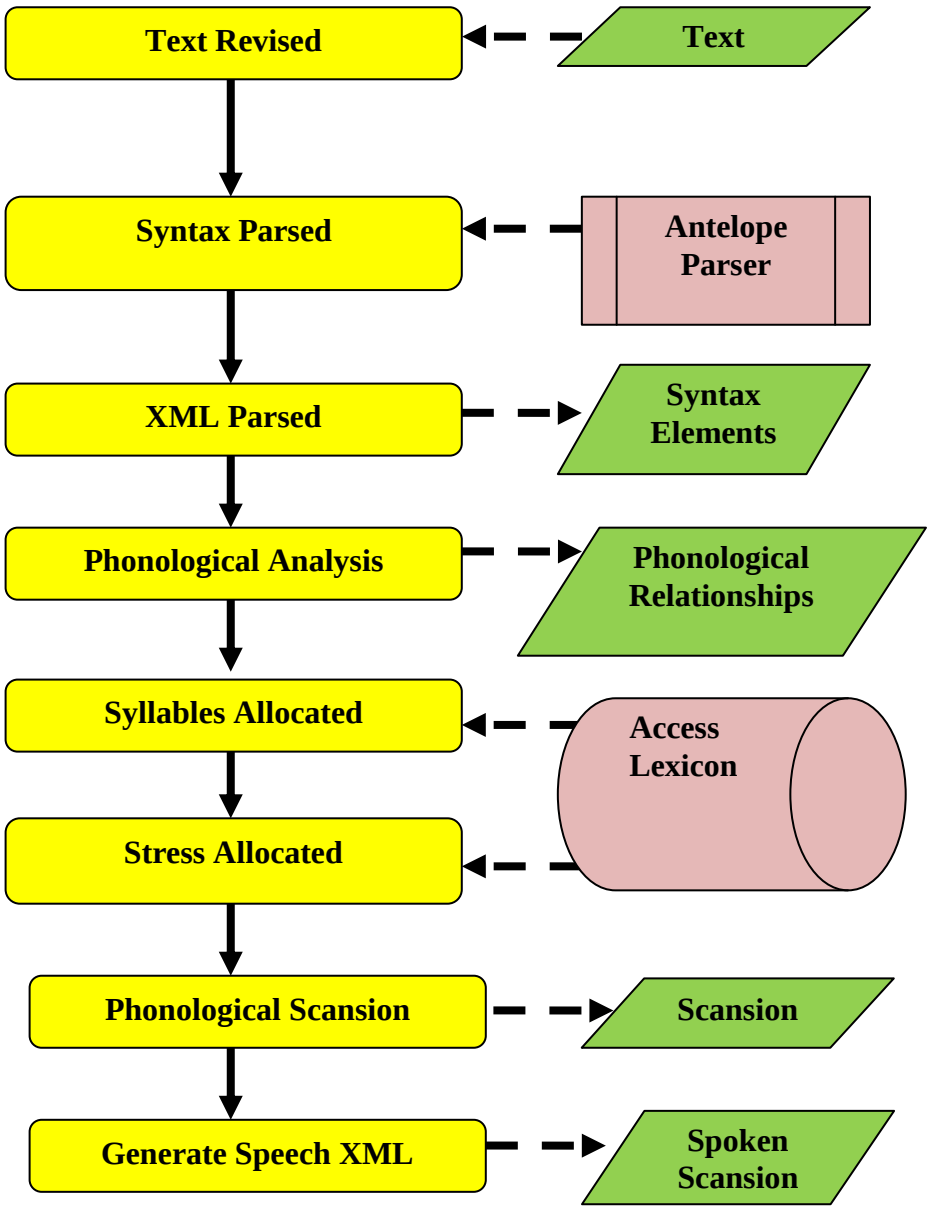


Figure 4: Calliope application program flow

An Antelope Natural Language Processor parses text into syntactic elements and relationships. Using this analysis, words are assigned stress from a lexicon, and are grouped into phonological units. The output is processed by a Phonological Scansion

algorithm, and the result is assessed for consistency: if the rhythm does not meet an arbitrary consistency, variant scansions of the line are assessed to determine the most consistent – see Table 30.

Type	Data
Original Text	Shall I compare thee to a summer's day?
Revised Text	Shall I compare you to a summer's day?
Syntactic analysis	Shall/VBD I/PRP compare/VB you/PRP to/TO a/DT summer/NN 's/POS day/NN ? (SQ Shall (NP I) (VP compare (NP you) (PP to (NP (NP a summer 's) day))) ?)
Phonological analysis	<{[Shall I compare thee]} {[to a summer's] [day]}>
Stress Patterns	shall(1) I(1) Com(0)pare(2) thee(1) To(1) A(1) Sum(2)mer(0)'s(-1) Day(2) nmwsnmnswm
All Stresses	<{[nm/ws\n]}mn/sw\]m]}>
KeyStresses	<{[xx/ws\n]}{[xx/sw\]s]}>
Meter	iambic pentameter (from the end back)
Stresses	(10 overall, 3 strong)
Analysis	Iamb(Reverse):5, 1IU, 2PW Trochee(Reverse):2, 1PP Dactyl(Reverse):3, 1IU Anapaest(Reverse):2, 1PW
Scansion	xx|ws|nx|xs|wm
Certainty	71% (5/7). The threshold is 55%, so no variants are tested.
Conclusion	The meter is iambic pentameter with a consistency of 71%

Table 30: Calliope analysis of Shakespeare, Sonnet 130.1

3.4.4 Application: Accuracy against Theory

(the results are given in 4.2.3)

A second experiment compared the application's phonological scansion with the theory's scansion for a corpus of test verses to determine how closely the application reproduces the theoretical results.

Category	Description
Corpus	Poetry corpus of 148 verses previously analysed by the Phonological Scansion theory, and representing the 8 categories of difficulty mentioned above
Test Subjects	Computer
Success Rate	Computer scansions match theoretical scansions 90% of time overall, and 100% of the time for categories 1 and 2.
Method	The computer will process the corpus using the application's Phonological Scansion, and compare to the Phonological Scansion's theoretical output.
References	Stauder (2000)
Assessment	The results are analysed for statistical confidence. The results are analysed within each difficulty category, and overall.

3.5 Phonological Scansion Application, Calliope

The main point of the research is to evaluate Phonological Scansion against Scandroid, using both Scandroid's native stress assignment system and the linguistic stress assignments.

The data for Scandroid's system is taken directly from the Scandroid program. In order to replicate the effects of scanning the revised stresses, I have reduced the four levels of stress to two levels depending on the relative weights of the stresses within a foot. I have then used words which Scandroid consistently scans as either strong ("test") or weak ("or") to reproduce the binary stress pattern, and had Scandroid scan these.

The reduction to binary stresses oversimplifies the procedure for Scandroid, but this is unavoidable – whilst ambiguous stresses could have been used for secondary and relatively weak stress (m and n), two consecutive ambiguous stresses in a foot (e.g. mn) might have been processed with incorrect relative stress whereas it is acceptable to most metrists to use relative stresses within feet.

3.5.1 Application: Speed against Scandroid

(the results are given in 4.3.2)

One criterion for success is efficiency, so the systems scanned the same verses on the same computer and the processing speed was recorded. The verses consisted of five iambic pentameter verses (since Scandroid copes best with iambic pentameter), of varying degrees of difficulty. Logs were used to indicate how long the program stayed in key functions. This quantitative data was assessed to determine whether the phonological scansion functions speeded up the scansion process or not. This test is similar to Zhang and Cercone (1999) who record the time taken for a CPU to process a stress assignment algorithm.

Speed tests were performed with Scandroid by processing large quantities of identical text, recording the time taken and calculating an average. Although it would have been preferable to use a computer timer procedure, C# versions of Scandroid proved unreliable, and the original Python program did not allow modifications. Moreover, since Scandroid is not consistent in the results it produces for a single text, an average at least calculates the most likely output.

Category	Description
Corpus	Poetry corpus of 5 iambic pentameter verses
Test Subjects	Computer
Success Rate	Calliope faster than Scandroid
Tests Objectives	A2
Method	the computer processes the corpus using the application's Phonological Scansion, and also using Scandroid with its own stress assignments. The speed of each of the processes is compared for the items in the corpus.
References	Zhang & Cercone (1999)
Assessment	The average speed over the 5 verses is compared to determine the faster application

3.5.2 Application: Accuracy against Scandroid

(the results are given in 4.3.1)

This test assessed how accurately the three systems replicate reliable expert scansion. The test was restricted by the features of the Scandroid program: the corpus was composed of representative verses of each common variant of iambic, anapaestic and trochaic verse, as well as some free verses – although the corpus was small, increasing it would not have extended the scope of the test though it might have increase the accuracy slightly. The three types of data which Scandroid displays are collected and compared against expert scansion. The output accurately replicated expert scansion if it matched the expert in each type of data.

Since the test assessed whether Calliope's stress assignments improved Scandroid's accuracy, this is a key test for evaluating the central proposition of the project – that stress assigned with reference to syntax improves scansion.

Category	Description
Corpus	Poetry corpus of 32 verses, each representing a common variant of anapaestic or iambic meters
Test Subjects	Computer
Success Rate	Calliope is more accurate than Revised Scandroid, if not Scandroid. Revised Scandroid is more accurate than Scandroid
Tests Objectives	A1, A3, B1, C2
Method	The computer processes a corpus of iambic pentameter and anapaestic meter using the application's Phonological Scansion, Scandroid and Revised Scandroid. The accuracy of each of the processes is compared for the items in the corpus against expert scansions for each meter. Data is collected for accuracy in identifying the correct number of feet, correct types of feet, and correct assignment of meter.
Assessment	The results are analysed for statistical confidence, and broken down by meter type.

3.5.3 Application: Accuracy against Non-Experts

(the test document is in the Appendices on pp. 200-203, and results are given in 4.4.3)

The output from the three systems was also compared to scansion produced by human beings to evaluate which is most effective. Although it was primarily compared to scansion produced by experts (who are more likely to identify the intended rhythm), since scansion is to some degree subjective, I also had non-experts scan verses to evaluate scientifically whether the systems are adopting non-expert assumptions in the scansion procedure.

For simplicity and consistency, these three tests were run on the same data which consisted of poems for which I have found expert scansions and which could be interpreted as iambic pentameter (that is, in meters with nine to twelve syllable patterns). This restriction tested Scandroid fairly since it only scans for ten syllable verse and is biased towards iambics.

There were at least 10 subjects for the non-expert test which was run on a computer (where subject chose the appropriate answer from drop-down lists or option buttons) or in hard-copy (where subjects marked the appropriate option from a list of alternatives) – this allowed subjects to use the format they were most comfortable with, so that they could focus on the test. Subjects were chosen with a variety of experience in poetry analysis to control against inexperienced subjects being led by external factors (such as question implication). In order to determine their relative experience, they were asked to assess their own experience and give details of their education background and exposure to poetry.

The subjects were given written verses with scansion produced by Scandroid with and without revised stress assignment, the phonological scansion procedure and other

theories (Base and Template Theory, Bracketted Grid Theory and AnalysePoems) where there is a difference in the scansions. Subjects were asked which scansion was best for a given verse, if any. This experiment was designed to test which scansion system matched subjects' assumptions best. The results were analysed quantitatively (how many matches for phonological scansion) and qualitatively (which verses produced the greatest variation among users).

The test makes two questionable assumptions. The first is that the expert opinion is reliable – where possible an opinion which is consistent and represents the principles if not the scansion of the majority experts is chosen, but in some cases the opinion may be faulty. Since I cannot avoid using the opinions chosen, to mitigate this error, I ran the test against opinions which are regularised according to accepted principles. The second assumption is that the subjects are representative of non-expert opinion as a whole – with such a limited sample this is unlikely, but, because I cannot use significantly larger numbers, it is unavoidable.

The test also deals with absolutes – whether there is a match or not with expert or non-expert scansion – whereas a test which assesses the degree of match would be more natural and potentially more revealing. Although Stauder's complexity measures could partly accomplish this by penalising any variations from meter, in practice metrists penalise particular variations more strongly, and some not at all – these measures do not approximate literary expert assessments well. However, a literary assessment test is described below.

Category	Description
Corpus	20 lines
Test Subjects	10 Subjects, Computer generates scansion
Success Rate	Categories 1-6, 100%, categories 7-8 80%. Improves on competing theories' success rates or analyses, eg/ Hayes (71%), Fabb, Scandroid, AnalysePoems, Groves
Tests Objectives	B2
Method	The computer processes the corpus using the application's Phonological Scansion, Scandroid and Revised Scandroid. The different scansions are presented, in a written format and electronic, to the test subjects, who chose which, if any, was more accurate. The results were compared to expert scansions and non-expert (subject) scansions, and the differences assessed.
References	Stauder (2000)
Assessment	The matches for each system to non-expert scansion were tested for statistical confidence. Matches to the most popular non-expert scansions are considered the most acceptable, but matches to expert scansion most accurate.

3.5.4 Application: Accuracy against Experts

This second test assesses the degree of variation from expert scansion. Assessments of non-expert scansion of lines from Pope by an expert in an online competition (McCaffery, n.d.) were analysed to determine the criteria used. Since the expert partly explains her judgements, critiques some of the scansions and gives her own scansion, it is possible to deduce how she decides on the winner. To guard against idiosyncratic judgements, her criteria were validated against the majority of expert opinions from Baker (1996), producing relatively universal benchmarks for expert scansion. These were applied to scansion produced by the three systems of the same material, and of one other poem (Shakespeare's Sonnet 130) which allowed AnalysePoems's scansion to be assessed. Although I am not aware of any other method which assesses or applies expert opinions, this one is designed to do no more than replicate an expert's judgment, so whilst unique it is probably reliable. The results are given in 4.5.

Category	Description
Corpus	Four poems, chosen to illustrate the scansion of Scandroid, AnalysePoems and human amateurs
Test Subjects	Computer generates scansion
Success Rate	Calliope matches the best human scansion, or exceeds it. Revised Scandroid produces a better scansion than Scandroid and AnalysePoems.
Tests Objectives	B2, B4
Method	the computer processes the corpus using AnalysePoems (where possible), the application's Phonological Scansion, Scandroid and Revised Scandroid. The expert's assessment of non-expert scansion in the online contest is used to derive the criteria employed, and these are applied to the computer scansions. The results are compared to expert and amateur scansions, and the differences assessed.
References	Stauder (2000)

Chapter 4 Results

Please note that the raw data for these results is given in the Appendices (p. 206).

Most of the data presented below is given an estimation of statistical confidence commonly used by statisticians (Becker, 1999) – the confidence value represents the range (confidence interval) above and below the given percentage into which 95% of all data, regardless of sample size, is likely to fall. Put another way, the lower the value, the more accurate the percentage. The confidence interval is formula is given in Equation 3, and assumes that the data fits a Bell Curve.

(for a confidence level of 95%)
Confidence Interval = $1.96 \times \sqrt{((p\ (100 - p)) / n)}$
where p is the percentage and n is the sample size

Equation 3: Confidence Interval

The data from subject tests is not given a confidence interval, because the sample taken is too small for confidence intervals to be of any real significance.

4.1 Application's Function Tests

(results from tests 3.2.1 (Stress), and 3.3.1 (Phonological phrase))

The function tests assess whether the phonological phrase and the most difficult stress assignment functions accurately produce the output expected in the literature. Successful implementation of these is a pre-requisite to the application's scansion functions and an assessment of the impact of syntax on scansion.

4.1.1 Summary

Type	Subtype	Match	Total	% Match	Target %	Confidence
Stress	**Compound Stress**	86	110	**78.2**	80	8%
Phonological Phrase	**Matched Phrases**	32	37	**86.5**	80	14% (11%)
	Breaks Correct		361 junctures	**91.1**	85	4%
	Junctures Correct		361 junctures	**90.0**	91	3%
	Juncture Insertions		361 junctures	**5.26**	3	2%

Key
Meets or Exceeds Target, Acceptably Close to Target, Fails to Meet Target

Table 31: computer processing results summary

The tests show that, in most aspects, the system is able to approximate Stress Assignments and Phonological Phrase analysis as well as, or better than, expected.

4.1.2 Phonological Phrase functions

The Phonological Phrase functions in general make their targets. The high figure for Juncture Insertions is still acceptable (judged by Atterer, 2000).

4.1.3 Compound Stress functions

The application's compound stress function almost makes its target. However, there are significant variations by the type of compound: Table 31 and Figure 5 below give a breakdown of these results.

type	Example	Correct	Total	percentage
agent	book seller	2	3	67%
empty	Farmer man	2	3	67%
lexical	Red herring	3	15	20%
material	Metal bridge	10	12	83%
measure	Pint jug	2	2	100%
place	City hall	23	26	88%
self	Self analysis	3	3	100%
time	Christmas Day	9	11	82%
no change		32	35	91%
Total		**86**	**105**	**78%**
Target				**80%**

Table 32: breakdown of compound stress assignment results

The function is unable to deal with compounds which require an understanding of the meaning for stress assignment (for example, agent and empty compounds), or those which are unpredictable exceptions to the normal rules and are required to be learnt by native speakers (termed 'lexical' compounds in the table). If these are excluded, the function makes its target. Of the remainder, the function performs less well with 'material' and 'time' compounds. This is probably because the database is less likely to mark the elements which produce a given compound type since there are so many elements. Overall, though, the compound stress function is acceptably accurate.

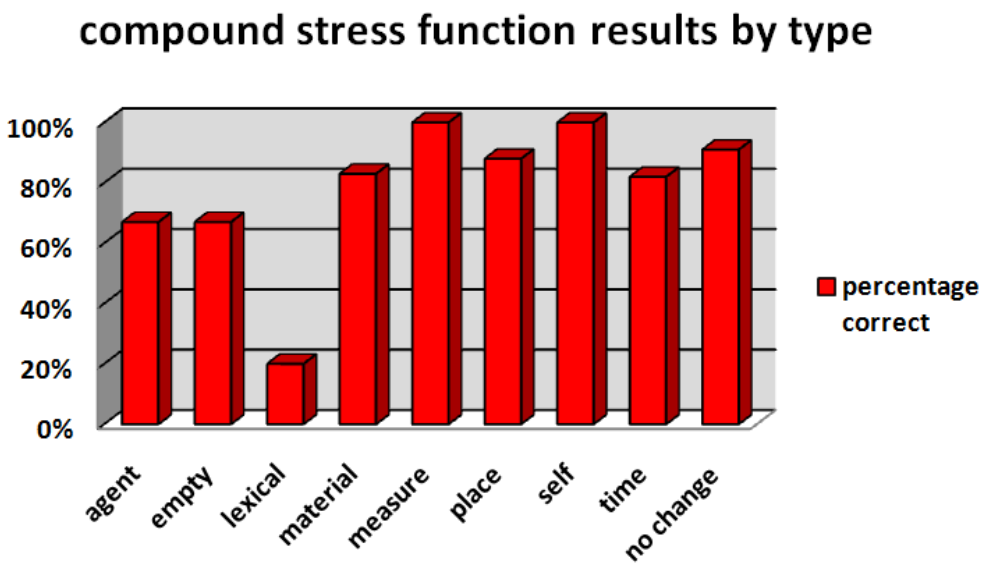


Figure 5: compound stress function results by type

4.2 Theory tests

The theory tests assess how accurately the Phonological Scansion theory reproduces expert scansion, compared to other scansion systems – and, consequently, whether the use of syntax improves scansion as suggested in the proposal. The tests also show how accurately the Phonological Scansion theory is implemented in Calliope. They are described in 3.4.2 and assess objectives T1, T2, T3 and T4.

4.2.1 Matches to expert scansion

(results from test 3.4.2)

The results of assessing Phonological Scansion and Scandroid against expert scansion of 313 iambic verses, and both systems against 14 verses scanned by AnalysePoems and experts are given below:

	count	all	%	Target	confidence
Scandroid Match	86	313	27.5%	71	10%
Phonological Scansion Match	194	313	62.0%	71	11%

Table 33: Phonological Scansion and Scandroid compared to expert scansions

***Sonnet* 130**	count	all	%	confidence
Scandroid Match	4	14	28.6%	24%
Phonological Scansion Match	6	14	42.9%	26%
AnalysePoems Match	5	14	35.7%	25%

Table 34 : Phonological Scansion, Scandroid and AnalysePoems*'s scansion* compared to expert scansion of Sonnet 130

The results indicate that AnalysePoems's scansion system is less accurate than the Phonological Scansion system, but more accurate than Scandroid in identifying expert

scansions of Sonnet 130. Since this is the only verse whose scansion Plamondon reveals, the assessment cannot be any more accurate.

4.2.2 Matches to expert scansion by Complexity Categories

The results of assessing the systems against regularised expert scansion of 372 verses are given in Table 35 (the scansion complexity categories are detailed above in Table 6):

Data	1	2	3	4	5	6	Grand Total
Scandroid Match	72	3	2	12	7	0	96
Phonological Scansion Match	173	10	4	37	16	1	241
Count	215	20	4	83	49	1	372
% Scandroid	**33.5%**	**15.0%**	**50.0%**	**14.5%**	**14.3%**	**0.0%**	**25.8%**
% Phonological Scansion	**80.5%**	**50.0%**	**100.0%**	**44.6%**	**32.7%**	**100.0%**	**64.8%**
Target	80	80	80	60	60	71	72
Scandroid confidence	6%	16%	49%	8%	10%	0%	4%
Phonological Scansion confidence	5%	22%	0%	11%	13%	0%	5%

Table 35: assessment of Scandroid and Phonological Scansion against ideal expert scansion by Scansion Complexity Categories

I have modified expert scansions to remove idiosyncrasies making them consistent internally, and with the agreed principles of other experts (Baker, 1996). Further arguments are given in the Appendices (p. 178). Although Phonological Scansion fails to meet its overall targets by 8%, in the simplest (Category 1) verses it succeeds (suggesting that the procedure shows promise). Moreover, it is over twice as effective as Scandroid. The confidence interval shows that the results are reliable.

4.2.3 Application Matches to Phonological Scansion theory

(results from test 3.4.4)

This test shows how accurately the application replicates the Phonological Scansion theory.

Category	1	2	3	4	5	6	7	8	total
Matched	37	19	0	46	25	0	1	6	134
Total	39	19	4	49	25	1	1	10	148
%	**95%**	**100%**	**0%**	**94%**	**100%**	**0%**	**100%**	**60%**	**91%**
confidence	**4%**	**0%**	**0%**	**4%**	**0%**	**0%**	**0%**	**8%**	**5%**
Target	100%	100%	100%	100%	100%	100%	80%	80%	90%
Total %	92.7%						63.6%		91%

Table 36: Calliope compared to Phonological Scansion theory

The application exceeds the target of replicating 90% of Phonological Scansions, and has a high probability of reliability (5% confidence interval overall).

4.3 Application Tests

The following tests assess how well the application compares to both the original Scandroid program, and the program using the stresses in lines assigned by the application. Assessments are made in terms of accuracy and speed. The accuracy test assesses one key claim of this proposal: whether stress assignments based on syntactic analyses improve scansion or not. If Scandroid scans better when it uses Calliope's stress assignments, then syntactic stress assignments will have improved the scansion.

4.3.1 Comparison to Scandroid with revised stresses

(results from test 3.5.2, and assessing objectives A1, A3, B1 and C2)

Absolute (Iambic)	Match Feet	Match Meter	Match Foot Length	Scandroid Error	Overall correct
Scandroid	25	26	22	5	20
Revised Scandroid	27	30	27	3	24
Calliope	30	31	31		29

Percentage (Iambic)	Match Feet	Match Meter	Match Foot Length	Scandroid Error	Overall Correct	Statistical Confidence
Scandroid	78%	81%	69%	16%	63%	17%
Revised Scandroid	84%	94%	84%	9%	75%	15%
Calliope	94%	97%	97%		91%	10%

Absolute (Anapaestic)	Match Feet	Match Meter	Match Foot Length	Scandroid Error	Overall correct
Scandroid	0	0	0	7	0
Revised Scandroid	0	0	0	7	0
Calliope	6	7	7		6

Percentage (Anapaestic)	Match Feet	Match Meter	Match Foot Length	Scandroid Error	Overall Correct	Statistical Confidence
Scandroid	0%	0%	0%	100%	0%	1%
Revised Scandroid	0%	0%	0%	100%	0%	1%
Calliope	86%	100%	100%		86%	12%

<u>Key</u>
"Match Feet" shows the % of lines with the feet matched to expert scansion
"Match Meter" shows the % of lines with the meter correctly identified (in this experiment, the most frequent foot determined the meter)
"Match Foot Length" shows the % of lines which have the correct number of syllables in all the feet
"Scandroid Error" shows the % of lines where Scandroid produced an error

Table 37: comparison of the accuracy of Calliope, Scandroid and Scandroid with revised stresses (Iambic and Anapaestic meters)

Absolute (Trochaic)	Match Feet	Match Meter	Match Foot Length	Scandroid Error	Overall correct
Scandroid & Revised Scandroid	0	0	0	15	0
Calliope	13	14	13		13

Percentage (Trochaic)	Match Feet	Match Meter	Match Foot Length	Scandroid Error	Overall Correct	Statistical Confidence
Scandroid & Revised Scandroid	0%	0%	0%	100%	0%	1%
Calliope	87%	93%	87%		87%	17%

Absolute (Free Verse)	Match Feet	Match Meter	Match Foot Length	Scandroid Error	Overall correct
Scandroid & Revised Scandroid	0	0	0	16	0
Calliope	5	5	7		5

Percentage (Free Verse)	Match Feet	Match Meter	Match Foot Length	Scandroid Error	Overall Correct	Statistical Confidence
Scandroid & Revised Scandroid	0%	0%	0%	100%	0%	1%
Calliope	31%	31%	44%		31%	23%

<u>Key</u>
"Match Feet" shows the % of lines with the feet matched to expert scansion
"Match Meter" shows the % of lines with the meter correctly identified (in this experiment, the most frequent foot determined the meter)
"Match Foot Length" shows the % of lines which have the correct number of syllables in all the feet
"Scandroid Error" shows the % of lines where Scandroid produced an error

Table 38: comparison of the accuracy of Calliope, Scandroid and Scandroid with revised stresses (Trochaic meter and Free Verse)

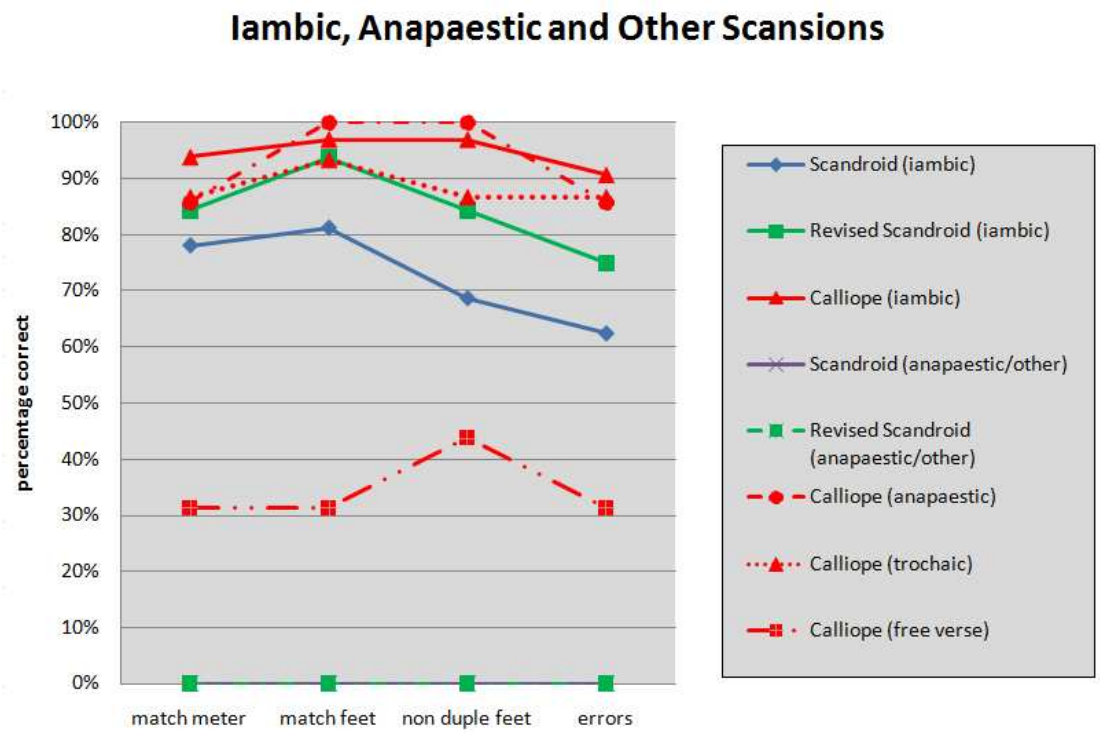


Figure 6: graph of the accuracy of Calliope, Scandroid and Scandroid with revised stresses

The results show:

1. Scandroid's procedure is unable to identify anapaestic verse, even when using revised stresses. This contradicts Hartman's claims (2005). It is also unable to identify trochaic or free verse.

2. Using the revised stress assignment function improves the accuracy of Scandroid's scansion: by 6% in identifying meter, 13% in identifying feet and 12% overall. This supports that claim that the revised stress patterns based on syntactic analyses are closer to the stress patterns perceived by native speakers and produce more accurate scansions.

3. Calliope is more accurate in producing expert scansion than Scandroid (even when revised stresses are used). It is also the only system which can identify anapaestic, trochaic and free verse.
4. Calliope is significantly less effective at identifying free verse. It is very effective at identifying iambic, anapaestic and trochaic verse (around 9/10 verses are classified correctly)

These tests are limited by the data inputted which represents common variations of different meters and produces a small sample size with a correspondingly low statistical confidence. However, this was necessary to accommodate changing the stress patterns in the Revised Scandroid application, as a pre-requisite to identifying the effect of syntax on stress assignments. The method for reproducing the revised stress patterns in Scandroid does not classify medial stressed syllables as ambiguous (which may alter the effectiveness of the system) on the grounds that these are identified as either weak or strong relative to the preceding stress, and the data from Scandroid screened out its (very limited) tendency to produce variant scansions from the same line on different runs. However, the conclusion that Revised Scandroid improves on Scandroid is probably secure – on no test did the Scandroid application improve on Revised Scandroid, and the margin of difference is (statistically) significantly large.

4.3.2 Comparison to Scandroid by speed of processing

(results from test 3.5.1 and assessing objective A2)

Process	average time (s)
Scandroid	0.174
Calliope (overall)	7.014
• stress allocation	1.074
• phonological units	0.148
• Syllables	5.048
• phonological scan	0.744

Table 39: Scandroid and Calliope processing speed

The results show that Scandroid is far faster than Calliope: it is 40 times faster than the entire process, and 4 times faster than the scansion function of the process. Scandroid is more efficient, but not necessarily more accurate than Calliope.

4.4 Non-expert Tests

The following results show non-expert assessment of scansion data already processed by the application and Scandroid (described in 3.5.3 to assess objective B2). The results are used to determine both which system more accurately reproduces expert scansion, and how acceptable the errors produced by the systems are to non-experts.

4.4.1 Summary

(results from test 3.5.3 and assessing objectives B2)

The following table shows a summary of the results from the non-expert tests.

% correct	Written Scansion Test
0.0%	0.0%
10.0%	9.1%
20.0%	18.2%
30.0%	27.3%
40.0%	18.2%
50.0%	27.3%
60.0%	0.0%
70.0%	0.0%
80.0%	0.0%
90.0%	0.0%
100.0%	0.0%
Total	100%

Table 40: test subject results summary

4.4.2 Experience of Non-experts

The following table summarises the experience of participants in the questionnaire, derived from conversations with the participants.

experience	level studied poetry at	currently read poetry	formally taught scansion	studied poetry post-secondary or formally taught scansion
total	**2 (undergrad)**	**1**	**2**	**4**
percentage	**11%**	**6%**	**11%**	**22%**
akini	school	no	no	no

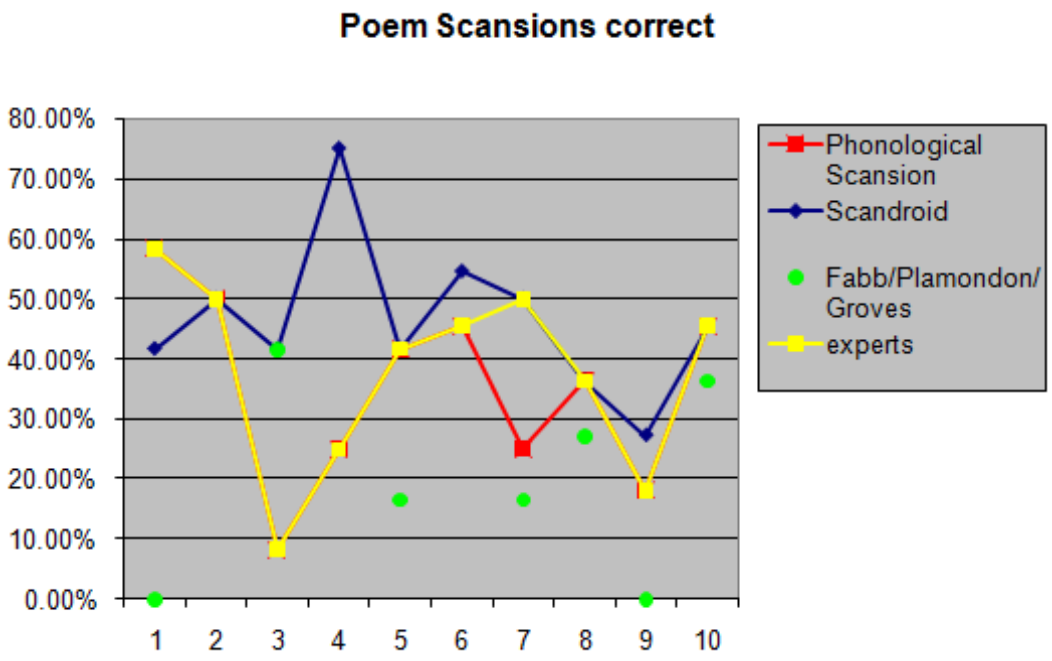
Table 41: levels of poetry experience of participants in the questionnaire

The table shows that the majority of participants (88%) have had no experience of poetry since school, and no experience of scansion. Although few have experience, this is predictable; and those that do, have a range of experience. The sample is probably as varied as could be expected and representative of the general population (given that it mirrors the assessments given in the literature in 1.3), implying that the results of the questionnaire are reasonably valid.

4.4.3 Scansion

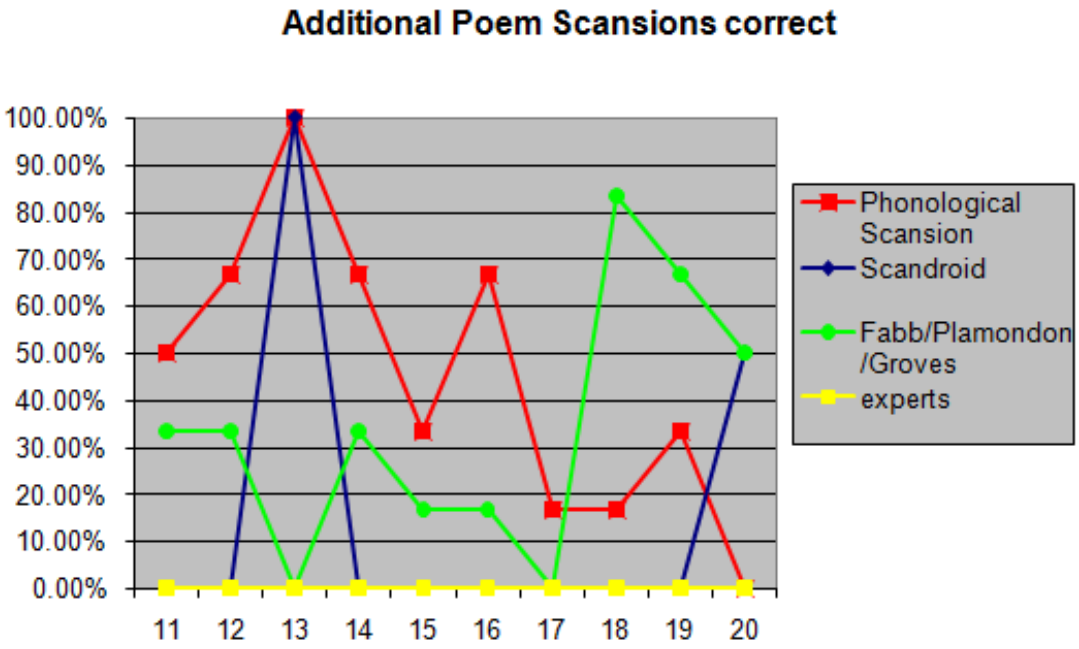
This section assesses the non-expert scansion of verses scanned by experts and by key systems. The results are used to assess the level of each of the computer systems, depending on the number of matches with expert scansion, and the acceptability to non-experts of any scansion which does not match with the experts' opinion. They will allow a comparison with Plamondon and Hartman's claims that their systems scan at undergraduate level, and will help to quantify how good computer scansion is with and without reference to syntactic information.

The following two figures show the matches to non-expert scansions for key systems scanning individual lines.



No	Line
1	Gone were but the Winter, come were but the Spring
2	The tide is full, the moon lies fair
3	The buzz saw snarled and rattled in the yard
4	No more to laugh, no more to sing
5	If hairs be wires, black wires grow on her head
6	A friend, a daisy, and a pearl
7	Only, from the long line of spray
8	than in the breath that from my mistress reeks
9	the Assyrian came down like the wolf on the fold
10	and yet, by Heaven, I think my love as rare

Figure 7: Scansion of Poems 1-10, showing Scandroid, Calliope, expert and other computer scansion against non-expert scansion



No	Line
11	and there him hideth and not appereth
12	when her lose gowne from her shoulders did fall
13	The emblem of two people being honest
14	Unruly Murmurs, or ill-tim'd Applause
15	By night he fled and at midnight return'd
16	Ay, some mad message from his mad Grandfather
17	Doth with his lofty and shrill-sounding throate
18	And those love-darting eyes must roll no more
19	Getting and Spending, we lay waste our pow'rs
20	When to the seasons of sweet silent thought,

Figure 8: Scansion of Poems 11-20, showing Scandroid, Calliope, expert and other computer scansion against non-expert scansion

The following two tables summarise the matches with expert and non-expert scansions for each of the systems.

System	% of non-expert agreement	Count	Total	confidence
Fabb	19.4%	7	116	7%
Groves	33.3%	20	116	9%
Plamondon	26.8%	9	116	8%
Scandroid	42.8%	62	116	9%
Calliope	40.2%	67	116	9%
Expert	37.9%	44	116	9%

Table 42: percentage of non-expert agreement with various scansion systems

System	% of expert agreement	Target	Count	total	confidence
Fabb	33.3%	71%	1	3	53%
Groves	50.0%	71%	1	2	69%
Plamondon	0.0%	71%	0	3	53%
Scandroid	27.3%	71%	3	11	26%
Calliope	75.0%	71%	9	12	25%
Non-expert	37.9%	71%	44	116	3%

Table 43: percentage of expert agreement with various scansion systems

The tests show:

1. Calliope is far better at approximating expert scansion than any of the other systems: 75% of expert scansions agree with the Calliope. Only 27% scansions agree with Scandroid. Plamondon does not agree with expert scansions at all, Fabb and Groves agree in 1/3 and 1/2 cases respectively. However, given that the scansion data for these three theories is limited, the figures give only a very general indication of their accuracy. Further more detailed and reliable indications of the superiority of Calliope over these theories are given in the Appendices. Calliope is the only application to exceed the accuracy achieved by Hayes for his text-setting algorithm.
2. Scandroid approximates a non-expert scansion better than any other system. This shows that the Scandroid system, where it succeeds, succeeds with a non-expert level of competence. It supports Plamondon's charge that Scandroid scans at the level of an undergraduate. However, Plamondon's implication that his system generates a more sophisticated scansion is not borne out by the (limited data): whereas his system does not match non-expert scansion, it does not match expert scansion either. The scansion it produces is quite idiosyncratic. In fact, with the significant exception of Calliope, all the other

systems are worse at approximating expert scansion than non-experts. As a means of improving the scansion of students, these systems are not effective. A comparison of Plamondon, Scandroid and Calliope with expert scansion given in the Appendices (p. 174) supports the conclusion that Calliope is the best system, but also indicates that Plamondon's system scans better than Scandroid, and at the level of an undergraduate.

3. Calliope did not match expert scansion in the two poems which contained contiguous strong stresses. It is not clear how the strong metrical stress is allocated in these contexts: it may be motivated by semantic interpretation or some other considerations. However, the application chose a scansion which, among non-experts, was ranked either first or second choice. Where it is mistaken, it produces a scansion which is still acceptable to non-experts. This conclusion is also supported by the additional evidence given in the Appendices (p. 174): where Calliope does not match expert scansion, its scansion is not penalised by the expert's criteria for assessing reasonable scansion.

The reliability of these conclusions is limited by the small sample size. However, the additional tests run also support them.

4.5 Expert Tests

The following tests (described in 3.5.4 to assess objectives B2 and B4) also determine how accurate Calliope, Scandroid and AnalysePoems are in reproducing expert scansion. However, they assess the degree of match to expert scansion using criteria deduced from an expert's marking of a scansion contest. The results will validate the previous conclusions, and show whether scansion is improved if syntactic data is used in stress assignment and scansion procedures.

4.5.1 Criteria for Expert Judgement

The tables below show the results of a scansion competition run by the Alsop Review (McCaffery, n.d.). The competitors were asked to email their scansion of Pope's Essay on Criticism 365-381:

> "The Sound must seem an Eccho to the Sense.
> Soft is the Strain when Zephyr gently blows,
> And the smooth Stream in smoother Numbers flows;
> But when loud Surges lash the sounding Shore,
> The hoarse, rough Verse shou'd like the Torrent roar.
> When Ajax strives, some Rocks' vast Weight to throw,
> The Line too labours, and the Words move slow;
> Not so, when swift Camilla scours the Plain,
> Flies o'er th'unbending Corn, and skims along the Main.
> Hear how Timotheus' vary'd Lays surprize,
> And bid Alternate Passions fall and rise!
> While, at each Change, the Son of Lybian Jove
> Now burns with Glory, and then melts with Love;
> Now his fierce Eyes with sparkling Fury glow;
> Now Sighs steal out, and Tears begin to flow:
> Persians and Greeks like Turns of Nature found,
> And the World's Victor stood subdu'd by Sound!"

The scansions were judged by an expert, A E Stallings, and the winner was declared to be Rodney Armstrong. The result is instructive: Peter Stewart Richards reproduces more of Stallings's scansion, but loses. By comparing Stallings's scansion and

comments with the competitors' scansion, it seems that some errors are less acceptable than others:

- feet of only weak syllables (coloured yellow) and are penalised (ww) – most prosodists agree with Stallings (Baker, 1996)
- feet of only strong syllables (coloured blue) are penalised (ss). However, since many prosodists accept spondees (ibid.), I have made the penalty half as severe as ww
- however, both patterns together (wwss) are not penalised. Stallings comments on the scansion of 'and(w) the(w) smooth(s) stream(s)':

  > "Most naturally for us, this would be read as two short syllables, two stressed... I might also, though have entertained an initial trochee"

  The wwss pattern, termed a "double iamb" is commonly accepted by prosodists (ibid.).

Stallings's formula is then:

Assessment = Verses Matching Expert Scansion – Verses containing Pyrrhics (except in Double Iambs) – ½ * Verses containing Spondees (except in Double Iambs)

Equation 4*: Stallings's scansion criteria*

Richards's scansion produces more of these unacceptable errors than Armstrong (in one line he produces both errors), and so his scansion appears more inaccurate.

Line	Peter S. Richards	Michael Pollick	Rodney Armstrong	Rachel Dacus	A.E. Stallings (expert)
Same	12	5	9	9	12
%	**71%**	**29%**	**53%**	**53%**	
Error	2.5	3.5	2.5	5	
%	**15%**	**21%**	**15%**	**29%**	
Total	**56%**	**9%**	**38%**	**24%**	
1	wswswswsws	wswswswsws	wswswswsws	wswswswwws	wswswswsws
2	swwswswsws	swwswswsws	swwswswsws	swwswswsws	swwswswsws
3	wwsswswsws	wwsswswsws	wwsswswsws	wwsswswsws	wwsswswsws
4	wsswswsws	wwssswsws	wwsswswsws	wswswswsws	wswswswsws
5	wsswswsws	wsswswsws	wswswswsws	wswswswsws	wswswswsws
6	wswswswsws	wswswswsws	wswswswsws	wswswswsws	wswswswsws
7	wsswwwsss	wsswwwsws	wswswwwsws	wswswwwsws	wswswswsws
8	wswswswsws	wswwwswsws	wswswswsws	wswswswsws	wswswswsws
9	wswswswswsws	wswwswswsws	wswwswswswsws	wswswswwws	wswswswswsws
10	wswswswsws	swswswswsws	wswswswsws	swwwswswsws	swwswswsws
11	wswswswsws	wwswwswsws	wswwswsws	wswwswsws	wswswswsws
12	swwswswsws	wwwswswsws	wwsswswsws	wwsswswsws	swwswswsws
13	wswswswsws	wswswwwsws	wswswwwsws	wswswwwsws	wswswswsws
14	wswswswsws	swsswswsws	wsswswsws	wwsswswsws	swwswswsws
15	wswswswsws	wswwwswsws	wswswswsws	wswswswsws	wswswswsws
16	swwswswsws	swwswswsws	swwswswsws	swwswswsws	swwswswsws
17	wwsswswsws	wwsssswsws	wwsswswsws	wwsswswsws	wwsswswsws

Table 44: Alsop Review scansions

The competitors' comments are helpful in establishing their relative experience (see Table 45) which is necessary when estimating the applications' levels by comparison.

Name	Academic	Practical	Comments	Category
Michael Pollick	High School	None	Pollick says "I haven't scanned a poem in anger since my Junior year in High School" (McCaffery, n.d.).	Secondary level education (school-leaver)
Rodney Armstrong	Engineer (Jones, 2001)	published some poems (Jones, 2001)		Amateur poet (undergraduate)
Peter Stewart Richards		posted poems for review online and implies elsewhere that he has tried to get published (Richards, 2006)	Stallings says "I'd have to say Peter's no amateur" (McCaffery, n.d.).	Amateur poet (experienced undergraduate)
Rachel Dacus	English graduate from Berkeley (Dacus, 2008)	published four books of her poetry as well as some poems in anthologies (Dacus, 2008)		Tertiary level education, poet (post-graduate)
A.E. Stallings	Studied Classics at Oxford and Georgia (Stallings, n.d.)	She describes herself as "a working metrical poet, not a scholar". Her published poetry has won some very prestigious awards (Stallings, n.d.)	Wikipedia notes "Stallings's poetry uses traditional forms, and she has been associated with the New Formalism, although her approach to formal verse is flexible, and she freely uses metrical substitution"	Expert (expert)
Key (Plamondon's categories – used in Plamondon, 2006a) Education categories				

Table 45: relative levels of experience of competitors in Alsop Review

I have used the same method to assess Scandroid and Calliope's scansion of Pope's lines and of Shakespeare's Sonnet 130 (which also assesses AnalysePoems).

4.5.2 Pope, Essay in Criticism: Comparing Scandroid and Calliope

Lines	Expert	Scandroid	Revised Scandroid	Calliope
Same		4	12	12
%	0%	24%	71%	71%
Error		4.5	0	0
%	0%	26%	0%	0%
Total		**-2%**	**71%**	**71%**
1	Wswswswsws	Wswswswsws	Wswswswsws	Wswswswsws
2	Swwswswsws	Sswswswsws	Swwswswsws	Swwswswsws
3	Wwsswswsws	Wwsswswsws	Swwswswsws	Swwswswsws
4	Wswswswsws	Wssswswsws	Wswswswsws	Wswswswsws
5	Wswswswsws	Wssssswsws	Wswswswsws	Wswswswsws
6	Wswswswsws	Wwsswsssws	Wswswswsws	Wswswswsws
7	Wswswswsws	Wssswswsss	Wswswswssw	Wswswswssw
8	Wswswswsws	Swwswswsws	Wswswswsws	Wswswswsws
9	wswswswswsws	Swswwswswsws	Swwswswswsws	Swwswswswsws
10	Swwswswsws	Ssswwswssw	Swwswswsws	Swwswswsws
11	Wswswswsws	Wswswswsws	Wswswswsws	Wswswswsws
12	Swwswswsws	Wwsswswsws	Swwswswsws	Swwswswsws
13	Wswswswsws	Sswswswsws	Wswswswsws	Wswswswsws
14	Swwswswsws	Swsswswsws	Swwswswsws	Swwswswsws
15	Wswswswsws	Sssswswsws	Wsswwswsws	Wsswwswsws
16	Swwswswsws	Swwswswsws	Swwswswsws	Swwswswsws
17	Wwsswswsws	Wwsswsswws	Swwswswsws	Swwswswsws

Table 46: Alsop Review scansions by Scandroid and Calliope

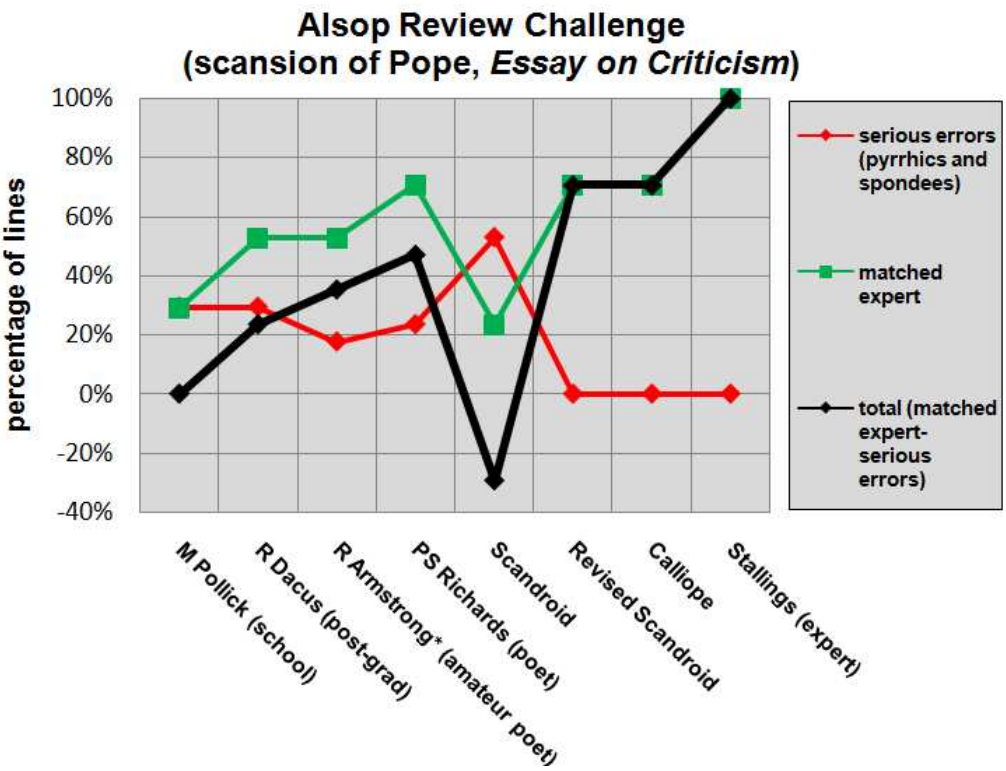


Figure 9: scansions in the Alsop Review Challenge

The results show:

1. Scandroid scans much worse than the competitors – it gets fewer scansions correct and makes many more errors by scanning sequences of strong stresses.

2. Calliope makes no errors, and matches the same number of expert scansions as the competitor who gets most correct – Richards.

3. When Scandroid uses the same stress pattern as Calliope (in Revised Scandroid), it produces an identical result.

4.5.3 Shakespeare, Sonnet 130: Comparing AnalysePoems, Scandroid and Calliope

The table below applies the same judgements to scansions of Shakespeare's Sonnet 130:

> "My mistress' eyes are nothing like the sun;
> Coral is far more red than her lips' red;
> If snow be white, why then her breasts are dun;
> If hairs be wires, black wires grow on her head.
> I have seen roses damask'd, red and white,
> But no such roses see I in her cheeks;
> And in some perfumes is there more delight
> Than in the breath that from my mistress reeks.
> I love to hear her speak, yet well I know
> That music hath a far more pleasing sound;
> I grant I never saw a goddess go;
> My mistress, when she walks, treads on the ground:
> And yet, by heaven, I think my love as rare
> As any she belied with false compare."

The expert scansion is by Raffel (1992).

	AnalysePoems	Scandroid	Calliope & Revised Scandroid	Expert
matched	9	6	9	14
%	64%	43%	64%	100%
errors	2.5	2.5	0	0
%	18%	18%	0%	0%
Total	**46%**	**25%**	**64%**	**100%**
1	wswswswsws	Wswswswsws	Wswswswsws	wswswswsws
2	wwsswswwss	Swwssswwss	Swwswswsws	swwswswsws
3	wswswswsws	Wswswswsws	Wswsswwsws	wswswswsws
4	Wswssssww s	Wswssssww s	Wswswswsws	wswswsswws
5	wswswswsws	Wssswswsws	Wswswswsws	swwswswsws
6	wswswswsws	Wssswswsws	Wswswswsws	wswswswsws
7	swwswswsws	Wswswswsws	Wswswswsws	swwswswsws
8	wwwswswsws	Wswswswsws	Wswswswsws	swwswswsws
9	wswswswsws	Wswswswsws	Wswswswsws	wswswswsws
10	wswswswsws	Wswswsssws	Wswswswsws	wswswswsws
11	wswswswsws	Wswswswsws	Wswswswsws	wswswswsws
12	wswswsswws	Wswswsswws	Wswswsswws	wswswsswws
13	wwwswswsws	Wswswwswsws	Wswswswsws	wswswswsws
14	wswswswsws	Wswswswsws	Wswswswsws	wswswswsws

Table 47: scansion of S*hakespeare's Sonnet 130 using A*lsop *Review's expert* assumptions

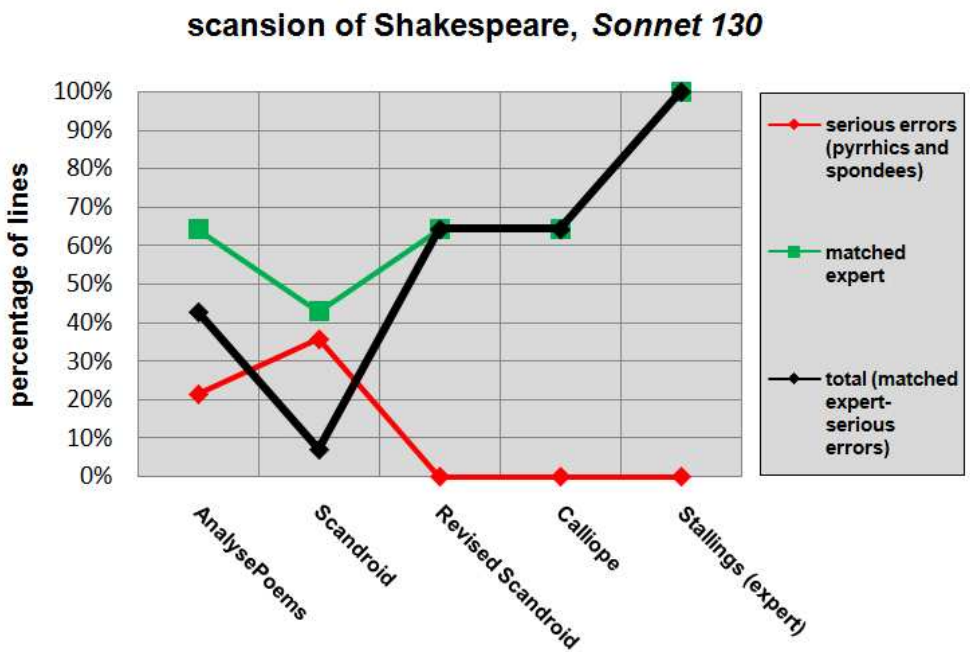


Figure 10: scansions of Shakespeare, Sonnet 130

The results show that:

1. Calliope produces the best scansion – it gets as many lines correct as AnalysePoems, but produces no significant errors. It exceeds the scansion of the best of the competitors.

2. AnalysePoems produces both types of scansion errors, one error in each of three lines. This indicates it scans as well as Richards, one of the better competitors in the Alsop Review, producing equally good and equally bad scansions.

3. Again, Scandroid produces the worst scansion, worse than any competitor in the Alsop Review.

4. Scandroid using the revised stresses produces a scansion identical to Calliope.

This test depends on the reliability of the scansion of the expert chosen. Stallings has excellent credentials and represents the majority of expert opinion (in Wallace, 1996) – although the weighting of particular errors may be wrong, it makes very little difference to the overall result (which is also confirmed by the two other tests). The algorithm derived by deduction is not a precise encapsulation of the expert's views; but its inaccuracy is unavoidable with such limited evidence, and acceptable since it provides the only means of assessing deviation from expert scansion.

4.5.4 Summary

1. Scandroid produces a scansion which is worse than the worst undergraduate student.
2. AnalysePoems produces a scansion which equates to that of a very accurate, if idiosyncratic amateur poet/undergraduate, but which is not the most acceptable scansion.
3. Both Scandroid and AnalysePoems accept spondees and pyrrhics where they are rejected by experts. This is partly due to inaccurate stress assignment procedures, though may also be due to an acceptance of some types of inaccurate scansion – see p. 179 in the Appendices for further discussion.
4. Calliope produces a scansion which betters the scansion of the most accurate amateur poet/undergraduate.
5. When Scandroid uses the stress assignments of Calliope, it replicates Calliope's results in most cases. The use of syntax dramatically improves stress assignments and its dependent scansion.

4.6 Results Summary

Table 48 and Table 49 assess syntactic procedures against the criteria for success developed in 1.2.3 (for the theory) and 2.7 (for the application).

no	criterion	Result
Successful criteria (1.2.3)		**The theory fails to meet the success criteria, but these may be set too high**
T1	Complexity Categories 1-6: 100% match experts	✘ The theory matches 92.7% of scansions in categories 1-6. However it does match 100% scansions in categories 2 and 5 (4.2.2)
T2	Complexity Categories 7-8: 80% match experts	✘ The theory matches 63.6% of scansions in categories 7-8 (4.2.2)
T3	Improves on competing theories' success rates	✓✓ The theory matches 64.8% of scansions for categories 1-6 compared to Scandroid's 25.8% (4.2.2)
T4	Replicates 71% of expert scansion	✘ The procedure replicates 64.8% of expert scansion (4.2.2)

Table 48: assessment of Phonological Scansion theory against success criteria

no	criterion	Revised Scandroid (stress determined by syntax)	Calliope (scansion determined by syntax)
Successful criteria (2.7)		**Revised Scandroid is successful, but Calliope is more successful**	
A1	Identifies acceptable lines which are rejected by Scandroid	✓ Using the syntactically-derived stress assignment procedure reduces Scandroid's errors by 7% (4.3.1) – that is, it scans 7% more lines.	✓✓ Calliope does not produce any errors (4.3.1) – it produces a viable scansion for evey line
A2	Quicker than Scandroid	N/A (same speed as Scandroid)	✘ Scandroid is 40x faster than the Calliope application (4.3.2)
A3	Corrects errors in Scandroid's scansion of lines	✓ Using the syntactically-derived stress assignment procedure increases Scandroid's accuracy by 12% (4.3.1)	✓✓ The syntactically-derived scansion procedure identifies 28% more iambic poems correctly than Scandroid (4.3.1)
Significantly successful criteria (2.7)		**Revised Scandroid is successful, but Calliope is more successful**	
B1	Identifies more meters than Scandroid	✘ Revised Scandroid only identifies the same meters as Scandroid: iambic and anapaestic. However, in practice neither is capable of identifying anapaestic lines (4.3.1).	✓ Identifies anapaestic meter (which Scandroid claims to) Identifies trochaic meter and free verse (which Scandroid cannot) (4.3.1)
B2	Identifies expert scansion more frequently than Scandroid	✓✓ Scandroid scans at the level of the worst undergraduate according to Stallings's criteria, whereas using syntactic data produces a scansion which betters the best undergraduate and approaches the level of expert scansion (4.5.4)	✓✓ Scans at a level close to expert (4.5.4) Agrees with around 40% more expert scansions than Scandroid (4.2.1, 4.4.3)
B3	Identifies more meters than AnalysePoems	As Phonological Scansion	✓ Agrees with 7% more expert scansion than AnalysePoems (4.2.1).
B4	Identifies expert scansion more frequently than AnalysePoems	As Phonological Scansion	✓ AnalysePoems scans at the level of a good undergraduate according to Stallings's criteria, whereas Calliope scans at a level close to expert (4.5.4),
Overall Success Criteria (2.7)		**Calliope is more successful than Revised Scandroid**	
C1	both applications are at least successful	✓	✓
C2	Calliope outperforms the Revised Scandroid application	✓ Calliope identifies more meters and more expert scansion than Revised Scandroid (4.3.1)	

Table 49: assessment of Revised Scandroid and Calliope against success criteria

The Phonological Scansion theory is much better than Scandroid's theory. Although it just fails to meet the high standards set for it, it is still significantly accurate.

Stress assignment which references syntactic information produces a much more realistic and accurate result than other computer scansion systems when compared to expert scansion.

The scansion procedure based in phonological theory is much more effective at identifying expert scansion and assessments of line meter than Scandroid, and three competing theories.

Chapter 5 Conclusions

5.1 Review of objectives

The two objectives of this study were to investigate the impact of syntax on stress assignments in lines of poetry, and on scansion.

These were assessed by implementing a linguistic stress assignment procedure, and a scansion procedure derived from Phonological Metrics (Phonological Scansion) in the Calliope application. Tests showed that both were implemented successfully.

To assess the effects of syntax on stress and scansion, three applications were tested: one which did not use syntax at all (Scandroid), one which used syntax for stress assignments only (Revised Scandroid), and one which used syntax for both stress and scansion assignments (Calliope).

Chapter 1 defined criteria for success for the Phonological Scansion theory involving matching expert assumptions more frequently than existing computer applications. Chapter 2 defined similar criteria for the Calliope application by comparing it to the Scandroid and Revised Scandroid applications.

Whilst the literature revealed methods to assess the accuracy of stress and phonological assignments, and the speed of processing (which were implemented in the study), there were no specific methods to assess the accuracy of scansions. However, I adapted methods to assess various aspects of scansion (and evaluated their principal weaknesses) in Chapter 3. Non-expert assumptions about the applications' scansions were gathered using a questionnaire; accuracy against expert assumptions was quantified from both absolute matches to expert scansions, and closeness of matches to the criteria deduced for one particular, representative, expert.

5.2 Review of similar studies

The literature review did reveal a number of previous computer applications of scansion: even though traditional methods of scansion produce subjective and unreliable results, these methods are used by the majority of computer scansion applications. Two of the most recent applications (Scandroid and AnalysePoems) have deficiencies in the assignment of stress to lines, and in the assignment of scansion.

Linguistic studies of meter have identified more scientific principles underlying the scansion process. This includes more natural stress assignment procedures (using syntactic theories such as the Nuclear Stress Rule and Compound Stress Rule), as well as a number of scansion theories based in Generative Metrics. Whilst most recent research is focussed on accurately reproducing variations in scansions (using Optimality Theory), observations which identify how English speakers identify meter using syntax (Phonological Metrics) have not been pursued, but offers opportunities for an improved computer scansion application.

5.3 Review of research methods

Unfortunately, there are no detailed assessments of computer applications in the literature, or studies of assessment methods. So the conclusions for this study are based on three generic research methods: a questionnaire, absolute matches to expert scansions and an algorithm to approximate one expert's assessment of scansions. Each has significant weaknesses.

The sample for the questionnaire is too small to generate reliable conclusions, and tested subjects who were uncertain about the accuracy of their responses.

Expert scansions were taken from the works of published literary scholars, but a given expert's scansions are not necessarily consistent with other experts, or even with their own scansions. Absolute matches to expert scansion only show whether a scansion precisely matches that expert's assumptions – it does not discriminate between non-matching scansions (usually the majority), some of which are likely to be closer to the expert scansion than others.

The algorithm used to assess scansion has inaccuracies: it was derived from one expert's judgements of a small set of data. The criteria had to be deduced from the expert's comments and assessments, and although it was verified against a wider range of expert opinion, the criteria are incomplete and may be inaccurate. This method has never been used before. However, it is the only way to evaluate the degree of inaccuracy of scansions.

The weaknesses of each method are in many cases balanced by the strengths of another – the relative subjectivity of expert scansions are balanced by the consistency of the expert algorithm. The uncertainty of the questionnaire assessments is balanced by the considered opinion of experts. The single expert opinion represented in both the algorithm and expert-matches is set against the range of non-expert opinions, allowing acceptable deviations to be identified by matches to native-speaker assumptions. The small size of the non-expert and single expert samples are balanced against the large size of the expert samples – and each percentage is assessed for statistical confidence. The three tests allow the accuracy of scansion to be quantified against both expert scansion and non-expert assumptions.

5.4 Findings

This study found that syntax improves stress assignments, producing a better approximation to natural stress patterns – the Revised Scandroid program produced better scansion of feet and identification of meter than the Scandroid program in the three tests.

The study also found that syntax improves scansion by using the Phonological Scansion procedure – specifically it is able to identify a much wider range of meters (including free verse) than Scandroid or Revised Scandroid. It is also able to approximate the expert scansion of individual feet significantly better than Scandroid, although it does not identify individual feet much better than Revised Scandroid since this part of scansion is controlled primarily by stress assignments which are identical in both programs.

In fact, the Calliope application (which implemented syntax for both stress assignments and scansion) reproduced expert scansion better than AnalysePoems, Bracketted Grid Theory and Base and Template Theory, and matched the scansion of the best English undergraduates. Syntax is, therefore, significantly successful at improving scansion.

5.5 Future Research

The Calliope could be improved by implementing a Stress Clash function using Youmans's model (discussed in the Appendices at p. 194). Functions could also be added to check for rhymes, and to identify possible rhetorical stress where words are

repeated within a line. The application currently tests lines for all possible metrical variants, prioritising syntactically permitted variants. It would be improved if an objectively verified weighting system could be developed to account for preferences of, for example, certain types of elision over certain types of insertions of syllables in lines, or to account for preferences across time, for example the low incidence of syllable-insertions in Jacobean poetry compared to Victorian.

It would be useful to determine more scientifically what the relative weights are for each of the phonological units used in the Phonological Scansion procedure, and interesting to see if it is as effective in other languages, such as Latin, Coptic, Aramaic or German (outlined in the Appendices on p. 151). There are also indications that it can define the conditions for loose meters more effectively than other theories – further investigation could validate the preliminary conclusions given on p. 171.

References and Index

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Note: I have indicated sources which I have not read, but which have been quoted in other sources, with *"(non vidi)"*

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Glossary

ACCENTUAL VERSE: verse with METER defined by the stress accent in syllables. This is a common system in more recent English verse

ACCENTUAL-SYLLABIC VERSE: verse which is both ACCENTUAL and SYLLABIC. This is the most common system for English verse from C14AD

ALCAIC: a Classical Greek METER applied to English poetry

AMPHIBRACH: one stressed syllable followed and preceded by one unstressed syllable (the adjective is amphibrachic; the word is sometimes given its Latinate form, amphibrachus)

ANAPAEST: two unstressed syllables followed by a stressed syllable (the adjective is anapaestic, the word can sometimes be spelled anapest)

BLANK VERSE: lines in IAMBIC PENTAMETER which have no rhyming scheme

CAESURA: a break in the middle of a verse. The plural is caesurae (sometimes the word is spelled cesura, and the plural cesuras).

CLASSICAL SCANSION: a method of SCANSION which uses FEET derived from Greek and Latin models, and identifies the METER by determining the dominant FOOT METER (usually the most frequent)

CLITIC PHRASE: a group of words consisting of a single CONTENT WORD and any unassigned non-content words which follow it, and any preceding non-contents words which are more syntactically dependent on it than any other content word

COMPOUND STRESS: the stress assignment in noun or verb phrases which consist of one of more words, each required to express the idea of the noun or verb

CONFIDENCE INTERVAL: a measure of the reliability of a percentage statistic – it expresses the range above and below the percentage value which where 95% of the data is likely to fall. The lower the interval, the more accurate the percentage

CONTENT WORD: a noun, adjective, verb, adverb or demonstrative pronoun

DACTYL: a stressed syllable followed by two unstressed syllables (the adjective is dactylic)

DIMETER: METERS with two FEET

DOUBLE IAMB: a PYRRHUS followed by a SPONDEE, and considered by some prosodists to be the equivalent of two IAMBS

DUPLE RHYTHM: FOOT METERS with two syllables – IAMBS, PYRRHICS, SPONDEES and TROCHEES

ELISION: when a word ending in a vowel is followed immediately by another word beginning in a vowel, the first vowel may be elided (that is, not sounded when the verse is scanned)

ENJAMBEMENT: lines which run across phrase breaks, usually sentence breaks (also spelled enjambment)

FOOT: groups of two or three syllables (as defined by the line METER) into which a line may be divided

FOOT METER: the METER that the RHYTHM of a FOOT matches. The most common foot meters are ANAPAEST, AMPHIBRACH, DACTYL, IAMB, PYRRHIC, SPONDEE and TROCHEE.

FREE VERSE: poetry which has no consistent METER

IAMB: an unstressed syllable followed by a stressed syllable (the adjective is iambic, and the word is usually given the Latin pronunciation 'yam' rather than Greek 'ee-am')

GENERATIVE METRICS: a linguistic approach to scansion which seeks to identify rules which generate all acceptable lines for a given METER

HEXAMETER: METERS with six FEET

INTONATIONAL UNIT: a phonological unit composed of syntactically connected PHONOLOGICAL PHRASES

LABELLING MISMATCH: occurs when the strength of a syllable's lexical stress is different from the stress expected in the METER

LEXICAL STRESS: stress assigned to a word from its pronunciation in prose, or natural speech

METER: the abstract pattern of stressed and unstressed syllables which all lines of metrical poetry conform to – there are many different individual meters. The word is sometimes spelled 'metre'

METRICAL SCHEME: see METER

METRICAL STRESS: alternations of strongly and weakly stressed syllables expected by the METER of the poem

OPTIMALITY THEORY: a theory used in linguistics which allows rules to be applied or omitted in particular contexts

PENTAMETER: METERS with five FEET

PHONOLOGICAL UNIT: a CLITIC PHRASE, PHONOLOGICAL PHRASE or INTONATIONAL UNIT

PHONOLOGICAL PHRASE: a PHONOLOGICAL UNIT composed of syntactically connected CLITIC PHRASES

PRAGMATICS: a theory in linguistics which details how phrases communicate their meaning

PYRRHIC: an unstressed syllable followed by another unstressed syllable (the adjective is also pyrrhic, although sometimes pyrrhus is used as the noun)

QUANTITATIVE VERSE: verse with METER defined by the quantity of syllables: whether syllables are open (consisting of short vowels not closed by a consonant), or closed (not open). This is the usual system for Greek and Latin verse

RHYTHM: the actual pattern of stressed and unstressed syllables in a line or FOOT

SAPPHIC: a Classical Greek METER applied to English poetry

SCANSION: the process of determining the RHYTHM and METER in a line of poetry, or the end result of the process

SPONDEE: a stressed syllable followed by another stressed syllable (the adjective is spondaic)

STRESS MAXIMUM: a GENERATIVE METRICS term to denote a strong stress between two weak stresses in the same syntactic phrase (a CLITIC PHRASE or PHONOLOGICAL PHRASE) – the plural is stress maxima

SYLLABIC VERSE: verse whose METER is defined by fixed numbers of syllables

TEMPORAL VERSE: verse whose METER is defined by the length of time taken to pronounce its lines

TEXTSETTING: the process of matching lyrics to musical beats

TETRAMETER: METERS with four FEET

TRIMETER: METERS with three FEET

TRIPLE RHYTHM: FOOT METERS with three syllables – AMPHIBRACHS, ANAPAESTS and DACTYLS

TROCHEE: a stressed syllable followed by an unstressed syllable (the adjective is trochaic)

Appendices

Appendix A - Phonological Scansion Procedure

In this section, I will discuss the theoretical Phonological Scansion procedure as derived from the conclusions of Hayes and Kaun (1996), and then the practical process derived from it and implemented in the Calliope application (as well as the patterns the process uses to match meters). I also detail some applications of the procedure in other languages and in solving some metrical problems as a demonstration of its effectiveness and validity.

A1 Theoretical Process

1. Mark the stress contours on a word/compound (0 – 5+)
2. Mark the relative stresses within the word (ws)
3. Mark the following syllables for Metrical Strength (the value the syllable receives with reference to the metrical scheme)

Type	Stress Level	Metrical Strength	Confidence Level
Polysyllabic compound/ phrasal verbs	Strong	Strong	High
Polysyllabic non-compound	Strong	Strong	Moderate
End of Phonological Phrase	Strong	Strong	Very High
End of Intonational Unit	Strong	Strong	Very High

Table 50: Phonological Scansion phonological units

4. Deduce the metrical schemes appropriate from:
    a. the position of the strong stresses,
    b. syllable number
    c. metrical schemes of other lines
5. Match strong metrical syllables to the metrical scheme of strong stresses (where weak stresses may be strong or weak)
6. Match the remaining syllables, rationalizing any elided syllables

A2 Practical Process

The following process describes the Phonological Scansion procedure. It is partly implemented in the Calliope application. Steps which are not implemented in the application are coloured blue and marked.

1. Get Stresses
    a. Mark the stresses in the line

b. Determine the relative stresses in each PP (w=weak, n=some stress, m=subordinate strong stress, s=strong stress)

2. Record Key Stresses
    a. Record all the stresses in a polysyllabic word (PW)
    b. Record the left-most stress in a Clitic Phrase (CP)
    c. Record the left-most stress and its preceding stress in a Phonological Phrase (PP)
    d. Record the left-most stress and its preceding stress in an Intonational Unit (IU)

3. Get the Syllable Count
    a. Count the syllables
    b. Match the stresses to patterns with the syllable count, else resolve missing or combined stresses and recount the syllables and get the new matching pattern.
    c. Final weak stresses
        i. Final weak stresses may be feminine endings i.e./ extra stress on pattern
        ii. Two final weak stresses are very likely to be double feminine endings (wn or ww)
    d. Resolve stresses
        i. resolve ww>W, or nw->N
        ii. stress clashes (ss, ms) > s-s,
        iii. add additional stress at intonational breaks or phonological phrase breaks

4. Record type of matches
    a. Find the number and type of stresses that match each pattern (allocate the greatest unit to the stress pattern where there is more than one unit).

5. Evaluate the matches
    a. Evaluate which pattern matches best – give greater priority to final units, then IU, then PP, then PW, then CP.
    Give the matches the following numerical values and choose the pattern with the highest total:
        IU=3
        PP=2
        PW3+ (a polysyllabic word with 3 or more syllables)=2
        CP=1
        PW2 (a polysyllabic word with 2 syllables)=1
    For triple rhythms:
        i. allocate patterns which do not match a triple rhythm to a None pattern, e.g./ mws, swm, sw, s)
        ii. assign dactyls and anapaests with strongly marked dipping rhythm (swm and mws) to None.
        iii. less marked dipping rhythms (e.g./ swn, smw or wms, nws) are permissible anapaests and dactyls

(NOTE: the following steps are not implemented in the Calliope application)

    b. Where there is a tie,

i. use surrounding patterns.
ii. If the syllable count is low, the line may be in no known meter.

c. Where the line is ambiguously trochaic or iambic, use the rhythm of polysyllabic words to determine the meter – the rhythm of the greatest number of words establishes the meter:
   i. Disyllabic words with a ws pattern are iambic, sw are trochaic
   ii. Use the pattern of the final two syllables of polysyllabic words
d. Accentual verse can be identified where the number of strong stresses in a line is constant, even if the number of syllables changes. There is likely also to be no dominant rhythmical pattern in the line.
e. Free verse can be identified where there is no dominant rhythmical pattern, and the number of strong stresses varies.

A2.1 Example

Line: Shall I compare thee to a summer's day?
Stresses: Shall(m) I(n) com(w)pare(s) thee(w) to(n) a(w) sum(s)mer's(w) day(s)?
Phonological Analysis: <{[/Shall I compare\ thee]} {[to a /summer's\] [day?]}>
Key Stresses: <{[/mnws\w]}{[xx/sw\]s]}>

Phonological Scansion Analysis:
10 syllables, 4 main stresses

Total duple units to match=1IU, 1PW2, 2PW4 = 8
Iambic Units (ws) = final IU, 1PW2, 1PW4 (total=6)
Trochaic Units (sw) =1PW4 (total=2)
Spondaic Units (ss) = none
Pyrrhic Units (ww) = none

Total triple units to match=2PW4, 1PW2 = 5
Anapaestic Units (wws) = none
Amphibrachic Units (wsw) = 1PW2 (total=1)
Dactylic Units (sww) =1PW4, 1PW4 (total=4)

The line matches only 2/3 units with a dactylic rhythm, and is not therefore dactylic.
The line is iambic with 6/8 units matched (75%)

The analysis indicates that the line is iambic pentameter with the scansion mn/ws/wx/xs/ws. The iambic rhythm is very strong (75% regular).

A3 Patterns

Syllable Count	Meter Name	metrical scheme
2	Iambic	Ws
2	Trochaic	Sw
2	Spondaic	Ss
2	Pyrrhic	Ww
3	Anapaestic	Wws
3	Dactylic	Sww
3	Amphibrachic	Wsw
7	Anacreontic	None
11/11/11/5	Sapphic	Swsxswwswss (x3) Swwss
11/11/9/10	Alcaic	s/sw/ss/sww/sx/s (x2) s/sw/ss/sw/sx sww/sww/sw/sx
Variable (13-17)	Classical Dactylic Hexameter	Sw(w)/sw(w)/s,w(,w)/sw(w)/sww/sw (final feet= sww/sw, caesura in 3rd foot)

Table 51: Phonological Scansion metrical patterns

A4 Application to Other Languages

One method of assessing the validity of Phonological Scansion in English poetry is to determine whether it works in similar contexts in other languages. In the section below I have applied the theory to accentual syllabic poetry in four languages, German, Neo-Aramaic, Coptic and Medieval Latin. However, it could also be applied to Dutch, Early Latin, Estonian, Modern Hebrew, Modern Greek, Old English, Old Icelandic and Norse dróttkvætt, Slovene and Russian, among others.

A4.1 German

A4.1.1 Analysis

German prosody is very much like English: it is derived from Classical Greek and Latin examples and applied to a stress-based language, resulting in accentual-syllabic meters.

No.	Line	Translation	Scansion	Comment
Lessing, Nathan der Weise 3.7.2043-2047 (from Lessing, 1893)			**Expected meter: hexameter (ws|ws|ws|ws|ws|ws)**	
2043	Es strebe von euch jeder um die Wette,	let each endeavour	ws|ww|ss|ws|ws	The inversion occurs in the clitic phrase {[von(w) euch(s)] je(s)der(w)}
2044	Die Kraft des Steins in seinem Ring an Tag	To vie with both his brothers in displaying	ws|ws|ws|ws|ws	No mismatch
2045	Zu legen, komme dieser Kraft mit Sanftmut,	The virtue of his ring; assist its might	ws|ws|ws|ws|wsw	No mismatch
2046	Mit herzlicher Verträglichkeit, mit Wohltun,	With gentleness, benevolence, forbearance,	ws|ww|ws|ws|wsw	The mismatched weak stress has a secondary accent, so fits the metrical pattern
2047	Mit innigster Ergebenheit in Gott	With inward resignation to the godhead	ws|ww|ws|ws|ws	The mismatched weak stress has a secondary accent, so fits the metrical pattern
Goethe, Hermann und Dorothea 4 (from Chisholm, 1995 – the translations are from Goethe, 1914)			**Expected meter: loose hexameter (sw(w)|sw(w)|sw(w)|sw(w)|sw(w)|sw)**	
4	Und es hingen herein Gutedel und Muskateller	There were the Muscatel, and there were the Chasselas hanging	sw|sww|ss|ww|sww| sw	Gu(s)te(w)del(w)] has a mismatch at the start of a clitic group
Hebbel, Mutter und Kind 370, 1035 and 1459 (from Chisholm, 1995 – the translations are my own)			**Expected meter: loose hexameter (sw(w)|sw(w)|sw(w)|sw(w)|sw(w)|sw)**	
370	Und der Jasmin vom Athos die Mitternachtsstunde verkündigt	And the jasmine of Athos the midnight hour announces	sww|sw|sww|sws|w ww|sw	die(w) Mit(w)ter(w)nachts(s)stun(w)de(w)} the mismatch occurs in the middle of a phonological phrase
1035	Brand, Viehsterben und Krieg euch wider Verhoffen betreffen	fire, livestock-death and war against your hopes concern you	ss|www|sw|sww|sw w|sw	Vieh(s)ster(w)ben(w)] The mismatch occurs at the start of a clitic group
1459	Und des plötzlichen Krähens? Der Hahn macht eben Visite	And of the sudden crow? The cock just makes a visit	sw|sww|sww|ss|ww w|sw	macht(s)}e(w)ben(w)] Vi(w)si(s)te(w)]}> the mismatch occurs at the end of a phonological phrase. However, only one unit shows a mismatch, the five others match the hexameter pattern – the rhythm is not affected by the mismatch
Key stress mismatches are in red, w is a weakly stressed syllable, s is a strongly stressed syllable				

Table 52: analysis of three German poems

A4.1.2 Discussion

In four of the five instances of mismatches, the mismatches occur at the start of phonological phrases – in the places predicted by the Phonological Scansion theory. However, in Hebbel, Mutter und Kind 1459, the stress mismatch occurs at the end of a significant phonological phrase. The full phonological scansion of the line is

the expected metrical pattern is loose hexameter:
sw(w)| sw(w)| sw(w)| sw(w)| sw(w)| sw(w)

[Und des plötzlichen] [Krähens]>? [Der Hahn]} [macht]} [eben] [Visite]}>
sw/\sww\]/\sw\>w/s}s}/ww]\w/sw\>
=xx\sww\sw>ws}s}xw]xsw>
(matches hexameter:1PW3, 2IU, 1PP, 1CP does not match hexameter:1PP)

The scansion shows that only one unit shows a mismatch, whereas five others match the hexameter pattern – the rhythm is probably not affected by the mismatch.

A4.1.3 Conclusions

The German data backs up the Phonological Scansion theory – even where a mismatch with the expected meter occurs in a significant phonological unit, the number of phonological units reinforcing the meter far outweighs the effect of the one deviant unit.

A4.2 Neo-Aramaic

This section analyses the following Neo-Aramaic poem “The Teakettle and the Boys” (translation and text in Yarre, 1957)

This discourse will well reveal
The colloquy between a teakettle and two boys.
The reader will be filled with astonishment.
Who are those two boys?

"The sun has risen; oh teakettle, what is doing?
Ascend onto the stove, sound thy voice!
And when thou hast boiled turn thy face (to us)!
That we may drink our tea and go to play."

"The whole company, come assemble!
See my steam and hear my sound!
Bring the tea-glasses and our sugar!
I will pour and you may drink."

"As thou hast ordered me have done, oh teakettle!
We have seated ourselves at the table, in order and as is our custom,
At our side the tray and with us our sugar;
In drinking tea we will enter a race."

"Get ready the tea-glasses,
The butter, the cake, the cheese, well understood!
Each one for himself may loud proclaim:
'I will drink three or four of those (glasses).' "

"The table is set, the appetite is craving;
We have not eaten, we all have not tasted anything.
Oh teakettle, teapot! Pour in all you can!
Our stomach is hungry and frightfully so."

"From the stove here I descend,
Above the glasses I hover;
My companion the teapot I have called to me,
In love and peace at its side I stand."

"Oh sweet pair, hail and welcome!
Come near, still closer!
Do your service in every respect!
So we will eat and drink with pleasure."

"Oh teapot, my companion, thou first start
And with colorful tea adorn the repast!
Then make a sign to me
That I may pour water with fervent zeal."

'(Here comes the tray with great beauty;
The spirit rejoices of those sitting at table.
He may come forward whose turn is called!
So we drink our tea and eat our food."

"The tray has risen and goes round the table.

It has completed their round, has returned to its place.
Then the spoon entered into the glass,
Stirred it well and prepared it nicely."

"Bravo! How is thy tea so sweet!
Come on, let us spread butter on the cake!
Mayest thou fare well, none compares with thee!
We hope thy provisions will never deplete."

"What kind of talk is this there?
Instead of glasses use bowls!
I am set for as many as seven (glasses for each);
I will pour until your stomach is satisfied."

"Thy words renew our appetite,
Thy tea is quenching our thirst,
But our gusto is not abated yet;
Smacking of lips has just started among us."

"Eat, eat, just cram it in!
The butter and the cake clean off from the table!
Now a dish of cheese pull over to you!
And when you are finished, then go strolling about!"

"We have eaten and are filled, we are so pleased;
Delight and pleasure we greatly derived;
From thee great favor we have received;
And now remain in peace, we do now go."

A4.2.1 Description

Neo-Aramaic is a Semitic language spoken in Iran and derived from Syriac which is closely related to Hebrew. A native-speaker of Neo-Aramaic, Yarre describes the metrical structure of the poem:

> "In its metrical arrangement our poem is following old Syriac tradition. We have here stanza division, each stanza in this case having four lines and each line consisting uniformly of four metrical feet of two syllables each. The metrical accent rests invariably on the first syllable of each foot, thus (reading from right to left) ws|ws|ws|ws. This metrical tone almost always falls on the penultimate, as it is this syllable which regularly carries the tone in modern Syriac. Of course, the metrical tone may also fall on any monosyllable"

The poem consists of 64 lines of iambic tetrameter, with the metrical accent coinciding with strong lexical stresses: it has a very similar metrical structure to English iambic verse.

A4.2.2 Exceptions to meter

I have detailed the cases where the stress accent does not coincide with the metrical accent. In the table below, the long syllables are marked with – (e.g./ ē), and metrically stressed syllables are marked with / (e.g./ é)

Line	Aramaic	Translation	Comment
1.1	á durásha spâi bit gáli	This discourse will well reveal	Monosyllable has stress
1.2	lmámla dcâidan wditrē yáli;	The colloquy between a kettle and two boys	di requires a metrical accent contrary to its lexical accent. Sw}sw} ws] sw}> Reversed foot in CP
1.3	qá qaryána bsír bit máli;	The reader will be filled with astonishment	Monosyllable has stress
1.4	máni- ’ína án trē yáli?	Who are those two boys?	Monosyllable has stress
3.3	l’íst.ikáni ulqándan mémun	Bring the tea-glasses and our sugar	Extra syllable in áni ulqándan swsw]wsw}sw> Extra syllable in CP
4.4	qa câi stáya b’órah. Lmâidan	In drinking tea we will enter a race	Reversed foot in CP Ws]sw}sw}sw>
6.1	súpra mlíli, ptíḥla ’ístâu	The table is set, the appetite is craving	Final long syllable treated as weak
6.4	stúmkan kpínta umári sâu	Our stomach is hungry and frightfully so	Final long syllable treated as weak
13.1	áha mud hábréli ávva?	What kind of talk is this there?	Final long syllable treated as weak
13.2	búdunli ’ístíkan qávva.	Instead of glasses use bowls!	Sw/w]s/sw}sw Reversed foot in CP
13.4	bdáryan hal dkísOkun sávva	I will pour until your stomach is satisfied	Sw}w]s/sw}sw Reversed foot in CP
14.1	ḥábrak ptihálun ’ístavan	Thy words renew our appetite	Sw]wsw}sww> Reversed foot in CP Extra syllable at end 1 missing stress Incorrect IU
14.2	cáyak matrúyúli sávan;	Thy tea is quenching our thirst	Sw]wsww}sw Too few stresses + PP has incorrect meter (unless stressed]wssw} as suggested by the notes, but not marked)
14.4	mircamírc zpíltēla gávan.	Smacking of lips has just started among us	Possibly too few feet? Wws}sww}sw Notes suggest sws}sww}sw
15.1	Kúlun, Kúlun, ḥa špâi dúsun	Eat, eat, just cram it in!	Sw>sw>w]s] sw} Reversal of foot in CP IUs have correct feet
15.4	kad príqlókun lbáddar ḥúsun	And when you are finished then go strolling about	W]ssw}sw]sw> Reversed foot in CP

6.3	câidan, câipaz, drímun há tâu	O teakettle, teapot! Pour in all you can!	IUs have correct feet
8.1	Zóga ḥílya, bséna wšláma	O sweet pair, hail and welcome!	IUs have correct feet
9.1	câipaz ḥbárti, sári qámta	O teapot, my companion, thou first start	IUs have correct feet
12.3	Básma gának! Hic lit táyak	Mayest thou fair well, none compares with thee!	IUs have correct feet
16.4	há pus bséna! dúvah zíllan	And now remain in peace we do now go.	IUs have correct feet

Table 53: Neo-Aramaic poem from Yarre (1957) analysed by Phonological Scansion

Type	Total	Total incorrect	%
Total CPs	9	7	77.8%
Total PPs	13	1	7.7%
Total IUs	10	1	10.0%
Total words	36		
Total breaks	32	9	28.1%

Table 54: summary of meter mismatches in phonological units for Neo-Aramaic poem

A4.2.3 Conclusions

1. Inverted feet occur most frequently in the less marked phonological unit (Clitic Phrase) (1.2, 4.4, 13.2, 13.4, 14.1, 15.4)
    - One Clitic Phrase contains an extra syllable (3.3)
2. The most marked phonological units (Intonational Units) have very few incorrect assignments (only 14.1)
    - The five verse-internal units show correct meter (6.3, 8.1, 9.1, 12.3, 16.4)
    - One intonational unit contains an extra syllable (14.1)
3. One Phonological Phrase with incorrect assignments has correct assignments if long syllables identified by Yarre are matched to strong stresses (14.4)
4. Stanza 14 is the most irregular: it shows missing syllables, three three-stress lines and irregular meter. This stanza may give an incorrect impression of the regularity of the poem. According to Yarre, it contains a large number of loanwords which may be difficult to assimilate to the natural Neo-Aramaic rhythms.

The probability that these results occurred by chance is very low (see Table 54): 28.1% of breaks are incorrect. This should reflect the percentage for all phonological units if the units had no effect on the assignments. However, there is a significant bias towards incorrect assignments in Clitic Phrases, and a bias away from incorrect assignments in higher phonological units. This replicates the key conclusions of Hayes and Kaun (1996) for English folk-songs.
The Neo-Aramaic poem's irregularities are explained by Phonological Scansion.

A4.3 Coptic

A4.3.1 Introduction

Coptic is the latest stage of Ancient Egyptian, still spoken in Egypt. What little is known about Coptic scansion, MacCoull (1989) indicates it is based on phonological units. She comments:

> "To quote Thomas Gelzer..: 'For the calculation of word-lengths not only single words, but also "word-groups" with internal relationships, such as preposition and noun, or epithet and proper name, must be taken into account.' This is pure Coptic philology. What is understood of Coptic meter (not much Coptic poetry survives) is indeed based on grasping the modifier group as a unity for prosodic purposes."

MacCoull (1999) describes the meter used in Coptic hymns, derived from the analysis by Junker (1908):

> "Both hymns are in four-line strophes (verses), the form universally employed for these compositions. The first hymn text of each pair is in the shorter quatrain-strophic meter known as "Adam" (from the first word of the first line of the pattern, "Adam was sad" [Adam eti efoi]); there are three stresses per line. The second of each pair is in the longer quatrain-strophic meter known as "Batos" (meaning "bush," from the pattern "The bush that Moses saw" [pi-Batos eta-Môusês nau erof]); there are four stresses per line."

Junker shows that Coptic poetry is not "metrical" (in the sense of conforming to a regular alternation of weak and strong syllables like trochaic or iambic meter), and he also rejects that idea that there is any significance in the number of weak syllables between strong, citing the first lines of 8 poems below (Table 55) as an example, because there is no consistent rhythm across all the lines. His conclusion is that Coptic poetry is purely accentual:

Line	Stress	Coptic	Transliteration	Stress Patterns
J1	3	Ⲉⲣϣⲁⲛⲟⲩϣⲁϫⲉ ⲉⲓⲉⲃⲟⲗ ϩⲛⲣⲱⲕ	Ershanoushaje eievol h^{e}nrôk	wwwswwwsws
J2	3	Ⲁⲓϫⲱ ⲉⲛⲟⲩϣⲁϫⲉ ⲛϩⲟⲩⲛⲉⲡⲁⲏⲓ	Aijô enoushaje ^{e}nhounepaêi	wswwswwwws
J3	3	Ⲉⲓⲥⲛⲉϫⲁⲧⲃⲉ ⲁⲩⲏⲣ ⲥⲁⲃⲉ	Eisnejatve auêr save	wwswwsws
J4	3	ⲙⲡⲉⲛⲉϩⲓⲟⲙⲉ ⲉⲣϩⲟⲧⲉ ⲛⲧⲟⲟⲩ	^{e}Mpenehiome erhote ^{e}ntow	wwwswwsws
J5	3	Ⲡⲣⲙⲙⲁⲟ ⲙⲡⲉⲣϣⲟⲩⲱⲟⲩ ⲙⲙⲟⲕ	^{e}Mp^{e}r^{e}mmao ^{e}mpershoushou ^{e}mmok	wwwswwsws
J6	3	Ⲁⲛⲁⲩⲛⲁⲕ ⲉⲛⲓⲙⲉⲩⲏ ⲡⲁⲧⲛⲁ	Anaunak enimewê patna	wwswwsws
J7	3	Ⲁϣⲡⲉⲫϩⲏⲩ ⲉⲛⲉⲭⲣⲏⲙⲁ ⲙⲡⲕⲁϩ	Ashpephêw enechrêma ^{e}mpkah	wwwswwsws
J8	3	Ⲁⲙⲟⲩ ⲙⲁⲣⲟⲛ ⲉⲡⲏⲓ ⲉⲡϣⲗⲏⲗ	Amou maron epêi epshlêl	wwwswsws

Table 55: first lines used as an example of stress patterns by Junker (1908)

> Es ist belanglos wieviele unbetonte Silben zwischen den Tonstellen liegen, und darum ist die Frage, ob rein jambische, trocäische u. s. w. Metren zu

> belegen sind, eine volkommen gleichgiltige, denn es handelt sich dabei um blosse Zufälligkeiten…
> Deutlich erkennt man, dass abgesehen von der Dreizahl der Tonstellen und dem rhythmischen Wechsel von Arsis und Thesis, keine Uebereinstimmung herrscht und in Sonderheit die Zahl der unbetonten Silben variert.
>
> "It is unimportant how many unstressed syllables there are between the stressed syllables, and therefore the question whether there is pure iambic meter or trochaic etc. is one that is completely irrelevant, because it concerns pure chance...It is clear to see that apart from the triple stresses and the rhythmic change of rise and fall of stress, no pattern prevails and, in particular, the number of unstressed syllables is variable." (my own translation)

Whilst MacCoull and Junker are correct about the patterns of strong stresses, a scansion using phonological units shows that there are additional rhythmic patterns in Coptic poetry. I will analyse three hymns below (from MacCoull, 1999) using a phonological analysis to show the rhythms. I have modified Junker's system to account for syllables which carry minimal stress (like -en in "heaven"), which are marked 'y'.

> Hymn for the feast of Gregory Thaumaturgus (from MacCoull, 1999)
> ϣⲟⲙⲧ ⲛ̀ⲣⲱⲙⲉ ⲁⲩⲉⲓ
> ϣⲁ ⲡⲉⲛⲉⲓⲱⲧ ⲁ̀ⲃ̀ⲣⲁⲁⲙ
> ⲙ̀ⲡⲛⲁⲩ ⲛ̀ϩⲁ̀ⲛⲁ̀ⲙⲉⲣⲓ
> ⲉϥϩⲉⲛ ϯⲥⲕⲏⲛⲏ̀
> Ⲅⲁⲃ̀ⲣⲓⲏⲗ ⲡⲓ-ⲛⲓϣϯ
> ⲛⲁⲣⲭⲁⲅⲅⲉⲗⲟⲥ
> ⲛⲉⲙ Ⲓⲱⲁⲛⲏⲗ
> ⲉⲣⲉ ⲡϫⲟⲉⲓⲥ ϩⲉⲛ ⲧⲟⲩⲙⲏϯ
> ϣⲟⲙⲧ ⲛ̀-ⲣⲁⲛ ⲉⲛⲭⲟⲥⲓ
> ϩⲉⲛ ⲧ̀-ⲫⲉ ⲛⲉⲙ ⲡⲕⲁϩⲓ
> ⲁⲥϫⲱⲗϩ ⲙ̀ⲱⲟⲩ ⲉ̀ⲃ̀ⲟⲗ
> ⲛ̀ϫⲉ ⲧⲓ-ⲥⲁⲃ̀ⲉ ⲙ̀Ⲡⲁⲣⲑⲉⲛⲟⲥ
> Ⲅⲉ ⲅⲁⲣ ⲁⲥ-ϯ-ⲙⲏⲛⲓ
> ⲙ̀-ⲡⲓ-ⲃ̀ⲁⲧⲟⲥ ⲉⲑⲟⲩⲁⲁⲃ̀
> ⲉⲣⲉ ⲡⲓ-ⲭⲣⲱⲙ ⲛ̀ϩⲏⲧⲥ
> ⲟⲩⲟϩ ⲙ̀ⲡⲉⲥ-ⲣⲱϕ
>
> from MacCoull (1999)
> Ⲑⲱⲟⲩϯ ⲛⲉⲙⲏⲓ ⲙ̀ⲫⲟⲟⲩ
> Ⲱ ⲛ̀ⲁⲓⲟϯ ⲛⲉⲙ ⲛ̀ⲁⲥⲛⲏⲟⲩ
> Ϩⲉⲛ ⲡ̀-ⲉⲣⲫ̀ⲙⲉⲩⲓ ⲉⲧⲥⲱⲧⲡ
> Ⲛ̀ⲧⲉ ⲡⲓ-ⲁⲣⲭⲓⲉⲣⲉⲩⲥ
>
> from MacCoull (1999)
> Ⲑⲱⲟⲩϯ ⲧⲏⲣⲟⲩ ⲛⲉⲙⲁⲛ ⲙ̀ⲫⲟⲟⲩ
> Ϩⲁ ⲛⲓⲗ̀ⲁⲟⲥ ⲛ̀ⲧⲉ ⲡ̀-ⲭⲣⲓⲥⲧⲟⲥ

Ⲛⲧⲉⲛⲑⲉⲗⲏⲗ ϩⲉⲛ ⲡⲓ-ⲉⲣⲫⲙⲉⲩⲓ
Ⲛⲧⲉ ⲡⲓ-ⲛⲓϣϯ ⲙ-ⲙⲁⲛⲉⲥⲱⲟⲩ

In the tables below:
w is a weak syllabic stress
s is a strong syllabic stress
y is a weak stress which does not count as a syllable (like '-le' in 'bottle')
] is a clitic phrase break
} is a phonological phrase break
/ marks a word break
- marks the join between a prefix and the word it modifies
\ marks a word break that is significant for rhythm

Type	Line	Transliterated Line	Translated Line	Stress	Stress Pattern	Stress Rhythm
A	1	Shomt n-rôme auei	3 men came	3	s]y/sw]ws	ssw\ws
A	2	sha peneiôt Abraam	to our father Abraham	3	s/ws]syw	s\wssw (s\wwsw to avoid stress clash?)
A	3	m-p-nau n-hanameri	at the time of noon	2	yys]ywwsw	s\wwsw
A	4	ef-hen ti-skênê	when he was in his tent:	2	ws]wsw	ws\wsw
A	5	Gabriêl pi-nishti	Gabriel the great	2	wys]wsw	ws\wsw
A	6	n-archangelos	Archangel	2	Yswws	Swws
A	7	nem Michaêl	with Michael	2	s/wws	Swws
A	8	ere p-Joeis hen toumêti	with the Lord in their midst.	3	ww/ys]s/wsw	wws\swsw
A	9	Shomt n-ran enchosi	3 names are exalted	3	s]ys]wsw	ss\wsw
A	10	hen t-phe nem p-kahi	in heaven and earth:	3	s/ys]w/ysw	ss\wsw
A	11	asjôlh mmôou ebol	there clothed herself in them	3	ws/ys]ws	wss\ws
A	12	nje tisabe m-Parthenos	the wise Virgin.	3	ys/wws]ywsw	swws\wsw
A	13	Ge gar as-ti-mêini	For she gave the sign	3	s/w]ws-wsw	swws\wsw
A	14	m-pi-batos ethouaab	of the holy bush	2	ywws]wys	wws\ws
A	15	ere pi-chrôm nhêts	with the fire in it	2	ww/ws]ys	wwws\s
A	16	ouoh mpes-rôkh	and it was not burned.[1]	2	s/yws	sws

Meter	Transliterated Line	Translated Line	Stress	Stress Pattern	Stress Rhythm
A	Thôouti nemêi mphoou	Gather with me today,	3	wsw]ys]ys	wsws\s
A	Ô naioti nem nasnêou	O my fathers and my brothers,	3	s]wsy]y/ws	s\wsws
A	hen p-erphmeui etsôtp	in the chosen commemoration	3	s]ysw]wsy	Sswws
A	nte pi-archiereus	of the high priest.	3	ys/ywsws	s\wsws
B	Thôouti têrou neman mphoou	Gather all with us today,	4	wsw]ws/ys]ys	wsw\wss\s
B	ha nilaos nte p-Christos	to the peoples of Christ,	4	s]wsw]ys/yws	swsw\sws
B	ntenthelêl hen pi-erphmeui	and let us rejoice in this commemoration	3	ywws]s/wysw	wws\swsw
B	nte pinishti m-manesôou	of the great shepherd.	4	ys/wsw]ysw-s	swsw\sws

Table 56: three Coptic poems analysed by phonological scansion

A4.3.2 Syntactic Analysis

The table demonstrates that the Adam meter (A) has roughly three strong stresses per line, and the Batos meter (B) has roughly four. However, the distribution of the stresses within words reveals a rhythm which pairs up contiguous lines (in the long Adam poem), and alternate lines (in the shorter poems) – the paired lines have the same colour in the table. For example, ef-hen ti-skênê | Gabriêl pi-nishti has the same pattern of weak and strong stresses across words: ws \ wsw (weak, strong, word break, weak, strong, weak). 14/24 lines pair up in this way (and probably more would, if we knew the correct accentuation and rules for stress clash in Coptic). This sort of analysis is similar to one performed on English poetry by Cureton (1996).

A4.3.3 Phonological Scansion Analysis

The phonological theories of Hayes and Kaun (1996) can also be applied to Coptic. In the examples above, there are (apparently) no deviations from the meter of each verse – paired verses show exactly the same stress pattern. However, in the following example from Junker (1908), there are slight differences in paired verses which can be explained with reference to phonological scansion:

ⲉⲣϣⲁⲛ-ⲟⲩⲣⲱⲙⲉ ϧⲱⲕ ⲉⲡϣⲉⲙⲟ
ⲧⲉϥⲉⲣ-ⲟⲩⲣⲟⲙⲡⲉ; ϣⲁⲕⲧⲟϥ ⲉⲡⲉϥ-ⲉⲓ
ⲁ-ⲁⲣⲭⲉⲗⲗⲓⲧⲏⲥ ϧⲱⲕ ⲉⲧⲁⲛϣⲏϥ
ⲓⲥ-ⲟⲩⲙⲏϣⲉ ⲛϣⲟⲟⲩ ⲉⲡⲓⲛⲁⲩ ⲉⲡⲉϥ-ϩⲟ

ⲉϣⲱⲡⲉ ⲧⲉⲕⲟⲛⲁϩ; ⲡⲁϣⲏⲣⲉ ⲙⲙⲉⲣⲓⲧ
ⲉⲣⲉ-ⲡϫⲟⲉⲓⲥ ⲛⲁⲥⲧⲟⲕ ⲉϫⲱⲓ
ⲉϣⲱⲡⲉ ⲟⲛ ⲁⲕⲕⲁ-ⲥⲱⲙⲁ ⲉϩⲣⲁⲓ
ⲙⲁⲣⲉ-ⲡϫⲟⲉⲓⲥ ⲉⲣ-ⲟⲩⲛⲁ ⲛⲉⲙⲁⲕ

Line	Transliterated Line	Strong Stresses	Stress Pattern (deviations in red)	Comment
A1	ershan-ourôme bôk epshemo	3	ww-wsw]s}wws	
A2	tefer-ourompe, shaktof epef-ei	3	ww-wsw}ws]ww-s	A2 has 3 initial weak stresses, A4 has 4: the variation occurs at the start of the phonological unit
A3	a-Archellitês bôk etansêf	3	w-wwsw]s}wws}	
A4	is-oumêshe nshoou epinau epef-ho	3	w-www/ys]wws]ww-s	
B1	eshôpe tekonah, pashêre mmerit	4	wsw]wsw} wsw]yws	B1 has an extra weak stress, which occurs at the start of a phonological phrase
B2	ere-pejoeis nastok ejôi	3	ww-ws/ws]ws	B2 has an extra weak initial stress, and B4 an extra medial weak stress – both occur at the start of phonological units
B3	eshôpe on akka-sôma ehrai	4	wsw/s}ww-sw]ws	
B4	mare-pjoeis er-ouna nemak	3	ww-s]w-ws}ws	

Table 57: analysis of one Coptic poem from Junker (1908)

The variations in the number of weak stresses in paired lines occur, as a rule, at the start of phonological units. Phonological theory predicts that this is a position which is metrically less significant, and helps to explain why these variations occur where they do in the poem.

A4.3.4 Conclusion

The analysis identifies a rhythm in Coptic poetry which has not, to my knowledge, been noticed before but is in agreement with the general principles of the Coptic language. The results indicate that Coptic poetry is accentual (having a fixed number of strong stresses), but also shows rhythms dependent on syntax and the numbers of weak stresses between strong. This makes it similar to modern English free verse with a fixed number of accents (for example, poems by Annie Finch in Finch et al., 2005).
The analysis shows that the phonological scansion system is valid for detecting rhythms apart from meter (as Hayes and Kaun suggest) and can identify rhythms which have a syntactic component (as Cureton suggests).

A4.4 Medieval Latin

A4.4.1 Description

Classical Latin poetry (C2BC-2AD) is based on a quantitative system – that is, the time taken to pronounce syllables in the line. However, the system is an import from contemporary Greek poetry. The pre-Classical Latin system, just like English poetry,

was based on stress accent. In fact, Quintilian, the earliest recorded Roman prosodist notes that the primary rhythm in early Latin poetry (C3-C2BC) was phrase-based (although this poetry also conformed to an accentual, and later quantitative, metrical pattern – Parsons, 1999):

> Quasi vero fecerint sint in compositione deprensi, sicut poema nemo dubitaverit spiritu quodam initio fusum … mox in eo repertos pedes …Neque vero tam sunt intuendi pedes quam universa comprensio, ut versum facientes totum illum decursum, non sex vel quinque partes ex quibus constat versus, aspiciunt: ante enim carmen ortum est quam observatio carminis, ideoque illud "Fauni vatesque canebant".
>
> "…poetry was originally the outcome of a natural impulse … while the discovery of feet came later… Further it is not so important for us to consider the actual feet but rather the general rhythmical effect of the phrase, just as the poet in writing a verse considers the metre as a whole, and does not concentrate his attention on the six or five individual feet that constitute the verse. For poetry originated before the laws which govern it, a fact which explains Ennius's statement that 'Fauns and prophets sang'"

The following table analyses three early Latin lines (including the line Quintilian quotes): the earliest is accentual and the later two quantitative, although the latest also has an additional, re-emerging accentual meter. All show Quintilian's accentual phrase-based rhythms.

Scansion System	Livius Andronicus, Odyssey fr. 1 (C3BC)	Ennius, Annales (C2BC)	Pervigilium Veneris 1 (C2AD)
Verses	**Virum mihi Camena insece versutum** "Tell me, O Muse, of the clever man"	**Versibus quos olim Fauni vatesque canebant.** "in verses which the Fauns and prophets once sang"	**Cras amet qui nunqu' amavit quiqu' amavit cras amet.** "tomorrow everyone who has never loved will love, and everyone who has once loved will love tomorrow"
Expected Meter	**Swsw|Sww|Swwsww (Saturnian)**	**Sww|Ss|Ss|S||s|Sww|Ss (hexameter)**	**Swss|Swss||Swss|Sws (trochaic septenarius)**
Quantitative (with Metrical Ictus)	**Ws/ws/|Wsw/|Sww/ssw** The quantitative system does not define consistent feet, nor does it match the expected metrical ictuses	**Sww|Ss|Ss|S||s|Sww||Ss** The quantitative system matches the expected stresses, including the metrical ictuses	**S/ws/s|/S'/wss||/S'/wss|/S/ws** The quantitative system matches the expected stresses, including the metrical ictuses
Accentual Phrasal Rhythm	**sw/sw>/sww>sww]sww** The phrasal rhythm shows a progressive pattern: 2x sw, 3x sww This rhythm is intentional.	**sww/s/sw]sw/||wsw]wsw** The phrasal rhythm shows a progressive pattern in the relative clause: A(sw)A(sw)B(wsw)B(wsw) This rhythm is intentional. The accentual rhythm of the final two feet (wsw|wsw) matches the ictuses of the quantitative rhythm	**s/sw]s/s'/wsw]/s'/wsw]/s/sw** The phrasal rhythm shows a chiastic pattern: A(ssw)B(swsw)B(swsw)A(ssw) This rhythm is intentional. The accentual rhythm of the final two feet (swsw|ssw) matches the ictuses of the quantitative rhythm
Accentual (with Metrical Ictus)	**Sw/sw/|Sww/|Sww/sww** The accentual system defines the expected feet, and the expected metrical ictuses.	**Sww|S/s|W/s|W/||w|Sw/w|Sw** The accentual system does not define consistent feet – it is not important for scansion. The metrical ictuses do not match the expected pattern	**S/sw/s|/S'/wsw||/S'/wsw|/S/sw** The foot is defined here as four accentual syllables, the second of which is not always weak, but the pattern approximates the expected meter so closely that an accentual rhythm is very probably intended to be heard. The first syllable in each foot bears the metrical ictus
Summary	**Livius's line is in accentual meter, but has a significant phrasal rhythm. There is no quantitative element to the meter**	**Ennius's line is in quantitative meter, but has a significant phrasal rhythm. There is no accentual element to the meter.**	**This line is also in quantitative meter with a significant phrasal rhythm. However, there is also an accentual meter which largely mirrors the quantitative.**

Expected scansion, Accentual scansion, Quantitative scansion, Accentual Phrase Rhythm, Valid System, Possibly Valid System
|| caesura - a syntactic pause in the line
| metrical foot boundary – marks the line into feet
/ word boundary
w = weak stress (either quantitative or accentual), s = strong stress (either quantitative or accentual)
S = strong stress in key metrical position (the "metrical ictus": it is the first stress in the foot)

Table 58: examples of early Latin poetry

Greenough (1893) summarises the view of modern scholarship (Browning, 1959, Poultney, 1978 and Holcombe, 2007):

> "It can only be that an early accentual feeling of rhythm was partially superseded among the learned by the purely quantitative Greek rhythm; but in the decline of scholarship or the levelling up of the lower classes, the old sense of accentual rhythm began to assert itself more and more. This is the view of … many others"

Oberhelman and Hall (1984) locate the shift to accentual rhythms in prose (if not poetry too) by C3AD. There was a further development in the Late Latin and Early Mediaeval periods (C5AD-C13AD) when the number of syllables in the native accentual poetry was fixed. First acknowledged by the English scholar Bede in C8AD (Fassler, 1987), this poetry is an exact parallel to the accentual-syllabic system of English from C14AD on, and provides a good test to see if the system of Phonological Metrics operative in English has a wider application.

A4.4.2 Phonological Scansion

Some Medieval Latin accentual-syllabic poetry shows the sort of substitutions found in early English accentual-syllabic poetry (for example strongly-stressed syllables where weakly-stressed syllables are expected, or iambs for trochees). Table 59 applies a phonological analysis to the variations in the C12AD Concilium Romarici Montis (from O'Donnell, 1994):

Line	Phonological Analysis	Comment
Concilium Romarici Montis 4-11 (Rhymed Catalectic Trochaic Tetrameter: sw|sw|sw|s||sw|sw|sw|s)		
Tale non audivimus nec fuisse credimus	sw}s]wsws}||swsw]swn}	No variations
in terrarum spatio a mundi principio.	s]wsw]swn}||sswwswn>	Inversion occurs at the start of a phonological unit
Tale numquam factum est sed neque futurum est.	Sw}sw]/sws}||ssw/wsws>	Inversion occurs at the start of a phonological unit
In eo concilio de solo negotio	Sswwswn}||ssw]wswn}	The first inversion occurs at the start of a phonological unit. The second occurs at the end of a clitic phrase
Amoris tractatum est, quod in nullo factum est;	Wsw]wsws>||sssw]/sws>	The variation occurs in a clitic phrase
sed de Evangelio nulla fuit mentio.	Ssnwswn}||swsw]swn>	The strong stress occurs at the start of a phonological phrase
Nemo qui vir dicitur illuc intromittitur.	Sw]ss]swn}||ws]nwswn>	The variations occur at the end of clitic phrases
Quidam tamen aderant qui de longe venerant.	Swsw]swn}||sssw]swn>	The variation occurs within a clitic phrase
The variations in this poem occur either at the start of phonological units, or at the end of the least significant unit, the clitic phrase.		
Key **Trochaic substitutions** **|| caesura** **] end of a clitic phrase** **} end of a phonological phrase** **> end of an intonational unit** **/ beginning of a phrasal verb**		

Table 59: Phonological Metrics applied to Medieval Latin verse

The table shows that the variations within Medieval Latin accentual-syllabic verse occur in the positions predicted by Phonological Metrics:

1. At the start of phonological units
2. At the end of the least significant phonological unit, the clitic phrase

A4.4.3 Conclusion

The Medieval Latin accentual-syllabic system closely models the English accentual-syllabic system, even to the point of allowing substitutions only in phonological insignificant locations in the verse.

Ironically, the Classical Latin system is least suited to the English system to which it has been applied. In fact, the same motivation imposed the (native Greek) system on Latin, as imposed the Classical Latin system on English – a perception cultural inferiority; and with the same results – a reversion to original accentual patterns after 500 years

A5 Application to Metrical Problems

Another indication of Phonological Scansion's usefulness would be if it was able to solve metrical problems more successfully than other theories. I have already shown how it identifies the meter in lines which conform to two metrical patterns (1.2.2). Further on, I will show how it identifies meter in lines which some experts consider impossible (p. 174). However, below I apply it to two other problems: the identification of lines which cannot be scanned (termed "unmetrical" lines), and the identification of the meter known as "Loose Iambics".

A5.1 Identification of unmetrical lines

A5.1.1 Introduction

One key motivation in the linguistic approach to scansion is the identification of lines which native speakers feel do not fit the expected metrical pattern –unmetrical lines. Groves (2007b) gives test lines which he classifies as unmetrical according to native speaker assumptions, and two main linguistic theories (Halle-Keyser and Kiparsky), concluding that only his own theory approximates native speakers' sensitivity – see Table 60.

Iambic Pentameter Lines (including a summary of whether the lines are considered metrical (o) or unmetrical (x) by four theories and a group of experts commentators).	Halle-Keyser	Kiparsky	Groves	Phonological	Experts
Give renew'd fire to our extincted Spirits (Othello 2.1.81)	x	x	o	o	o
This line is incorrectly classified by Kiparsky's theory, but correctly by Groves' theory. It is also correctly classified by Phonological Scansion: 3 phonological units are iambic, only 1 is trochaic, unambiguously predicting the expected iambic rhythm					
Though Death doth consume, yet Virtue preserves (from Gascoigne, 1575)	o	x	x	x	x
The line is incorrectly classified by Halle-Keyser, but correctly by the other theories, including Phonological Scansion: only 1 phonological unit is iambic but 3 are trochaic, indicating a trochaic rhythm which is at odds with the iambic metrical pattern					
Your meaning I understand by your eye (Gascoigne, 1575)	x	x	x	x	x
The line is correctly classified by all the theories, including Phonological Scansion: 2 phonological units are iambic, and 2 are trochaic, indicating that the rhythm has a tendency towards trochaic, but is uncertain, and therefore opposes the metrical pattern.					
How many bards gild the lapses of time (Keats)	x	x		o	o
The line is incorrectly classified by Halle-Keyser and Kiparsky. Phonological Scansion identifies 3 iambic units and 2 trochaic units, indicating that the rhythm has a tendency towards iambic, supporting the poem's meter, but is uncertain.					
Ode to the West Wind, by Percy Bysshe Shelley	x	o		o	o
The line is incorrectly classified by Halle-Keyser and Kiparsky. Phonological Scansion identifies 3 trochaic units and 1 iambic unit (after assigning 'to the' to one metrical stress), indicating that the rhythm is regular trochaic.					

Table 60: Phonological Scansion identification of unmetrical lines (after Groves, 2007b)

A5.1.2 Discussion

The table shows that both Groves's theory and Phonological Scansion match expert assumptions equally well, although Phonological Scansion identifies these in two additional examples. It works, very broadly, by identifying which type of metrical foot fits the greatest number of stresses at the end of phonological units. If two or more feet types have a similar number of stresses, then the resultant rhythm is ambiguous, and appears unmetrical. Similarly the line appears unmetrical if the resultant rhythm is clear, but not a rhythm expected by the metrical pattern. The last two lines analysed ('How many bards gild the lapses of time' and 'Ode to the West Wind, by Percy Bysshe Shelley') are used by Halle-Keyser as examples of unmetrical lines, and have become benchmarks for metrical theories since. However, they have been variously accepted by Attridge (1982), Hayward (1996a) and Wright (1988) as metrical. For 'How many bards..' Phonological Scansion indicates that the line has an uncertain rhythm, but one that conforms to the iambic meter of the poem, and is therefore not unmetrical. For 'Ode to the West Wind..', the line is strongly trochaic.

A5.1.3 Conclusion

Phonological Scansion performs better than the two theories tested by Groves (2007b), and is equally as good as Groves's own theory when compared to expert intuitions. It also sides with recent expert opinion on two of Halle-Keyser theory's benchmark lines.

A5.2 "Loose Iambics"

Robert Frost, an influential metrical poet of the early C20, once wrote (Frost, 1939):

> "All that can be done with words is soon told. So also with meters—particularly in our language where there are virtually but two, loose iambic and strict iambic."

His term "loose iambic" refers to the meter of many poems written in C19 and early C20, which whilst appearing to have a iambic rhythm also permits not only non-iambic duple feet (like trochees), but also triple feet (typically anapaests and dactyls). An example is given in Table 61. Poems in loose meters form a category on the continuum between strict metrical poetry and free verse, but it is difficult to find an objective definition of where one type begins and the other ends.

A5.2.1 Previous Analyses

A few scholars have proposed definitions of loose iambic meter. Steele (2001) extends the types of permitted substitutions within loose iambic meter from duple to triple feet. However, the conditions he imposes on when these can be applied, although descriptive, are ultimately subjective:

> "the ear can easily locate the … metrical beats in each line, and because rhymes point the line-endings, the additional light syllables do not obscure the measure"

Halle and Keyser (1999), followed by Fabb (2003), reworked their stress maximum theory by formulating particular rules and procedures to identify the meter. Tarlinskaja (1993) also defines a separate metrical pattern to categorise the meter. She calls it "dolnik" and also finds it in Russian and German poetry. However, both these approaches require additional rules to accommodate the pattern. It would be much more satisfactory to be able to define loose meters within commonly accepted metrical rules.

A5.2.2 Phonological Metrics Analysis

Phonological Scansion has been successful in identifying meter of the strict type (whether iambic or other varieties), even when the line rhythm is irregular. It may also be able to produce a good definition of Frost's loose meters. Table 61 gives an analysis of one of Frost's poems (which is not strict iambic, and so must be an example of Frost's loose iambic poetry). The metrical analysis by Tarlinskaja (1993) is given a phonological analysis, and conclusions are drawn.

Line	Metrical Analysis (from Tarlinskaja, 1993)	Phonological Analysis	Comment
Frost, "The Red Man" 1-12			
He is said to have been the last Red man	wns|wns|ws|ms	wns}|wns|ws]|m]s>	"He's said to've been"
In Acton. And the Miller is said to have laughed--	ws|wnws|wns|wns	w/s|w>nw/s|w}ns]|/wns>	"'n' the Miller's said to've" The unexplained syllable ends an intonational unit
If you like to call such a sound a laugh.	nws|ws|mws|ws	nws]|ws}|mws]|ws>	"If y' like". The syllable "such" begins a clitic phrase
But he gave no one else a laugher's license.	nws|wms|ws|wsw	nws]|wms}|ws|w]sw>	"But" and "no" both begin clitic phrases
For he turned suddenly grave as if to say,	wns|s|wns|wnws	wns}|s|wn]s}|wnws>	"For" and "as" both begin phonological units. The others can be elided: "Suddn'ly" and "t' say", and also perhaps "'s if"
'Whose business,--if I take it on myself,	ns|wnws|ws|ws	ns|w>nws|w]s|ws>	"if" begins a clitic phrase. The unexplained syllable ends an intonational unit.
Whose business--but why talk round the barn?--	ns|wws|ms|ws	ns|w>ws|m]s|ws}	The unexplained syllable ends an intonational unit.
When it's just that I hold with getting a thing done with.'	nws|nws|ws|wnmsw	nws}|nws}|ws|w}nm]sw>	"gett'n'" allows elision, as may "when's just". "that" and "a" begin phonological units
You can't get back and see it as he saw it.	ws|ws|ws|wnmsw	ws|ws}|ws|w]nmsw>	"See't 's he saw" is possible. Otherwise "as" begins a phonological unit.
It's too long a story to go into now.	wns|ws|wns|mws	wns]|ws}|wns|mw]s>	's too long" allows elision, as does "t' go" and perhaps "int' now"
You'd have to have been there and lived it.	ns|ws|ns|wsw	ns]|ws|ns}|wsw}	Regular iambic tetrameter (with a feminine ending)
Then you wouldn't have looked on it as just a matter	nws|ns|nwns|wsw	nws]|ns|nw}ns]|wsw}	"Then y' wouldn't" and "on't" allow elision. "as" may ("'s just") but certainly begins a clitic phrase
Of who began it between the two races.	ws|ws|wws|wmsw	ws|ws|w}ws|wm]sw>	"'tween th' two" allows elision
Where the poem deviates from strict iambic meter, in most cases the syllables could either be elided or are at the start of phonological units. The remaining deviations occur at the end of intonational units.			
Key **Syllables which could be elided** **Syllables at the beginning of phonological units** **Syllables which could be elided and are also at the beginning of phonological units** **Unexplained deviations**			

Table 61: Phonological Analysis of Loose Iambic Meter

A5.2.3 Conclusions

In the table above, deviations from Frost's strict iambic meter (that is triple feet where duple feet are expected) fall into three categories:

1. Syllables which could be elided in normal speech (for example "b'tween"): elision would resolve the disruption, by removing the excess syllable converting the triple feet to the expected duple.
2. Syllables which occur at the beginning of phonological units (for example "it's" in "It's too long]"). According to Phonological Metrics, these syllables are less significant metrically than syllables at the end of units, so the deviation is less likely to disrupt the iambic flow. In addition, syllables in these positions are more likely to be pronounced more quickly than others, and lead to elisions (Kim, 1999). In fact many of the elisions also fall into this category (marked in pink in Table 61).
3. Syllables which occur at the end of intonational units. This is a significant location according to the theory. However, Wright (1988) has observed that syllables here often show metrical deviations, because they are able to shift the focus onto the beginning of the next phrase, particularly where the following phrase is in apposition to the previous phrase – in other words, the syllables marked in green in Table 61, which are at the end of intonational units and form appositional phrases, are less significant metrically. These account for all of the deviations in the poem.

It seems that by referencing elision and phonological units Frost's loose iambics do conform to the strict iambic pattern.

A5.2.4 Further Examples

This is also the case for the following lines used by Tarlinskaja (1993) to demonstrate her dolnik theory of loose meter, and for the following loose iambic poems which I have analysed: Anon. Arden of Feversham 2.84, Massinger, A New Way To Pay Old Debts 2.3.143, Frost, The Road Not Taken, Hardy, The Oxen, Hardy, The Wound and Hardy, The Walk

Anon. Arden of Feversham 2.84
For(n) I(s) must(m)} to(n) the(w) Isle(s)] of(n) She(s)ppy(w)} **with(n)** speed(s)}

Massinger, A New Way To Pay Old Debts 2.3.143
Though(m) they(n) paint(s) her(w)>, so(n) she(w) catch(s)] the(w) lord(s)} I'll(m) thank(s) 'em(w)}

Frost, The Road Not Taken, 1-5
Two(m)] roads(s)} di(w)verged(s)} **in(w)** a(w) yel(s)low(w)] wood(s)}
And(w) sor(s)ry(w)] **I(w)** could(s) not(w) trav(s)el(w)] both(s)}
And(w) be(s) one(m) tra(s)**vel(w)**er(n)] long(s)} I(w) stood(s)}
And(w) looked(s) down(w)] one(s)} as(w) far(s)] **as(w)** I(w) could(s)}
To(w) where(s) it(w) bent(s)] in(w) **the(w)** un(s)der(w)growth(s)>.

Hardy, The Oxen 1-8
Christ(s)mas(w)] Eve(m)>, and(w) twelve(s)] **of(w)** the(w) clock(s)>.

“Now(s)] they(n) are(m)} all(s)] **on(w)** their(w) knees(s)”>,
An(w) el(s)der(w)] said(s)} **as(w)** we(w) sat(s)] **in(w)** a(w) flock(s)}
By(w) **the(w)** em(s)bers(w)] **in(w)** hearth(s)side(m)}ease(s)>.

We(w) pict(s)ured(w)} **the(w)** meek(s)] mild(w)] creat(s)ures(w)} where(s)}
They(w) dwelt(s)} **in(w)** their(w) straw(s)y(w)] pen(s)>,
Nor(w) did(s) it(w) oc(s)cur(w)} **to(w)** one(s)] **of(w)** us(w)} there(s)}
To(w) doubt(s)} they(w) **were(w)** kneel(s)ing(w)] then(s)}

Hardy, The Wound 1-8
I(w) climbed(s)] **to(w)** a(w) crest(s)}
And(w) fog(s)-fes(w)tooned(s)]
The(w) sun(s)} lay(m)] west(s)}
Like(w) a(w) crim(s)son(w)] wound(s)}
Like(w) that(w) wound(s)] of(w) mine(s)}
Of(w) which(s)} none(w)] knew(s)]
For(w) I’d(m) giv(s)**en(w)]** no(w) sign(s)}
That(w) it(w) pierced(s) me(w)] through(s)}

Hardy The Walk
You(w) did(s) not(w) walk(s)] with(w) me(s)}
Of(w) late(s)} **to(w)** the(w) hill(s)-top(w)] tree(s)}
    By(w) the(w) gat(s)ed(w)] ways(s)>,
    As(w) **in(w)** ear(s)**li(w)**er(w)] days(s)>;
    You(w) **were(w)** weak(s)] and(w) lame(s)>,
    So(w) **you(w)** nev(s)er(w)] came(s)>,
And(w) I(w) went(s)] a(w)lone(s)>, **and(w)** I(w) did(s) not(w) mind(s)>,
Not(w) think(s)ing(w)] **of(w)** you(s)} as(w) left(s)] be(w)hind(s)>.

I(w) walked(s)] up(w) there(s)} to(w)-day(s)}
Just(s)} in(w) the(w) form(s)er(w)] way(s)>;
    Sur(w)veyed(s)] a(w)round(s)}
    The(w) fam(w)il(s)**i(w)**ar(w)] ground(s)}
    By(w) my(w)self(s)] a(w)gain(s)>:
    What(w) diff(s)**e(w)**rence(w)>, then(s)?>
On(s)ly(w)} that(w) un(s)der(w)ly(s)ing(w)] sense(s)}
Of(w) the(w) look(s)] **of(w)** a(w) room(s)} **on(w)** re(w)turn(s)ing(w)] thence(s)}

All the deviations are accounted for using either elision or placement at the start of phonological units (coloured blue, red or purple). There are no unexplained deviations (coloured green).

A5.2.5 Summary

Phonological Metrics offer the possibility of producing a definition of loose meters without requiring additional rules.

A6 Conclusions on Phonological Scansion Theory

Phonological Scansion theory is applicable to at least three other languages. It is also able to solve metrical problems more succinctly than competing theories. Both these findings indicate that the theory is valid.

Appendix B – Calliope Assessed

The research has assessed Calliope primarily against Scandroid. This section compares Calliope to expert scansion and the systems of Plamondon, Raabe, Fabb and Groves to determine how effective it is against more recent sophisticated theories. It also assesses Calliope against a Scandroid prototype, revealing some of Hartman's assumptions about scansion.

B1 Expert scansion

B1.1 Identification of Meter in Difficult Lines

Calliope is able to identify the meter in lines which are classified as being difficult or impossible to scan successfully. For example, Steele (1999) comments:

> "were we to encounter anywhere but in The Song of Hiawatha ('Introduction',34),
> The blue heron, the Shuh-shuh-gah
> it is unlikely that we would emphasize the two definite articles at the expense of 'blue' and the first syllable of 'Shuh-shuh-gah.' But that is what Longfellow wishes us to do, since he is writing in trochaic tetrameter:
> The(s) blue(w) her(s)on(w), the(s) Shuh(w)-shuh(s)-gah(w)
> … why can we not consider Longfellow's line about the heron as being trochaic with an iambic substitution in the first foot and an iambic substitution after a mid-line pause?
> The(w) blue(s) her(s)on(w), the(w) Shuh(s)-shuh(s)-gah(w)
> To this question, I would respond that, unless we stress the first syllable of Longfellow's line we lose the meter. It dissolves."

However, from a phonological perspective, the line shows only a very minor deviation from regular trochaic tetrameter. The two syllables ending the intonational unit and line ending are both regular trochees, only the first Shuh of polysyllabic word Shuh-shuh-gah indicates a iambic pattern: {[The(w) blue(s)] [her(s)on(w)]}>, [the(w) Shuh(m)-shuh(s)-gah(w)]}. Calliope has no difficulty in identifying a trochaic rhythm without adjusting the linguistic stress pattern.

B1.2 Assessed against Human Expert Scansion and Scandroid

The following analysis is of a poem by E. Woolley, given online (Woolley, 2006). The poem is scanned by four contributors (scottish jenny, brian w, Michael Wren, and boromir). I have also scanned it using Stallings's principles (detailed in 4.5.1), and with Scandroid and Calliope.

> Extraction Woes, by E. Woolley
> "I wonder why we call them "wisdom" teeth.
> Vestigial structures crowd, impact, occlude,
> make mastication painful; often food
> like steak becomes a hazard to the mouth.
> Oh, bloody day! Extractions take their toll;
> You'll slurp potato soup. No straws! Suck gauze.

Relief sought in a Hydrocodone haze,
and packs of Sonic ice fall short. You'll still
be numb from cheek to chin; cry out for Mom
who'll barely recognize your puffy face.
Soft tissue swells, dry sockets slow the pace
of healing -- weeks of agony to come.
I had the surgeon put mine in a jar
so I could see them when I start to whine
about the boss' temper or the swine
who grabs my ass each time I pass his chair.
I ask myself how does a long lunch line
compare to plastic props that wretched my jaw?
Do traffic-jams seem half as irksome now
when oozing orifice memories remain?"

Line	scottish jenny	Brian w	Michael Wren	boromir	Stallings (expert)
same	9	15	16	17	20
%	**43%**	**75%**	**80%**	**85%**	**100%**
error	8.5	1	1	1.5	0
%	**43%**	**5%**	**5%**	**8%**	
Total	**3%**	**70%**	**75%**	**78%**	**100%**
1	wswswswsws	wswswswsws	wswswswsws	wswswswsws	wswswswsws
2	wswswsswws	wswswswsws	wswswswsws	wswswswsws	wswswswsws
3	sswwwswsws	wswswswsws	wswswswsws	wswswswsws	wswswswsws
4	wswswswwws	wswswswsws	wswswswsws	wswswswsws	wswswswsws
5	sswswswsws	sswswswsws	wswswswsws	wswswswsws	wswswswsws
6	wswswswsws	wswswswsws	wswswswsss	sswswswsws	wswswswsws
7	wswswwswws	wswswwsss	wsswwwsss	wsswwswsws	wsswwswsws
8	wswswswsws	wswswswsws	wswswswsws	wswswswsws	wswswswsws
9	wswswswsws	wswswswsws	wswswswsws	wswswswsws	wswswswsws
10	wswswwwsws	wswswswsws	wswswswsws	wswswswsws	wswswswsws
11	wswswswsws	wswswswsws	wswswswsws	wswswswsws	wswswswsws
12	wswswswwws	wswswswsws	wswswswsws	wswswswwws	wswswswsws
13	wswswsswws	wswswswsws	wswswsswws	wswswsswws	wswswsswws
14	wswswwwsws	wswswswsws	wswswswsws	Wswswswsws	wswswswsws
15	wswsswwss	wswswswsws	wswswswsws	Wswswswsws	wswswswsws
16	wswswswsws	wswswswsws	wswswswsws	Wswswswsws	wswswswsws
17	wswswswsws	wswswswssw	wswswswsss	Wswswswsws	wswswswsws
19	wswswswsws	wswswswsws	wswswswsws	Wswswswsws	wswswswsws
20	wswsswwsws	wswswswsws	wswswswsws	Wswswswsws	wswswswsws
21	wswswwswwws	wswswwswsws	wswswsswsws	Wswswwswsws	wswswsswws

Table 62: scansion of Extraction Woes by human contributors (Woolley, 2006)

Line	Scandroid	Calliope	Stallings (expert)
Same	4	15	20
%	**19%**	**75%**	**100%**
Error	5.5	2	0
%	**28%**	**5%**	
Total	**-8%**	**70%**	**100%**
1	Wswswswsws	wswswswsws	wswswswsws
2	Swwswsswsw	wswswswsws	Swwswsswsw
3	Swwswswsws	swwswswsws	swwswswsws
4	Wswswswsws	wswswswsws	wswswswsws
5	Sswswswsws	sswswswsws	Sswswswsws
6	Ssswwsssss	wswswswsws	Ssswwsssss
7	Wsswwwsws	wsswwswsws	wsswwwsws
8	Wswswssss	wswssswsws	Wswswssss
9	Wswswsssws	wswswswsws	Wswswsssws
10	Sswswswsws	wswswswsws	Sswswswsws
11	Sswssswsws	wswswswsws	Sswssswsws
12	Wswswwsws	wswswswsws	wswswwsws
13	Wswswwsswws	wswswsswws	wswswwsswws
14	Wswswswsws	wswswswsws	wswswswsws
15	Wswsswsws	wswswswsws	Wswsswsws
16	Wswssswsws	wswswswsws	Wswssswsws
17	Wswswswsss	wswsswwsws	Wswswswsss
19	Wswswswsws	wswswswsws	Wswswswsws
20	Wswssswsws	wswswswsws	Wswssswsws
21	Wswswswsws	wswswsswsws	Wswswswsws

Table 63 scansion of Extraction Woes by Scandroid and Calliope

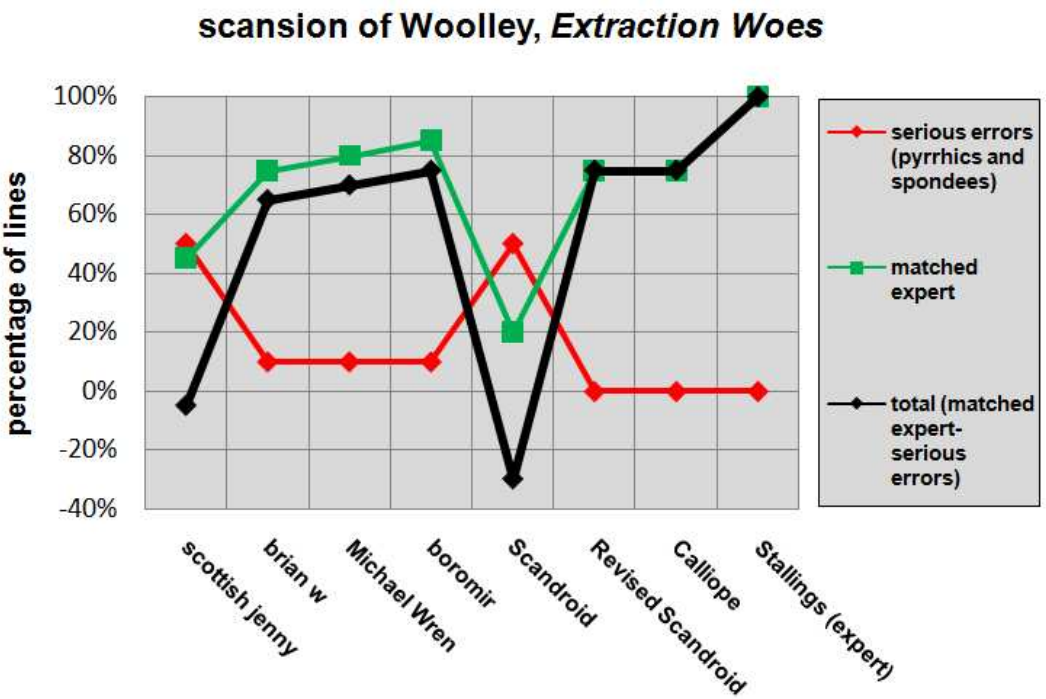

Figure 11: scansions of Woolley, Extraction Woes

From the scansion of "Extraction Woes":

1. Scandroid does far worse than the worst contributor
2. Calliope matches the value of one of the better contributors. However, two better scansions were contributed, although they were not much better.
3. The data replicates the results from the scansion of Pope's lines described above (4.5.2)

B1.3 Assessed against Unmodified Expert Scansion

The results of assessing regularised expert scansion against Scandroid and Phonological Scansion theory's scansion of 372 verses, categorised by complexity, are given in 4.2.2. The scansion was modified to screen out the subjectivity of expert scansions described in 2.1.3. In particular, the following:

1. A high percentage of expert scansions do not pick up on scansions like 'Shall(s) I(w) com(w)pare(s)' where there is a deviation from the meter in a phonologically insignificant part of the line (category 2). Instead experts produce the incorrect pattern 'Shall(w) I(s) com(w)pare(s)'. This is not so surprising, since Hayes and Kaun (1996) predict that it is precisely at these points that deviations from the standard meter are least likely to be detected.
2. Some expert scansions also permit sequences of strong syllables where one or more are subordinated to a main stress, although a greater number of experts prefer to subordinate one strong stress where possible (Baker et al, 1996). I have modified the expert scansion by replacing subordinate strong stresses with weak metrical stresses. This allows contiguous strong stresses where the stresses are not co-dependent.

Tests against unmodified scansions are given below as a comparison:

Category	**1**	**2**	**3**	**4**	**5**	**6**	**Grand Total**
Scandroid Match	77	3	2	19	14	1	116
Phonological Scansion Match	172	6	4	40	17	0	239
Count	215	20	4	83	49	1	372
% Scandroid	**35.8%**	**15.0%**	**50.0%**	**22.9%**	**28.6%**	**100.0%**	**31.2%**
% Phonological Scansion	**80.0%**	**30.0%**	**100.0%**	**48.2%**	**34.7%**	**0.0%**	**64.2%**
Target	80	80	80	60	60	71	72
Scandroid confidence	6%	16%	49%	9%	13%	0%	5%
Phonological Scansion confidence	5%	20%	0%	11%	13%	0%	5%

Table 64: assessment of Scandroid and Phonological Scansion against expert scansion by Scansion Complexity Categories

Assessed against unmodified expert scansions, Phonological Scansion performs better than, or as good as, Scandroid in all but one category. However, it falls far short of meeting the targets. I have accepted the modified expert scansion data as more accurate for the following reasons:

1. The modified scansion is consistent with the agreed principles of a representative sample of prosodists (Baker, 1996)

2. The data produced by modified expert scansion shows the expected decline in accuracy across Complexity Categories – the unmodified scansion shows an aberrant decrease in accuracy in Category 2 verses.
3. Additionally, the main difference between the modified and unmodified scansion data occurs in Category 2 verses, where there are variations in phonologically insignificant syllables. In other words, most people will not hear any significant difference between modified and unmodified scansions, but they do produce a disproportionately great effect on the accuracy of all the procedures being tested. It seems reasonable to regularise scansion in this category, giving the regular and most supported scansion the benefit of the doubt.

B2 Hartman's Scandroid prototype

B2.1 Introduction

Hartman (1996) describes a prototype of Scandroid using the scansion of Shakespeare's Sonnet 116 as a demonstration of its capabilities. I have compared this with Scandroid and Calliope's scansion of the same sonnet, in order to reveal some of Hartman's assumptions about appropriate scansions.

> "Let me not to the marriage of true minds
> Admit impediments. Love is not love
> Which alters when it alteration finds,
> Or bends with the remover to remove:
> O no! it is an ever-fixed mark
> That looks on tempests and is never shaken;
> It is the star to every wandering bark,
> Whose worth's unknown, although his height be taken.
> Love's not Time's fool, though rosy lips and cheeks
> Within his bending sickle's compass come:
> Love alters not with his brief hours and weeks,
> But bears it out even to the edge of doom.
> If this be error and upon me proved,
> I never writ, nor no man ever loved."

Hartman (1996) comments:

> "there's nothing in the computer's scansion that I'd mark wrong in a student paper, though there are lines that I would scan differently. (I read the fourth line 'Or bends with the remover to remove')"

B2.2 Analysis

Line	Scandroid prototype (1996)	Scandroid (2005)	Calliope & Revised Scandroid	Stallings (expert)	Hibbison
Same	6	4	8	14	6
%	**43%**	**29%**	**57%**	**100%**	**43%**
error	4	5	0	0	0
%	**29%**	**36%**	**0%**	**0%**	**0%**
Total	**7%**	**-9%**	**57%**	**100%**	**43%**
1	Swswwswwss	swswwswwss	swswwswsws	swswwswsws	wswswswsws
2	Wswswsswss	Wswswwssss	wswswsswws	wswswsswss	wswswswsws
3	Wswswswsws	wswswswsws	wswswswsws	wswswswsws	wswswswsws
4	Wswswswsws	wswswswsws	wsswwswsws	wsswwswsws	wswswswsws
5	Sswswswsws	sswswswss	wswswswsws	ssswwswsws	wswswswsws
6	Wswswswswsw	wswswswswsw	wswswswswsw	wswswswswsw	wswswswswsw
7	Wswswswswws	wswswswswws	wswswswsws	swwswswsws	wswswswsws
8	Wswswswswsw	wswswswsssw	Wswswswswsw	wswswswswsw	wswswswswsw
9	Sssswswsws	ssssswsws	Swwswswsws	sswswswsws	wswswswsws
10	Wswswswsws	wswswswsws	Wswswswsws	wswswswsws	wswswswsws
11	Sswswwssws	sswswwssws	Swwsswwsws	sswswswsws	wswswswsws
12	Wswsswwwsws	wswsswwwsws	Wswswswsws	wswsswwsws	wswsswwsws
13	Wswswswsws	wswswswsws	Wswswswsws	wswswswsws	wswswswsws
14	Wswswsssws	wswswsssws	Wswswsswws	wswswsswws	wswswswsws

Table 65: scansions of Sonnet 116

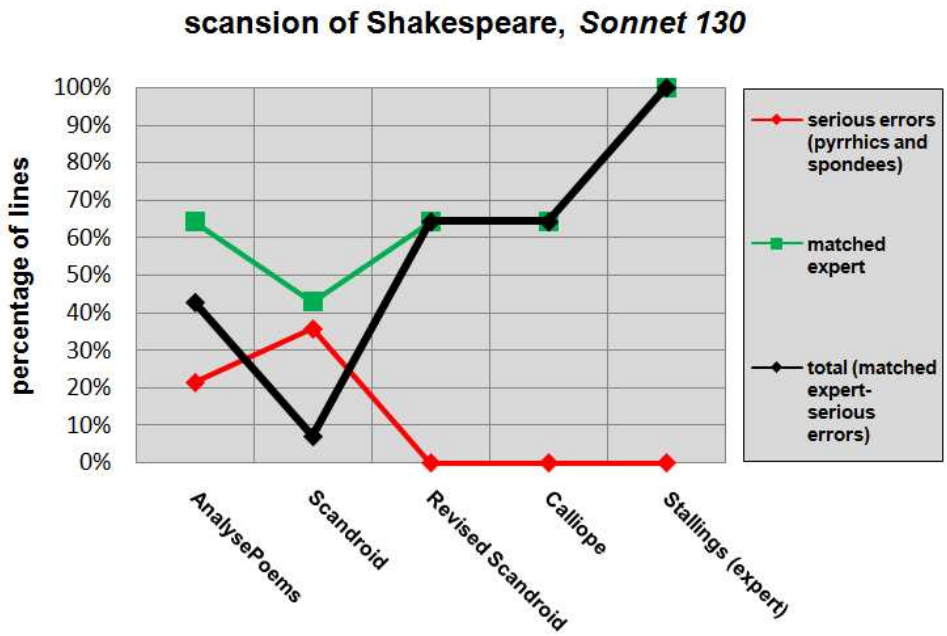

Figure 12: scansions of Shakespeare, Sonnet 116

The results show:

1. The Scandroid prototype matches more lines than the Scandroid program
    a. The prototype makes fewer errors than the program because the program assigns strong stress to the verb "to be" more frequently, sometimes ignoring a scansion which agrees with Stallings
    b. The prototype is more accurate than the program, but both frequently make (what Stallings considers) serious errors in their scansions. If Hartman allows these in undergraduates' work, then he is allowing errors which are avoided by the majority of prosodists.
2. Calliope performs better than a literary expert (Hibbison, 2005) when measured against Stallings's scansion. Hibbison allows relatively stressless syllables to be stressed to meet iambic norms - for example "not" (line 1), and "the" (line 4) - whereas Calliope avoids this.
3. Calliope differs from Stallings's scansion partly because it demotes spondees (ss) too frequently (lines 2, 5, 9, 11 and 12). For some prosodists (Baker, 1996) this practice is preferable to allowing spondees, and would probably be acceptable to Stallings. The other deviations arise from an acceptable difference of emphasis on words – Stallings stresses "it" in line 7, Calliope "is". Whereas Scandroid produces serious scansion errors, Calliope's errors are minor, if they are even errors at all.
4. Scandroid using the revised stresses produces an identical scansion to Calliope.

B2.3 Summary

This test again shows that Calliope outperforms Scandroid and approximates a very good scansion. It also shows that Hartman allows Scandroid to produce what at least one prosodist considers to be serious scansion errors. This has limited the effectiveness of the program.

B3 Plamondon's AnalysePoems

Calliope is compared to AnalysePoems (see 2.6.5) using two criteria:

1. Accuracy in identifying the meter of poems
2. Accuracy in identifying the rhythm of individual feet

B3.1 Identification of Meter

The table below compares the Calliope application with AnalysePoems in the identification of meter. The lower the cumulative difference from expert rankings the better the application approximates expert scansion.

Poem	AnalysePoems	Diff	Rank	Meter	Inversion	rank	Calliope	%	Rank
To Anthea, Herrick	87	16	1	Iambic	97	1		100	2
A daughter of Eve, Rossetti	87	21	2	Iambic	92	3	60/63	95.2	4
Amy Margaret, Allingham	86	7	3	mixed iambic / trochaic	95	2	90/90	100	2
Ulysses, Tennyson	84	21	4	Iambic	91.9	4	96/104	92.3	6
To the memory of Mr Oldham, Dryden	84	23	5	Iambic	88.3	5	96/99	97	3
Sonnet 130, Shakespeare	83	39		Iambic	90				
My Last Duchess, Browning	83	27		Iambic	80				
When I consider how my light, Milton	79	19	6	Iambic	80	7	46/55	83.6	7
Ode to a Nightingale, Keats	73	22	7	Iambic	88	6	78/82	95.1	5
Out, Out, Frost	73	25		free verse					
Dover Beach, Arnold	69	20	8	mostly iambic	77	8	54/66	81.8	8
Cumulative difference from expert rankings			4			0			10

Table 66: An assessment of AnalysePoems (yellow data) against Calliope (blue data) compared to expert scansions (white data).

B3.1.1 Regularity of line rhythm

The 'Inversion' column gives the percentage of feet which do not fit the overall metrical pattern. This is the benchmark against which to assess AnalysePoems and the phonological scansion. The poems are ranked from the highest percentage of inversions to the lowest (white rank column).

AnalysePoems's scansion ranks the poems in the same order as the inversion information (yellow rank column).

Calliope does not. The data in the column 'Calliope' gives the weighting of patterns matched in different phonological units (where Clitic Phrases have less weight in the total than Intonational Units). The ranking (in the blue rank column) does not match the inversion rankings. This may not be surprising since it is meant to approximate the perception of the listener, who will ignore some inversions as less relevant. It is,

though, difficult to measure whether this ranking corresponds to the actual perception of listeners – that is, do the poems of Keats and Dryden sound more rhythmical than those of Milton and Rossetti?

B3.1.2 Discrimination of meters

However, Calliope is able to identify that there is a distinct difference in the meters of Out, Out and Ode to a Nightingale. AnalysePoems classifies both as duple meter with a confidence level of 73%, and a difference between first and last assessments of around 23%. However, Out, Out is free verse, according to both experts and the Calliope. Calliope also makes the type of duple meter explicit, where AnalysePoems probably has the data, but does not reveal it.

B3.1.3 Conclusion

AnalysePoems is better at identifying the regularity of meter. However, Calliope is better at identifying meters (such as iambic or free verse), and perhaps at distinguishing between subtypes of meter (such as iambic pentameter as opposed to iambic tetrameter).

B3.2 Identification of Feet

Plamondon (2006a) gives a detailed scansion of Shakespeare's Sonnet 130. I have compared this to Calliope's scansion and Raffel's (1992) expert scansion of the same sonnet in order to determine which matches the expert scansion of individual feet more accurately. A summary of the differences is given on p. 185, and conclusions drawn.

B3.2.1 Analyse*Poems's* Scansion

my(1) MISTRESS'(2) –(0) eyes(2) are(1) no(2)thing(0) LIKE(2) the(0) sun(2)
coral(0) –(0) is(1) far(1) more(1) red(2) than(0)her(0) lips'(2) red(2)
if(0) snow(2) be(0) white(2) why(1) THEN(2)her(0) breasts(2) are(1) dun(2)
if(0) hairs(2) be(0) WIRES(2) black(2)WIRES(2) grow(2) on(0) her(0) head(2)//
i(1) HAVE(2) seen(1) ro(2) ses(0)DAMASK'D(2) –(0) red(2) and(1) white(2)
but(1) NO(2) such(1) ro(2) ses(0) see(2) i(1)IN(2) her(0) cheeks(2)
AND(2) in(0) some(1) PERFUMES(2) –(0) IS(2)there(1) MORE(2) de(0) light(2)
than(0) in(0) the(0) breath(2) that(1) FROM(2)my(1) MISTRESS(2) –(0) reeks(2)//
i(1) love(2) to(0) hear(2) her(0) speak(2) yet(1)WELL(2) i(1) know(2)
that(1) mu(2) sic(0) HATH(2) a(0) FAR(2)more(1) plea(2) sing(0) sound(2)
i(1) grant(2) i(1) ne(2) ver(0) SAW(2) a(0)GODDESS(2) –(0) GO(2)
my(1) MISTRESS(2) –(0) WHEN(2) she(1)walks(2) treads(2) on(0) the(0) ground(2)//
and(1) yet(1) by(0) heav'n(2) i(1) think(2)my(1) love(2) as(1) RARE(2)
as(1) a(2) ny(0) SHE(2) belied(0) –(2) with(0)false(2) com(0) pare(2)

ns|ws|ws|ws|ws
ww|nn|ns|ww|ss
ws|ws|ns|ws|ws
ws|ws|ss|sw|ws

B3.2.2 Phonological Scansion from Calliope

{[MY(m) mis(s)tress'(w)] [eyes(s)]} {[are(n)]} {[noth(s)ing(w)] [like(s) the(w) sun(s)]}
Xsw]s}n]sw]xws} = (10) (Iambic = final PP, 3CP, 1PP) = Iambic) -> ms/ws/ns/ws/ws

[Co(s)ral(w)]} [is(n) far(m)] [more(n)] [red(s)]} [than(n) her(w) lips'(s)] [red(m)]}>:
sw}nm]n]s}xxs]m> = (10) (Iambic=1CP, 1PP, Trochaic=final IU, 1PP) =Trochaic -> sw/nm/ns/nw/sm
wn->W, sm->s-m = sW}m]n]s}xxs]-m> = (10) (Iambic=final IU, 1CP, Trochaic=2PP, 1CP) =Trochaic ->sW/mn/sn/ws/-m

[If(n) snow(s)] [be(n) white(s)]}>, [why(s)] [then(n) her(w) breasts(s)]} [are(m) dun(s)]};
=xs]ns>s]xws}ms} =(10) (Iambic=final PP, 1PP, 1IU, 1CP, Trochaic=1CP) =iambic ->ns/ns/sn/ws/ms

{[If(n) hairs(s)] [be(n)]} {[wires(s)]}, {[black(s)] [wires(m)]} {[grow(s)]} {[on(w) her(w) head(s)]}
Xsn}s}>sm}s}xws}= (10) (Iambic=finalPP, 1PP, 1IU , Trochaic=2PP) iambic?
(stress clash ssms> resolve (ww>w, s>s> s>-s = xsm}s}-sm}s}Ws}) (Iambic = final PP, 3PP(ms,ms, -s) 1CP) Iambic -> ns/ns/-s/ms/Ws
(no stress clash > (s>s) (Iambic final PP, 1PP, 1IU (ns), Trochaic = 1PP(sm), 1PP) = Iambic ->ns/ns/sm/sw/ws

ms|ws|ns|ws|ws
sw|mn|sn|ws|-m
ns|ns|sn|ws|ms
ns|ns|sm|sw|ws
ms|ws|ns|ws|ws

B3.2.3 Scansion by Raffel (1992)

My(w) mis(s)tress'(w) eyes(s) are(w) no(s)thing(w) like(s) the(w) sun(s)
Co(s)ral(w) is(w) far(s) more(w) red(s) than(w) her(s) lips'(w) red(s):
If(w) snow(s) be(w) white(s), why(w) then(s) her(w) breasts(s) are(w) dun(s);
If(w) hairs(s) be(w) wires(s), black(w) wires(s) grow(s) on(w) her(w) head(s).

ws|ws|ws|ws|ws
sw|ws|ws|ws|ws
ws|ws|ws|ws|ws
ws|ws|ws|sw|ws

B3.2.4 Summary

AnalysePoems	differences	Calliope	differences	Raffel
ns/ws/ws/ws/ws	All match	ms/ws/ns/ws/ws	All match	ws|ws|ws|ws|ws
ww/nn/ns/ww/ss	3 pyrrhic feet + 1 spondaic foot (v. unusual), 1 foot matches	sW/mn/sn/ws/-m	3 feet match, elision (not unusual), missing syllable (unusual)	sw|ws|ws|ws|ws
ws/ws/ns/ws/ws	All match	ns/ns/sn/ws/ms	4 feet match	ws|ws|ws|ws|ws
ws/ws/ss/sw/ws	4 feet match, 1 spondaic foot (unusual)	ns/ns/sm/sw/ws	4 feet match, 1 trochee (v. common)	ws|ws|ws|sw|ws
ns/ws/ws/ws/ws	All match	ms/ws/ns/ws/ws	All match	ws|ws|ws|ws|ws

Table 67: comparison of the scansions of Sonnet 130 produced by AnalysePoems, Raffel and Calliope

AnalysePoems and Calliope both match about the same number of feet as Raffel's expert scansion. However, when trying to find the best scansion, AnalysePoems produces more unusual patterns (pyrrhics and spondees) than Calliope (which resolves these using elision and insertion of syllables). In fact, a line containing four pyrrhic (ww) or spondaic (ss) feet is unique in English poetry (so far as I can tell). Both AnalysePoems and Calliope cannot reference rhetorical or contrastive stress, and so are unable to determine the scansion of 'her' in "Co(s)ral(w) is(w) far(s) more(w) red(s) than(w) her(s) lips'(w) red(s)". However, Calliope's four-level stress system is able to identify the relative strengths of "is(w) far(s) more(w) red(s)" where AnalysePoems's three-level stress system produces "is(w) far(w) more(s) red(s)" .

B3.2.5 Conclusion

Although neither AnalysePoems's system nor Calliope can reference key pragmatic information, Calliope produces less unlikely scansions by using four levels of stress, and by employing elision and insertion.

B4 Raabe's Frost program

B4.1 Description

Raabe (1975) develops an automated scansion system, implemented in SNOBOL on an IBM 360/365, with a dictionary to assign three levels of stress to lines of poetry (including one ambiguous stress which can be resolved according to the values of contiguous stresses). He uses the system to scan Frost's entire corpus, primarily to verify expert assumptions by identifying statistical patterns in Frost's poetry (along the lines of Slavic Metrists like Tarlinskaja).

He himself identifies some improvements to the system (which he does not implement) including regularising the stress patterns by raising secondary stresses to primary stresses if surrounded by weak stresses, and lowering secondary stresses to weak stresses if surrounded by primary stresses. He is also aware that sequences of three weak stresses conform to the metrical pattern by promoting appropriate weak stresses to strong.

B4.2 Assessment

I have assessed Raabe's system's scansion of Frost's Desert Places against Calliope and Scandroid, compared to an expert scansion of the poem using the criteria identified for Stallings's assessments. The expert scansion consists of the linguistic stresses of each line fitted into disyllabic feet. I have assessed Raabe's original system as well as the system modified by the improvements he suggests which are detailed above. The results of the assessment are given in Table 68 below.

Desert Places
"Snow falling and night falling fast, oh, fast
In a field I looked into going past,
And the ground almost covered smooth in snow,
But a few weeds and stubble showing last.
The woods around it have it--it is theirs.
All animals are smothered in their lairs.
I am too absent-spirited to count;
The loneliness includes me unawares.
And lonely as it is that loneliness
Will be more lonely ere it will be less--
A blanker whiteness of benighted snow
With no expression, nothing to express.
They cannot scare me with their empty spaces
Between stars--on stars where no human race is.
I have it in me so much nearer home
To scare myself with my own desert places"

Lines	Expert	Scandroid	Raabe	Raabe Revised	Calliope
Same	16	5	1	7	14
%	100%	31%	6%	44%	87.5%
Error	0	8.5	10	4.5	0
%	0%	53%	63%	28%	0%
Total	**100%**	**-22%**	**-56%**	**16%**	**81%**
1	wswsswwsws	sswwsswsss	sswwsswsss	sswwsswsss	wswswswsws
2	swswswwsws	wwsssswsws	wwswwswsws	wwsswswsws	swswswwsws
3	wwsswswsws	wwsswswsws	wwsswswsws	wwsswswsws	swswwswsws
4	swwswswsws	wwsswswsws	wwsswswsws	wwsswswsws	swwswswsws
5	wswswswsws	wswswswsws	wswswwwws	wswswswsws	wswswswsws
6	wswswswsws	swswwswsws	sswwwswwws	sswwwswsws	wswswswsws
7	swwswswsws	wsssswswsws	wwsswswwws	wwsswswsws	swwswswsws
8	wswswswsws	wwswswswsws	wswwwswsws	wswswswsws	wswswswsws
9	wswswswsws	wswwswswssw	wswwwwwsww	wswswswsws	wswswswsws
10	wswswswsws	wsssswswsws	wwsswswwws	wwsswswsws	wswswswsws
11	wswswswsws	wswswswsws	wswswwwsws	wswswswsws	wswswswsws
12	wswswswsws	wswswswsws	wswswswwws	wswswswsws	wswswswsws
13	wswswswswsw	wswswswswsw	wswswwwswsw	wswswswswsw	wswswswswsw
14	wsswswwswsw	wsswsssswsw	wsswswsswsw	wsswswsswsw	wsswswwswsw
15	wswswswsws	wswswssws	wwwwwwssws	wswswssws	wswswswsws
16	wswsswwswsw	wswswwsswsw	wswswwsswsw	wswswwsswsw	wswsswwswsw

Table 68*: assessment of Raabe*'s system against Scandroid, Calliope expert scansion *of Frost's Desert Places*

Raabe's original system produces the worst scansion, even worse than Scandroid. It makes frequent significant errors, and only matches the expert scansion once. When it is revised with Raabe's suggested improvements it matches more expert scansion than Scandroid, and makes fewer significant errors – mainly because all the sequences of weak stresses are resolved. Calliope again produces the greatest number of matches to expert scansion, and no significant errors.

B4.3 Conclusion

Raabe's original system is the worse than Calliope and even worse than Scandroid – perhaps the worst system studied in this project. However, when it is revised with Raabe's suggestions it is much better than Scandroid (and perhaps as good as AnalysePoems). Nevertheless, Calliope is still significantly better.

B5 Fabb's Bracketted Grid Theory

B5.1 Christina Rossetti, Spring Quiet

Fabb uses Bracketted Grid Theory to determine the meter and rhythm of poems (see 2.5.2). Calliope and expert scansion agree with Bracketted Grid Theory in 4/5 of the test poems. However, Fabb's theory fails to identify the rhythm in Christina Rossetti's Spring Quiet.

Poem	Expert	Bracketted Grid Theory	Calliope
Barrett-Browning, Sonnets from the Portuguese 21	Iambic pentameter	Iambic pentameter	Iambic pentameter
Browning, The Lost Leader	Dactylic Trimeter	Dactylic Trimeter	Dactylic Trimeter
Charlotte Bronte, Diving	Anapaestic Tetrameter	Anapaestic Tetrameter	Anapaestic Tetrameter
Christina Rossetti, Spring Quiet	Trochaic Trimeter	**Irregular Dimeter**	Trochaic Trimeter
Christina Rossetti, Up-Hill	Iambic Pentameter	Iambic Pentameter	Iambic Pentameter

Table 69: comparison of Calliope with expert scansion and Bracketted Grid Theory

It fails in Spring Quiet because Fabb uses Stress Maxima to identify feet in "loose iambic" verse (and only in loose iambics). Where the procedure encounters a stress which is a stress maximum, it will assign that stress and the stress immediately before it to a foot, even if this causes the foot to contain more than two syllables. In effect, the foot structure is built around stress maxima.
For example in "arching high over" Fabb notes a stress maximum in 'high(w) o(s)ver(w)'. When the procedure comes to assign feet, instead of making o- (of 'over') the start of the foot, it also includes the entire stress maximum in the foot, '|high over|'.

Line	Fabb	Calliope	Expert
Gone were but the Winter	wws|[wsw]|	sm]|xxsw} (2 units = trochaic) = sm|nw|sw	sw|sw|sw
Come were but the Spring	wws|ws	sm]|xxs} (2 units= truncated trochaic) = sm|nw|s-	sw|sw|sw
I would go to a covert	wws|[wsw]| ('to a' = 1 stress)	xms]x[sw]} ('to a' = 1 stress) (2 units=trochaic) = sm|snw|sw	sw|sww|sw
Where the birds sing	ws|ws|	m}xs]xs} (stress clash of 'birds(s) sing(s)' resolved by inserting a stress) (3 units = truncated trochaic) = mw|s-|s-	sw|sw
Where in the whitethorn	ws|[wsw]|	m>xxxsw] (insert a stress between 'where' and 'in' because of the pause break) (2 units = trochaic) m-|nw|sw	s|sw|sw

Table 70*: analysis of 'Spring Quiet' 1*-5 by Bracketted Grid Theory and Calliope

The table shows that Fabb's analysis fails for two reasons: firstly, his stress assignments are incorrect (perhaps motivated by the expected outcome). For example, Fabb stresses 'Gone were but' as wws with no stress on the two content words 'gone were'. Calliope stresses both 'gone' and 'but'. Secondly, Fabb's insistence that stress maxima can distort the meter is unnecessary. With a normal stress assignment, the lines are quite easily identified as trochaic trimeter by Calliope: there are no feet with an irregular rhythm, although two syllables need to be merged into one stress, and two unsounded stresses have to be inserted.

B5.2 Duple Meters with Ternary Rhythms

B5.2.1 Introduction

Malof (1964 and 1970), Kiparsky (1975), Attridge (1982), Hayes (1984), Groves (1998) and Fabb (2002) all note that there are some lines in which both duple and triple (or ternary) rhythms operate simultaneously. They attempt to identify how one or other of the rhythms can be shown to be dominant. Hayes excludes some of Attridge's scansions on linguistic grounds. Of the remaining lines, both his and Kiparsky's solution is to reference the rhythm of contiguous lines to fix the meter. Attridge, Groves and Fabb use their own systems to help choose a dominant rhythm, but in most cases are unsuccessful.

B5.2.2 Analysis

The table below (Table 71) details the comparison with Calliope – matches to expert scansion are coloured green:

Theory	Scansion	Comment
a. Strangers to slander, and sworn foes to spite (John Pomfret, The Choice 3)		
Stran(s)gers(w) to(w) sla(s)nder(w) and(n) sworn(m) foes(s) to(w) spite(s) (the expected rhythm is iambic pentameter to fit the poem's meter)		
Iambic	Strangers| to slan|der, and| sworn foes| to spite	
Anapaestic	Stran|gers to slan|der, and sworn| foes to spite	
Expert	**Iambic**	
Fabb	Strangers to slander and sworn foes A.)* *) * *) * *) * *) * *) * * * * * Sw/ws/ws/ws/ws B.)*) * * *) * * *) * * *) * * * * s/wws/wws/wws	Fabb notes that the line is in a iambic pentameter poem, and that it has one rhythm (A) which implies iambic pentameter. However, he comments "the arrangement of monosyllables also permits a different rhythmic explicature (B)…this …resembles…**iambic**…only weakly…deriving the conclusion that the line is **anapaestic** tetrameter (with a short initial foot)." He does not resolve the issue.
Calliope	=/sw]x/sw>xm]s}ws} (Iambic=fPP,1PP,1IU, Trochaic=1PW2, Anapaestic=1CP,1IU, No Triple=1CP,1PP) iambic (7/8) or anapaestic(4/7) =**iambic (87.5%)**	Calliope identifies a iambic rhythm in 3 out of 4 units (with a weighting of 7/8). It identifies an anapaestic rhythm in 2 out of 4 units (with a weighting of 4/7) – the final foot "foes to spite" produces a pattern of sws (an amphimacer) which cannot be construed as an anapaest. The overall rhythm is iambic with a consistency of 87.5%.

b. When to the sessions of sweet silent thought (Shakespeare, Sonnet 30)		
When(s) to(n) the(w) ses(s)sions(w) of(w) sweet(s) si(m)lent(w) thought(s) (the expected rhythm is iambic pentameter to fit the poem's meter)		
Anapaestic	When| to the ses|sions of sweet |silent thought|	
Iambic	When to| the sess|ions of| sweet sil|ent thought|	
Expert	Iambic Pentameter	
Attridge	B ŏ B ŏ B ô B o B	Attridge inserts a beat between "sweet" and "silent", and merges "to the" and "-sions of" into single beats. This creates a **iambic line with anapaestic rhythms**. However, it is very unusual for Shakespeare to insert a beat, and unheard of for him to merge syllables into one beat (except using elision)
Groves	A-O_ o-A-o O_--a--A-o A a.S w w S w w Ṣ w w S b.Ś w w S w W s S w S	Groves tests both an **anapaestic** (a) and a **iambic** (b) reading of the line. Although his theory excludes pentameter readings of some tetrameter lines, it does not exclude either reading here. Neither reading has a weak lexical stress syntactically subordinated to a strong lexical stress, but placed in strong metrical position.
Calliope	=xxx/sw}xs/mw]s} (Iambic=final PP,1PP, Trochaic=1PW3, Anapaestic=2PP, No Triple=1CP, final PP) Iambic (4/6=66.6%) or anapaestic (4/7=57.1%) =**iambic (66.6%)**	Calliope does not insert or merge any syllables (keeping with Shakespeare's practice). It determines that the iambic rhythm matches 2 of 3 units, with a weighting of 4/6 or 66.6% consistency. It also determines that the anapaestic rhythm matches 2 of 4 units, with a weighting of 4/7 or 57.1% consistency. One unit has a rhythm of mws ("silent thought") and so is rejected – if it were accepted as anapaestic, the line would be anapaestic; the other unit is the initial "When". The line is iambic, since iambic rhythm has a higher consistency.

c. I look on the sky and the sea (Barrett-Browning, *The Runaway Slave at Pilgrim's Point* 7)		
I(n) look(s) on(n) the(w) sky(s) and(n) the(w) sea(s) (the expected scansion is uncertain: previous lines are anapaestic or iambic; the number of strong stresses in each line is 4,4,3,3,4 and 3)		
Anapaestic	I look| on the sky| and the sea|	
Iambic	I look| on the| sky and| the sea|	
Expert	Accentual	
Fabb	I look on the sky and the sea)* *) * *) * *) * *))* *] * (* *] * (* *]	The first scansion (with foot breaks marked by ')') is iambic, the second is anapaestic (with foot breaks marked by ']' because stress maxima are used to fix the meter). Fabb is unable to decide on a dominant rhythm in the line: he concludes "Neither option thus maintains the uniformity of the stanza. Again, we have disruption, exploiting a metrical ambiguity...which cannot be fully resolved…**the metre is iambic-anapaestic**"
Calliope	=ws}xws]xws} =(8 syllables, 3 strong stresses) (Iambic=1PP, finalPP, Trochaic=1CP, Anapaestic=fPP, 1CP No Triple=1PP) Iambic(4/5) Or Anapaestic(3/5) **= iambic (80%)** (but previous lines are anapaestic and iambic, with strong stresses of 4 433 43, indicating **accentual Ballad Meter)**	Calliope determines that 2 of 3 units have a iambic rhythm – with a weighting of 4/5. 2 of 3 units also have an anapaestic rhythm, but with a lower weighting of 3/5. Therefore, the line is iambic with a consistency of 80%. Three previous lines have iambic rhythm, and three have anapaestic rhythm – so, the poem may be a unique mixed anapaestic/iambic meter. However, the strong stress pattern is 4 433 433 (including this line's 3 strong stresses), indicating that the poem is composed in the popular contemporary Ballad Meter of alternating 4 and 3 strong stresses per line. Calliope determines **the meter is neither iambic nor anapaestic but accentual**.

Table 71: Ternary and Duple rhythms resolved

B5.2.3 Discussion

Calliope is able to determine the meter in lines which Fabb determine as irresolvably ambiguous (a and c). In one case (b), where Attridge resorts to processes which the poet would very probably not have used, Calliope reaches the same conclusion without modifying the line at all. In every case Calliope determines that there is a significant subordinate rhythm, but also identifies a dominant meter which fits with the expected pattern of the poem. In one case (c) it is able to determine that an accentual reading is the appropriate one, rejecting both anapaestic and iambic readings.

B5.2.4 Conclusion

Contrary to the assumption of Kiparsky and Hayes, the lines themselves in isolation are able to indicate the expected meter. In fact, in 30 test verses (14 verses from non-triple rhythm poems identified in the literature, and 16 verses from the anapaestic poems The Destruction of Sennacherib by Byron and Diving by Charlotte Bronte) Calliope identifies a dominant rhythm which is also the expected rhythm for the verse - whether anapaestic, iambic or accentual. The procedure appears to have identified a

previously unnoticed scientific principle for consistently determining the meter in these cases. It also resolves the problem posed by Hayes: why poets writing iambic pentameter verse “scrupulously avoid” some formats of lines which have significant triple rhythms, but allow other formats with an even greater tendency for the same rhythms – from a phonological perspective, verses in the second category have dominant iambic rhythms.

B6 Groves's Base Template Theory

Calliope produces very similar results to Groves's Base-Template theory (2.5.1). However, there are two differences – the treatment of Stress Clash and the assignment of meter in some contexts. Both are discussed in detail below.

B6.1 Extended Stress Clash

Groves (1998) extends Hayes's Stress Clash theory to make the line conform to metrical criteria.

B6.1.1 Hayes's theory

Selkirk (1984) and Hayes (1984a) investigated cases where the stress in the first word of a compound is further back in the word than expected. They determined that where two contiguous syllables in different words or phrases contain stresses on a given phonological level not separated by at least one stress on the next lowest level on an intervening syllable, there is a stress clash. The clash is resolved by retracting the left stress back leftwards to the next most stressed syllable. An example is given in Table 72.

A hundred thirteen men	
x 1 0 x x x x x x x x x x x x x A hundred thirteen men	intial stress grid: clash between 1 and 0
x 1 x x 2 0 x x x x x x x x x x A hundred thirteen men	1 retracted to next available syllable stressed on the level below, clash between 2 and 0
x x x x 2 0 x x 3 x x x x x x x A hundred thirteen men	2 retracted to next available syllable stressed on the level below, no clash (the syllables 2 and 0 are separated by an intervening syllable stressed at the next lowest level, 3)
A(w) hun(s)dred(w) thir(s)teen(w) men(s)	result in terms of binary stresses

Table 72: Stress Clash in "a hundred thirteen men" from Hayes (1995)

More recently Hayes (1995) has identified that the stress can only be retracted to stressed syllables in the same phonological phrase.

B6.1.2 Stress Clash in Verse

The effect of stress clash on verse is less certain. Some think that it is matched to metrical stress, but can be reversed to fit the meter (Kiparsky, 1975 and Groves, 1998), or that different rules apply in different genres (Hanson, 2006). However, it seems more likely that compound dependents are assigned metrical positions

determined by the primary stress in the (following) governing word (Koelb, 1979 and Youmans, 1983), and not by their own stress patterns. Whatever the process is, if two contiguous strong lexical stresses occur within a dependent noun, only the most metrically consistent one is heard by people reciting the verse. If stress clash cannot be resolved linguistically, then it is resolved by changing the word order – see Table 73.

Type		**Analysis**
A. In our well-found successes, to report (Shakespeare, Coriolanus 2.2)		
Stress Clash	…well(m)-found(s) suc(w)cess(s) -> …well(s)-found(m) suc(w)cess(s)	stress on 'found-' retracts to 'well'
Metrical (and heard) Stress	**..well(W)-found(S) suc(W)cess(S)**	'well' has weaker stress than 'found'
B. After a well-graced actor leaves the stage (Shakespeare, Richard II 5.2)		
Stress Clash	…well(m)-graced(s) ac(s)tor(w) -> …well(s)-graced(m) ac(s)tor(w)	stress on 'graced-' retracts to 'well'
Metrical (and heard) Stress	**..well(S)-graced(W) ac(S)tor(W)**	'well' has stronger stress than 'graced'
comment	Despite the same lexical stress pattern in A and B, B has a completely different metrical stress pattern, apparently due to the position of the primary stress in 'actor'. Moreover, A does not allow a lexically accurate and metrically acceptable trochee ('well(S)-found(W) suc(W)cess(S)') but prefers the compound to follow the stress pattern of 'success'.	
C. Buy terms divine in selling hours of dross; (Shakespeare, Sonnet 146)		
Stress Clash	…di(w)vine(s) terms(s) -> (cannot retract) -> **…terms(s) di(w)vine(s)**	Stress clash in divine(s) terms(s), but stress cannot retract on divine
comment	There is a stress clash between 'divine' and 'terms', but divine does not allow stress retraction. To avoid these sorts of clashes, divine is placed after its noun in most English poetry.	

Table 73: Stress Clash in verse

If Groves and Kiparsky are correct, according to data given by Fitzgerald (2003) the word order in lines would be changed to avoid the stress clash. Since very little word order change is apparent, this undermines Groves's theory. If Koelb and Youmans are correct, then the lexical stress of compound adjectives is irrelevant to their metrical position, though the lexical stress of compound nouns should correspond to the metrical positions – a theory supported by Table 73 and data in Hanson (2006). It seems that the second theory is more likely to be correct.

In any case, Phonological Scansion easily identifies the meter correctly without using Groves's stress analysis.

B6.2 Incorrect Assignment of Meter

Groves's theory identifies whether a line fits a particular meter; it can sometimes force an incorrect reading, where the line fits an alternate pattern but is identified with the pattern being tested for. One example, discussed below, is Wyatt's "They fle from me".

B6.2.1 Thomas *Wyatt's "They fle from me"*

"They fle from me that sometyme did me seke
with naked fote stalking in my chamber
I have sene theim gentill tame and meke
that nowe are wyld and do not remembre
that sometyme they put theimself in daunger
to take bred at my hand and nowe they raunge
besely seking with a continuell chaunge

Thancked be fortune it hath ben otherwise
twenty tymes better but ons in speciall
in thyn arraye after a pleasaunt gyse
When her lose gowne from her shoulders did fall
and she me caught in her armes long and small
therewithall swetely did me kysse
and softely said dere hert howe like you this

It was no dreme I lay brode waking
but all is torned thorough my gentilnes
into a straunge fasshion of forsaking
and I have leve to goo of her goodenes
and she also to use new fangilnes
but syns that I so kyndely ame served
I would fain knowe what she hath deserved"

Groves identifies the poem as loose form of iambic pentameter - in agreement with the assessments of Raffel (1992), Adams (1999) and others. If the poem is iambic, then it is very loose; so loose, in fact, that Wyatt's first editor, Tottel, felt the need to rewrite the poem to make it more regular. Daalder (1972) notes "Tottel's lines scan, or are meant to, where they do not in Wyatt".

However, Schwartz (1963) follows an argument of C.S. Lewis that C15 poems which appear to be irregular iambic pentameter are actually imitating the older tradition of accentual verse. He identifies a regular four-beat accentual pattern in Wyatt's poem:

> "we might suppose the poem to be written in irregular iambics. But if the lines are supposed to be pentameters (as most of them seem to be) we have difficulty accounting for seven lines…which will not yield more than four feet. And there are lines which sound very odd to an ear trained in iambic movements: lines seven, nine, twelve and seventeen…If we assume an accentual-syllabic norm for the poem, we get a slipshod and badly handled meter, and one which cannot produce the very real rhythmic power of the poem. If, however, we assume a four-stressed accentual norm, we get a firm and skilfully managed meter and we do not violate any rhetorical emphases...the scansion makes good rhythmical sense precisely at those points where violence is done by an iambic norm."

Calliope identifies this accentual pattern, rejecting a iambic or trochaic rhythm in some lines which Schwartz indicates do not fit the pentameter pattern. In these lines, Groves identifies a pentameter rhythm – see Table 74 – and rejects Schwartz's idea for three reasons: because it proposes a meter that is too simple and does not reference phonology in a rigorous way, but mainly because Wyatt tends towards using ten-syllable lines, implying a pentameter pattern. The following analysis is based in phonological theory, and supports Schwartz's conclusions:

Type	Analysis
Line	**With naked fote stalking in my chambre**
Groves	o------A--o A ^ A---o O o-----A---o w-----S w---S w-S w----S w-----S o
Calliope	with (w) na(s)ked(w)] fote(s)]} stal(s)king(w)] in(n) my(w) cham(s)bre(w)]} =x/sw]s}/sw]xx/sw} = (10,4) (Iambic=1PP, 1CP, Trochaic=1PP, 1CP) (/3/6) trochaic?? =(inserting weak stress ss-> s-s) x/sw]s}/-sw]xx/sw} = (11,4) (Iambic=fPP, 1PW2, 1PP, 1CP, Trochaic=) (6/6) iambic
Comments	Groves inserts a weak stress and creates a regular iambic rhythm. Calliope does the same thing, ignoring a possible accentual reading of the line.
Line	**That nowe are wyld and do not remember**
Groves	o------A o----Ao O O O o---A---o w------S w---S w—S [w-S] w---S o
Calliope	that(w) now(s)] are(w) wyld(s)]} and(w) do(s) not(m) re(w)mem(s)ber(w)]} xs]ws}x/smwsw}= (10,4) (Iambic=1PW5, 1PP, 1CP, Trochaic=fPP, 1PV5) (7/13) iambic???
Comments	Groves makes the word 'wyld' disyllabic (sw). Phonological Scansion is unable to identify a dominant rhythm for the line, and chooses an accentual reading.
Line	**That sometime they put theimself in daunger**
Groves	^O | A---b o---A O O o---A----o wS S---w w---S w----S w—S o
Calliope	{[That(w) /some(s)time(w)\] [they(n) put(s) /theim(w)self(s)] [in(n) /daun(s)ger(w)]} x/sw]ns\ws]xsw}= (10,4) (Iambic=2PW2, Trochaic=fPP, 2PW2) trochaic rhythm (4/6)
Comments	Groves inserts a weak stress at the start of the line Calliope identifies a weak trochaic rhythm, which may indicate that the line should be read as accentual.
Line	**Therewithall swetely did me kysse**
Groves	^A-----o O ^ A---o O o----A w-S w--S w-S w—S w---S
Calliope	{[There(s)with(w)all(n)] [swete(s)ly(w)] [did(m) me(n) kysse(s)]} /swn]/sw]/mns} = (8,4) (Iambic=1PW2, 1PW2, 1PP, Trochaic=1PW3) Iambic rhythm (4/6)
Comments	Groves inserts a weak stress at the start of the line, and after the first word. Calliope identifies a weak trochaic rhythm, which may indicate that the line should be read as accentual.

Table 74: Groves*'s analysis of ambiguous lines in Wyatt's "They fle from me"* compared to Phonological Scansion

Since there is a variation of between 8 and 11 syllables in each line, but there are always four strong stresses, and since the meter of the lines varies and in many lines no rhythm is dominant, Calliope prefers to read the poem as four-beat accentual. Groves's theory allows him to read the poem as (an idiosyncratic) iambic pentameter.

B6.2.2 Conclusion

Calliope is able to assess a far wider range of scansion possibilities than Groves's analysis which has a tendency to produce false readings.

Appendix C – Test Documents

This section details the test documents given to subjects as part of the assessment of the Calliope application. The documents were provided in either a web form or printed format, depending on the preference of the subject.

C1 Written Scansion Tests

Verse Scansion Tests

Lines of poetry have beats in them – that is, some syllables are more strongly emphasised than others. This gives the line a rhythm, like the musical beats in songs.

For example, in 'Humpty Dumpty had a great fall,' emphasised syllables are written in capitals: 'HUMPty DUMPty HAD a great FALL'. The line has four beats. Note that the beats do not necessarily match strongly stressed syllables - 'great' is stressed but does not carry a beat.

Below are different combinations of beat in **10** verses, with the emphasised syllables written in CAPITALS. **Please tell me which version of each verse sounds most natural by ticking the appropriate box**.

Try pronouncing each verse, putting emphasis only on the syllables in capitals - this is to add rhythmical beats to the line, but should not eliminate the natural stress of the sentence. Ideally, the verse will have a regular rhythm without sounding unnatural.

I have given the verses before and after the test verse in each case - the test verse rhythm should fit the rhythm of the surrounding verses.

For example, in 'Gone were but the Winter, come were but the Spring':

- If you think the **first** version sounds most natural, **tick the letter next to it**.
- If you are not sure which sounds best, tick the letter next to **'?'**.

1	**Gone were but the winter, come were but the spring** in Gone were but the winter, come were but the spring I would go to a covert, where the birds sing. Where in the whitethorn, singeth a thrush And a robin sings in the holly-bush.	A	GONE were BUT the WINter, COME were BUT the SPRING
		B	GONE were but the WINter, COME were but the SPRING
		C	gone were BUT the WINter, come were BUT the SPRING
		D	?

2	**No more to laugh, no more to sing** in Talk what you please of future spring And sun-warm'd sweet tomorrow:- Stripp'd bare of hope and everything, No more to laugh, no more to sing, I sit alone with sorrow.	A	No MORE to LAUGH, no MORE to SING
		B	NO MORE to LAUGH, NO MORE to SING
		C	?
3	**The buzz saw snarled and rattled in the yard** in The buzz saw snarled and rattled in the yard And made dust and dropped stove-length sticks of wood, sweet-scented stuff when the breeze drew across it	A	The BUZZ SAW SNARLED and RATTled in the YARD
		B	The BUZZ saw SNARLED and RATTled IN the YARD
		C	The BUZZ saw SNARLED and RATTled in the YARD
		D	The BUZZ SAW SNARLED and RATTled IN the YARD
		E	?

please turn over…

4	**The tide is full, the moon lies fair** in The sea is calm tonight. The tide is full, the moon lies fair Upon the straits; on the French coast the light Gleams and is gone; the cliffs of England stand;	A	The TIDE is FULL, the MOON lies FAIR
		B	The TIDE is FULL, the MOON LIES FAIR
		C	?
5	**If hairs be wires, black wires grow on her head** in If snow be white, why then her breasts are dun; If hairs be wires, black wires grow on her head. I have seen roses damask'd, red and white. But no such roses see I in her cheeks;	A	If HAIRS be WIRES, BLACK WIRES GROW on her HEAD
		B	If HAIRS be WIRES, BLACK wires GROW on her HEAD
		C	If HAIRS be WIRES, black WIRES GROW on her HEAD
		D	?
6	**A friend, a daisy, and a pearl** in Which are you, Amy Margaret? A friend, a daisy, or a pearl, A kindly, simple, precious girl - Such howsoe'er the world may twirl, Be ever, - Amy Margaret!	A	a FRIEND, a DAISy, AND a PEARL
		B	a FRIEND, a DAISy, and a PEARL
		C	?

7	**Only, from the long line of spray** in Come to the window, sweet is the night-air Only from the long line of spray Where the sea meets the moon-blanched land, Listen! you hear the grating roar.	A	onLY, from THE long LINE of SPRAY
		B	ONly, FROM the long LINE of SPRAY
		C	ONly, FROM the LONG LINE of SPRAY
		D	ONly, FROM the LONG line of SPRAY
		E	?
8	**than in the breath that from my mistress reeks** in But no such roses see I in her cheeks; And in some perfumes is there more delight Than in the breath that from my mistress reeks. I love to hear her speak, yet well I know	A	than in the BREATH that FROM my MIStress REEKS
		B	THAN in the BREATH that FROM my MIStress REEKS
		C	?
9	**The Assyrian came down like the wolf on the fold** in The Assyrian came down like the wolf on the fold, And his cohorts were gleaming in purple and gold; And the sheen of their spears was like stars on the sea, When the blue wave rolls nightly on deep Galilee.	A	the AssYRian CAME DOWN like the WOLF on the FOLD
		B	the AssYRian CAME down like the WOLF on the FOLD
		C	the AssYRian came DOWN like the WOLF on the FOLD
		D	THE AssYRian came DOWN LIKE the WOLF on the FOLD
		E	?
10	**And yet, by heav'n, I think my love as rare** in I grant I never saw a goddess go; My mistress, when she walks, treads on the ground: And yet, by heaven, I think my love as rare As any she belied with false compare	A	AND YET, by HEAV'N, i THINK my LOVE as RARE
		B	and YET, by HEAV'N, i THINK my LOVE as RARE
		C	and yet, by HEAV'N, i THINK my LOVE as RARE
		D	?

The lines are from these poems:
Matthew Arnold, Dover Beach; Christina Rossetti, Daughter of Eve; Robert Frost, Out, Out
Christina Rossetti, Gone were but the Winter; William Shakespeare, Sonnet 130;
William Allingham, Amy Margaret, Lord Byron, The Destruction of Sennacherib

C2 Additional Written Scansion Tests

More Verse Scansion Tests

Lines of poetry have beats in them – that is, some syllables are more strongly emphasised than others. This gives the line a rhythm, like the musical beats in songs.

For example, in 'Humpty Dumpty had a great fall,' emphasised syllables are written in capitals: 'HUMPty DUMPty HAD a great FALL'. The line has four beats. Note that the beats do not necessarily match strongly stressed syllables - 'great' is stressed but does not carry a beat.

Below are different combinations of beat in **10** verses, with the emphasised syllables written in CAPITALS. **Please tell me which version of each verse sounds most natural by ticking the appropriate box**.

Try pronouncing each verse, putting emphasis only on the syllables in capitals - this is to add rhythmical beats to the line, but should not eliminate the natural stress of the sentence. Ideally, the verse will have a regular rhythm without sounding unnatural.

I have given the verses before and after the test verse in each case - the test verse rhythm should fit the rhythm of the surrounding verses.

For example, in 'and there him hideth and not appeareth:

- If you think the **first** version sounds most natural, **tick the letter next to it**.
- If you are not sure which sounds best, tick the letter next to **'?'**.

1	**and there him hideth and not appeareth** in Wherewithall, unto the he*art's forest he flee*th, Leaving his enterprise with pain and cry, And there him hideth and not appeareth. What may I do when my master feareth, But in the field with him to live and die?	A	and THERE him HIdeth AND not APPEARreth
		B	and THERE him HIdeth AND NOT appEAReth
		C	and there him HIdeth and NOT appEAReth
		D	?
2	**when her loose gown from her shoulders did fall** in Thanked be fortune it hath been otherwise twenty times better but once in special	A	WHEN her loose GOWN FROM her SHOULders did FALL
		B	WHEN her loose GOWN from HER SHOULders

	in thine array after a pleasant guise When her loose gown from her shoulders did fall and she me caught in her arms long and small.		did FALL
		C	?
3	**The emblem of two people being honest** in Talking in bed ought to be easiest Lying together there goes back so far An emblem of two people being honest. Yet more and more time passes silently	A	The EMblem OF two PEOple BEing Honest
		B	The EMblem of TWO PEOple BEing Honest
		C	?

please turn over…

4	**Unruly murmurs, or ill-tim'd applause** in And let your rash, injurious clamours end: Unruly murmurs, or ill-*tim'*d applause, Wrong the best speaker, and the justest cause. Nor charge on me, ye Greeks, the dire debate:;	A	unRUly MURmurs, OR ill-TIM'D applause
		B	unRUly MURmurs, OR ILL-tim'd applause
		C	?
5	**By night he fled and at midnight return'd** in When Satan who late fled before the threats Of Gabriel out of Eden, now improv'd In meditated fraud and malice, bent On mans destruction, maugre what might hap Of heavier on himself, fearless return'd. By Night he fled, and at Midnight return'd;	A	By NIGHT he FLED and AT MIDnight reTURN'D
		B	By NIGHT he FLED and AT midNIGHT reTURN'D
		C	By NIGHT he FLED and at MIDNIGHT reTURN'D
		D	?
6	**Ay, some mad message from his mad Grandfather** in Demetrius, here's the son of Lucius; He hath some message to deliver us. Ay, some mad message from his mad grandfather. My lords, with all the humbleness I may, I greet your honours from Andronicus-	A	ay, SOME mad MESSage FROM his MAD GrandFATHer
		B	ay, SOME mad MESSage FROM his MAD GRANDfathER
		C	?

7	**Doth with his lofty and shrill-sounding throat** in Upon a fearful summons. I have heard, The cock, that is the trumpet to the morn, Doth with his lofty and shrill-sounding throat Awake the god of day; and, at his warning,.	A	Doth WITH his LOFty AND shrill-SOUNDing THROAT
		B	DOTH with his LOFty AND SHRILL-sounding THROAT
		C	DOTH with his LOFty AND shrill-SOUNDing THROAT
		D	Doth WITH his LOFty and shrill-SOUNDing THROAT
		E	?
8	**And those love-darting eyes must roll no more** in Cold is that breast which warmed the world before, And those love-darting eyes must roll no more. Thus, if eternal justice rules the ball, Thus shall your wives, and thus your children fall	A	And THOSE love-DARTing EYES must ROLL no MORE
		B	And THOSE LOVE-darting EYES must ROLL no MORE
		C	?
9	**Getting and Spending, we lay waste our pow'rs** in The world is too much with us; late and soon, Getting and spending, we lay waste our powers; Little we see in Nature that is ours; We have given our hearts away, a sordid boon!	A	GETting and SPENDing, WE lay WASTE our POW’RS
		B	GETting and SPENDing, we LAY waste our POW’RS
		C	?
10	**When to the sessions of sweet silent thought,** in When to the sessions of sweet silent thought I summon up remembrance of things past, I sigh the lack of many a thing I sought, And with old woes new wail my dear time's waste	A	WHEN to the SEssions OF sweet SIlent THOUGHT,
		B	WHEN to the SEssions of SWEET silent THOUGHT,
		C	WHEN to the SEssions of SWEET SIlent THOUGHT
		D	?

Appendix D – Test Data

This section details all the data used to test the research, including tests for Stress Clash which were not used in the final version of the application.

D1 Compound Stress and Stress Clash

The following table shows the output of Calliope's stress algorithms as applied to examples derived from the literature. The Answer column shows the result given in the literature.

Example	Answer	Type	Match	Type Results	Stress Results	Syllable Results
afternoon sun	afternoon sun(1)	compound stress – time	TRUE	; time;	afternoon(2) sun(1)	Aft(2) er(0) noon(2) Sun(1)
apartment garden	apàrtment gárden ?	compound stress - material	TRUE	;locations; place; location;	apartment(2) garden(1)	Ap(0) art(2) ment(0) Gar(1) den(0)
apricot jam	apricot jam(1)	compound stress	TRUE	;materials;	apricot(2) jam(1)	A(2) pri(0) cot(0) Jam(1)
baggage claim	baggage(1) claim	compound stress	TRUE		baggage(1) claim(2)	Bag(1) gage(0) Claim(2)
Black belt	Black(1) belt	compound stress	TRUE		Black(1) belt(2)	Black(1) Belt(2)
blind date	blínd date	compound stress	FALSE		blind(1) date(2)	Blind(1) Date(2)
blind spot	blínd spot	compound stress	TRUE		blind(1) spot(2)	Blind(1) Spot(2)
blood money	blóod money	compound stress	TRUE		blood(1) money(2)	Blood(1) Mon(2) ey(0)
blood red	blóod réd	compound stress	TRUE		blood(1) red(2)	Blood(1) Red(2)
blue moon	blue moon(1)	compound stress	FALSE		blue(1) moon(2)	Blue(1) Moon(2)
book seller	book(1) seller	compound stress	TRUE		book(1) seller(2)	Book(1) Sell(2) er(0)
Boston University Psychology Department	Boston(2) University Psychology Department(1)	compound stress	TRUE	; place;	Boston(1) University(2) Psychology(2) Department(1)	Bo(1) ston(0) U(3) ni(0) ver(2) si(0) ty(0) Psy(0) chol(2) o(0) gy(0) De(0) part(1) ment(0)
boy actor	boy ac(1)tor	compound stress – empty	TRUE	empty	boy(2) actor(1)	Boy(3) Ac(2) tor(0)
bread roll	bread roll(1)	compound stress	FALSE		bread(1) roll(2)	Bread(1) Roll(2)
bread shop	bread(1) shop	compound stress	TRUE		bread(1) shop(2)	Bread(1) Shop(2)

brown bear	brówn bear	compound stress	FALSE		brown(1) bear(2)	Brown(1) Bear(2)
brown sauce	brówn sauce	compound stress	TRUE		brown(1) sauce(2)	Brown(1) Sauce(2)
Buckingham Palace	Buckingham Palace(1)	compound stress	TRUE	; place;	Buckingham(2) Palace(1)	Buc(2) king(0) ham(3) Pal(1) ace(0)
Cashiers check	Cashiers(1) check	compound stress	TRUE		Cashiers(1) check(2)	Cas(1) hi(0) ers(0) Check(2)
china doll	china doll(1)	compound stress – substance	TRUE	;materials;	china(2) doll(1)	Chi(2) na(0) Doll(1)
chocolate santa	chocolate sa(1)nta	compound stress	TRUE	;materials;	chocolate(2) santa(1)	Choc(2) o(0) late(0) san(1) ta(0)
Christmas cake	Chrís(1)tmas cake	compound stress	FALSE	; time;	Christmas(2) cake(1)	Christ(2) mas(0) Cake(1)
Christmas card	Chrístmas card	compound stress	FALSE	; time;	Christmas(2) card(1)	Christ(2) mas(0) Card(1)
Christmas day	Christmas day(1)	compound stress – time	TRUE	; time;	Christmas(2) day(1)	Christ(2) mas(0) Day(1)
Christmas Eve	Chrístmas Éve	compound stress	TRUE	; time;	Christmas(2) Eve(1)	Christ(2) mas(0) Eve(1)
Christmas pudding	Christmas púd(1)ding	compound stress – time	TRUE	; time;	Christmas(2) pudding(1)	Christ(2) mas(0) Pud(1) ding(0)
city employee	city employee(1)	compound stress	TRUE	locations; place; location;	city(2) employee(1)	Cit(2) y(0) Em(3) ploy(0) ee(1)
city hall	city hall(1)	compound stress - location	TRUE	locations; place; location;	city(2) hall(1)	Cit(3) y(0) Hall(2)
corn flakes	corn(1) flakes	compound stress	TRUE		corn(1) flakes(2)	Corn(1) Flakes(2)
cotton shirt	cotton shirt(1)	compound stress – substance	TRUE		cotton(1) shirt(2)	Cot(1) ton(0) Shirt(2)
dog house	dog(1) house	compound stress	TRUE		dog(1) house(2)	Dog(1) House(2)
dog house	dóg house	compound stress	TRUE		dog(1) house(2)	Dog(1) House(2)
dollar bill	dollar bill(1)	compound stress – measure	TRUE	; measure;	dollar(2) bill(1)	Dol(2) lar(0) Bill(1)
double-take	dóuble-táke	compound stress	FALSE		double(1) take(2)	Dou(1) ble(0) Take(2)
double-talk	dóuble-tàlk	compound stress	TRUE		double(1) talk(2)	Dou(1) ble(0) Talk(2)
downstairs	downstairs	compound	FALSE		downstairs(1)	Down(1) stairs(0)

bedroom	bed(1)room	stress – location			bedroom(2)	be(2) dro(0) om(0)
drinking water	drin(1) king-water	compound stress	TRUE		drinking(1) water(2)	Drink(1) ing(0) Wa(2) ter(0)
Easter bunny	Easter(1) bunny	compound stress	TRUE		Easter(1) bunny(2)	East(1) er(0) Bun(2) ny(0)
Easter feast	Easter feast(1)	compound stress – time	TRUE		Easter(2) feast(1)	East(3) er(0) Feast(2)
evening primrose	evening primrose(1)	compound stress – time	TRUE	; time;	evening(2) primrose(1)	E(2) ven(0) ing(0) Prim(1) rose(3)
foul-smelling	fóul-smélling	compound stress	FALSE		foul(1) smelling(2)	Foul(1) Smell(2) ing(0)
freedom fighter	fréedom fighter	compound stress	TRUE		freedom(1) fighter(2)	Free(1) dom(0) Fight(2) er(0)
freight car	fréight càr	compound stress	TRUE		freight(1) car(2)	Freight(1) Car(2)
French Canadian	Frènch-Canádian ?	compound stress	TRUE	; place;	French(2) Canadian(1)	French(2) Ca(0) na(1) di(0) an(3)
French horn	Frènch hórn ?	compound stress	TRUE	;locations; place; location;	French(2) horn(1)	French(2) Horn(1)
full employment	fúll emplòyment	compound stress	TRUE		full(1) employment(2)	Full(1) Em(0) ploy(2) ment(0)
full-blooded	fúll-blòoded	compound stress	TRUE		full(1) blooded(2)	Full(1) Blood(2) ed(0)
full-length	fúll-lèngth	compound stress	TRUE		full(1) length(2)	Full(1) Length(2)
garden apartment	gárden apàrtment	compound stress	TRUE		garden(1) apartment(2)	Gar(1) den(0) Ap(0) art(2) ment(0)
general delivery	gèneral delivery	compound stress	FALSE		general(1) delivery(2)	Gen(1) er(0) al(0) De(0) liv(2) er(0) y(0)
gentleman farmer	gentleman far(1)mer	compound stress – empty	TRUE	; empty;	gentleman(2) farmer(1)	Gen(2) tle(0) man(0) Farm(1) er(0)
gift certificate	gíft certificate	compound stress	TRUE		gift(1) certificate(2)	Gift(1) Cer(0) tif(2) i(0) cate(0)
glove compartment	glóve compartment	compound stress	TRUE		glove(1) compartment(2)	Glove(1) Com(0) part(2) ment(0)
great-aunt	gréat-àunt	compound stress	TRUE		great(1) aunt(2)	Great(1) Aunt(2)
great-grandmother	grèat-grándmother	compound stress	FALSE		great(1) grandmother(2)	Great(1) gran(2) dmot(0) her(0)
green grocer	green(1) grocer	compound stress	TRUE		green(1) grocer(2)	Green(1) Gro(2) cer(0)
hard liquor	hard liq(1)uor	compound stress	FALSE		hard(1) liquor(2)	Hard(1) Liq(2) uor(0)

horse shoe	horse shoe(1)	compound stress	FALSE		horse(1) shoe(2)	Horse(1) Shoe(2)
house dog	hóuse dòg	compound stress	TRUE		house(1) dog(2)	House(1) Dog(2)
Irish stew	Irish stew(1)	compound stress	TRUE	;locations; place; location;	Irish(2) stew(1)	I(0) rish(2) Stew(1)
iron maiden	iron mai(1)den	compound stress – substance	TRUE	materials	iron(2) maiden(1)	I(3) ron(0) Maid(2) en(0)
iron-monger	iron(1) – monger	compound stress	TRUE		iron(1) monger(2)	I(1) ron(0) Mon(2) ger(0)
jumble sale	jum(1) ble-sale	compound stress	TRUE		jumble(1) sale(2)	Jum(1) ble(0) Sale(2)
kitchen towel	kitchen to(1)wel	compound stress - location	TRUE		kitchen(1) towel(2)	Kitch(1) en(0) Tow(2) el(0)
low-grade	lòw-gráde	compound stress	FALSE		low(1) grade(2)	Low(1) Grade(2)
Madison Avenue	Madison Av(1)enue	compound stress – place	TRUE	; place;	Madison(2) Avenue(1)	Ma(2) di(0) son(3) Av(1) e(0) nue(0)
Madison Street	Má(1)dison Street	compound stress	TRUE		Madison(1) Street(2)	Ma(1) di(0) son(3) Street(2)
Mars bar	Mars bar(1)	compound stress	FALSE		Mars(1) bar(2)	Mars(1) Bar(2)
Medical supplies	Me(1)dical supplies	compound stress	TRUE		Medical(1) supplies(2)	Med(1) ic(0) al(0) su(2) ppli(0) es(0)
memory cache	memory(1) cache	compound stress	TRUE		memory(1) cache(2)	Mem(1) o(0) ry(0) Cache(2)
metal bridge	metal bridge(1)	compound stress	TRUE	materials	metal(2) bridge(1)	Met(3) al(0) Bridge(2)
metal fatigue	me(1)tal fatigue	compound stress	TRUE		metal(1) fatigue(2)	Met(1) al(0) Fa(0) tigue(2)
Mississippi Valley	Mississippi Valley(1)	compound stress	TRUE	; place;	Mississippi(2) Valley(1)	Miss(2) iss(0) ip(3) pi(0) va(1) lley(0)
morning coffee	morning coffee(1)	compound stress	TRUE	; time;	morning(2) coffee(1)	Morn(2) ing(0) Cof(1) fee(0)
morning prayers	morning prayers(1)	compound stress	TRUE	; time;	morning(2) prayers(1)	Morn(2) ing(0) Prayers(1)
morning sickness	morning(1) sickness	compound stress	FALSE	; time;	morning(2) sickness(1)	Morn(2) ing(0) sic(1) kness(0)
motor bike	motor bike(1)	compound stress	TRUE	materials	motor(2) bike(1)	Mo(3) tor(0) Bike(2)
mountain railway	mountain rai(1)lway	compound stress	TRUE	locations; place; location;	mountain(2) railway(1)	Moun(3) tain(0) Rail(2) way(1)
Murray Hill	Murray Hill(1)	compound stress	TRUE	place	Murray(2) Hill(1)	Mu(3) rra(0) y(0) Hill(2)
musical chairs	musical	compound	FALSE		musical(1)	Mu(1) sic(0) al(0)

	chairs(1)	stress			chairs(2)	Chairs(2)
opera buff	opera(1) buff	compound stress	TRUE		opera(1) buff(2)	Op(1) er(0) a(0) Buff(2)
paper shop	paper(1) shop	compound stress	TRUE		paper(1) shop(2)	Pa(1) per(0) Shop(2)
Philadelphia lawyer	Philadelphia lawyer(1)	compound stress	FALSE		Philadelphia(1) lawyer(2)	Phi(1) la(0) de(0) lphi(0) a(0) Law(2) yer(0)
pint jug	pint jug(1)	compound stress – measure	TRUE	; measure;	pint(2) jug(1)	Pint(2) Jug(1)
post office	post office(1)	compound stress	FALSE		post(1) office(2)	Post(1) Of(3) fice(0)
red herring	red her(1)ring	compound stress	FALSE		red(1) herring(2)	Red(1) Her(2) ring(0)
rice pudding	rice pudding(1)	compound stress	TRUE	;materials;	rice(2) pudding(1)	rice(2) Pud(1) ding(0)
road end	road end(1)	compound stress	TRUE	;locations; place; location;	road(2) end(1)	Road(2) End(1)
rogue trader	rogue tra(1)der	compound stress	FALSE		rogue(1) trader(2)	Rogue(1) Trad(2) er(0)
Safe house	Safe(1) house	compound stress	TRUE		Safe(1) house(2)	Safe(1) House(2)
salt water	salt water(1)	compound stress	FALSE		salt(1) water(2)	Salt(1) Wa(2) ter(0)
school dinner	school din(1)ner	compound stress	TRUE	locations; place; location;	school(2) dinner(1)	School(3) Din(2) ner(0)
sea water	sea water(1)	compound stress	TRUE	;materials; locations; place; location;	sea(2) water(1)	Sea(2) Wa(1) ter(0)
self analysis	self ana(1)lysis	compound stress – self	TRUE	; self;	self(2) analysis(1)	Self(2) A(0) nal(1) y(0) sis(0)
self promotion	self promo(1)tion	compound stress – self	TRUE	; self;	self(2) promotion(1)	Self(2) Pro(0) mo(1) tion(0)
self-obsessed	sèlf-obséssed	compound stress – self	TRUE	; self;	self(2) obsessed(1)	Self(2) ob(1) ses(0) sed(0)
Sick room	Sick(1) room	compound stress	TRUE		Sick(1) room(2)	Sick(1) Room(2)
silk shirt	silk shirt(1)	compound stress – substance	TRUE	;materials;	silk(2) shirt(1)	Silk(2) Shirt(1)
South street	South(1) street	compound stress	TRUE		South(1) street(2)	South(1) Street(2)
Staten Island	Staten(2)	compound	TRUE		Staten(1)	Sta(1) ten(0) Is(2)

Ferry	Island Ferry(1)	stress			Island(2) Ferry(2)	land(0) Fer(2) ry(0)
Straight edge	Straight(1) edge	compound stress	TRUE		Straight(1) edge(2)	Straight(1) Edge(2)
summer fruit	summer fruit(1)	compound stress	TRUE	time	summer(2) fruit(1)	Sum(3) mer(0) Fruit(2)
summer vacation	summer vaca(1)tion	compound stress – time	TRUE	time	summer(2) vacation(1)	Sum(3) mer(0) Va(0) ca(2) tion(0)
Surprise Lake	Surprise Lake(1)	compound stress	TRUE	; place;	Surprise(2) Lake(1)	Sur(0) prise(2) Lake(1)
valley floor	valley floor(1)	compound stress	TRUE	;locations; place; location;	valley(2) floor(1)	va(2) lley(0) Floor(1)
village shop	village shop(1)	compound stress	TRUE	;locations; place; location;	village(2) shop(1)	Vil(2) lage(0) Shop(1)
wood floor	wood floor(1)	compound stress – subtance	TRUE	measure; locations	wood(2) floor(1)	Wood(3) Floor(2)
A hundred thirteen men	A hun(2)dred thir(3)teen(4) men(1)	stress clash	TRUE		A hun(2)dred thir(3)teen(4) men(1)	
Four new mugs	Four(1) new(2) mugs(1)	stress clash	FALSE		new(1) mugs(3)	New(1) Mugs(2)
fourteen women	four(2)teen wo(1)men	stress clash	FALSE		fourteen(2) women(1)	

D2 Prosodic Analysis

The following table gives the output of Calliope's phonological analysis algorithms when processing examples found in the literature – the expected output is given in the "expert analysis" column.

Source	Text	Expert Analysis	Accept	Application Analysis
Selkirk/Cureton	Given the chance, rabbits reproduce quickly	<{[Given][the chance]}>, <{[rabbits]} {[reproduce] [quickly]}>	ok	{[Given*]* [the chance*]}{[rabbits*]*}{[reproduce*]* [quickly*]}
Gee and Grosjean	more than fifteen carpenters are working in the house	<{[More than fifteen carpenters]}{[are working]}{[in the house]}>	ok	{[More*] [than fifteen*] [carpenters*]*}{[are working*]*}{[in the house*]*}
Cureton, Williams	unless the mind change, unless	[unless] [the mind]} {[change]}>, <{[unless]	TRUE	{[unless the mind*] [change*]*}{[unless
Cureton, Williams	the stars are new measured, according	[the stars]} {[are new] [measured]}>, <{[according]}	TRUE	the stars*]*}{[are*]* [new*]}{[measured*]*}{[according*]*}
Cureton, Williams	to their relative positions, the	{[to their relative] [positions]}>, <{[the	TRUE	{[to their relative*] [positions*]*}{[the
Cureton, Williams	line will not change, the necessity	line]} {[will not] [change]}>, <{[the necessity]}	TRUE	line*]*}{[will not*]* [change*]}{[the necessity*]*}
Cureton, Williams	will not matriculate: unless there is	{[will not] [matriculate]}>: <{[unless there is]}	TRUE	{[will not*]* [matriculate*]}. {[unless there is*]*}
Hayes(1989)	A piece of pie	[A piece] [of pie]	TRUE	{[A piece*]*}{[of pie*]*}
Hayes(1989)	As shallow as Sheila	[As shallow] as Sheila	TRUE	{[As shallow*]*}{[as Sheila*]*}
Hayes(1989)	Give Maureen some	[Give] [Maureen] [some]	ok	{[Give*]* [Maureen* some]}
Hayes(1989)	He wouldn't do it this week, but he would have last week	He wouldn't do it this week, [but he would have] [last] [week]	FALSE	{[He would n't*]* [do* it] [this week*]}{[but*]*}{[he would have*]* [last*] [week*]}
Hayes(1989)	His shadow	[His shadow]	TRUE	{[His shadow*]*}

Hayes(1989)	Is Sheila coming	[Is Sheila] coming	TRUE	{[Is Sheila*]*}{[coming*]*}
Hayes(1989)	It was thought of constantly	[It was thought of] [constantly]	FALSE	{[It was thought*]} {[of constantly]*}
Hayes(1989)	John would have left	[John] [would have left]	TRUE	{[John*]*}{[would have left*]*}
Hayes(1989)	Please leave them alone	[Please] [leave them] [alone]	FALSE	{[Please*]*}{[leave*]*}{[them alone*]*}
Hayes(1989)	We'll save those people a seat	[We'll save] [those people] [a seat]	TRUE	{[We 'll save*]* [those people*]}{[a seat*]*}
Hayes(1989)	Will you save me a seat	[Will you save me] [a seat]	TRUE	{[Will you save* me]*}{[a seat*]*}
Hayes(1989) In Cureton (1992, pp 51)	On Tuesdays, he gives the Chinese dishes	[On Tuesdays]}>, <{[he gives] [the Chinese]} {[dishes]}>	TRUE	{[On Tuesdays*]*}{[he gives*]* [the Chinese*] [dishes*]}
Hayes(1989), Nespor and Vogel (1983)	The frog ate a fly for lunch	<{[The frog]} {[ate a fly] [for lunch]}>	TRUE	{[The frog*]*}{[ate*]* [a fly*]}{[for lunch*]*}
Hayes(1989), Nespor and Vogel (1983)	This is the cat that caught the rat that stole the cheese	{This is the cat} {that caught the rat} {that stole the cheese}	ok	<{[This is the cat*]*} {[that caught*]*} {[the rat*]*} {[that stole*]* [the cheese*]}>
Hayes(1989), Selkirk (1978)	In Pakistan, Tuesday, which is a weekday, is a holiday	{In Pakistan}, {Tuesday}, {which is a weekday},{ is a holiday}	TRUE	<{[In Pakistan*]*}><{[Tuesday*]*}><{[which is a weekday*]*}><{[is a holiday*]*}>
Hayes(1989), Selkirk (1978)	Emmet, alias the Rat, eats only cheese	Emmet}, {alias the Rat}, {eats only cheese}	ok	<{[Emmet*]*}><{[alias*]*} {[the Rat*]*}><{[eats*]* [only*] [cheese*]}>
Cureton (1992)	Laura's shadow	Laura's] shadow	TRUE	<{[Emmet*]*}><{[alias*]*} {[the Rat*]*}><{[eats*]* [only*] [cheese*]}>
Cureton (1992)	Those fellas shafted him	Those fellas] shafted him	TRUE	{[Those fellas*]*}{[shafted* him]*}
Cureton (1992)	Mrs Shaftow	Mrs] Shaftow	TRUE	{[Mrs*] [Shaftow*]*}
Cureton (1992)	{The invisible worm}	{[The invisible] [worm]}	TRUE	<{[The invisible*] [worm*]*}>
Cureton (1992)	{That flies in the night}	{[That flies] [in the night]}	TRUE	<{[That flies*]*} {[in the night*]*}>
Cureton (1992)	{In the howling storm}	{[In the howling] [storm]}	TRUE	{[The invisible*] [worm* That]*}{[flies*]*}{[in the night*]*}{[In the howling*] [storm*]*}

Cureton (1992)	The curfew tolls the knell of parting day	[The curfew] [tolls] [the knell] [of parting] [day]	TRUE	{[The curfew*]*}{[tolls*]*}{[the knell*]*}{[of parting*] [day*]*}
Selkirk (1981)	The absent-minded professor has been avidly reading the latest biographyof Marcel Proust	/<<{The absent-minded professor}>><<{has been avidly reading}> <{the latest biography}{of Marcel Proust}>>/	TRUE	{[The absent-minded*] [professor*]*}{[has been avidly*] [reading*]* [biographyof*] [the latest*]}{[Marcel*] [Proust*]*}
Selkirk (1981)	In Pakistan, Tuesday is a holiday	{[In Pakistan]}, {[Tuesday] [is a holiday]}	TRUE	<{[In Pakistan*]*}><{[Tuesday*]*} {[is a holiday*]*}>
Selkirk (1981)	Tuesday, which is a weekday, is a holiday	[Tuesday]}, {[which is a weekday]}, {[is a holiday]}	TRUE	<{[Tuesday*]*}><{[which is a weekday*]*}><{[is a holiday*]*}>
Cureton, Yeats When You Are Old	WHEN you are old and gray and full of sleep	<{[WHEN you]} {[are old]} {[and gray]}> <{[and full]} {[of sleep]}>	FALSE	<{[When*]*} {[you are old*]*} {[and gray* and]*} {[full*]*} {[of sleep*]*}>
Cureton, Frost Nothing Gold	Nature's first green is gold,	<{[Nature's] [first] [green]} {[is gold]}>,	TRUE	<{[Nature* 's] [first*] [green*]*} {[is gold*]*}>
Cureton, Gilbert Major General	I am the very model of the modern Major General	/<{[I am]} {[the very] [model]}> <{[of the modern] [Major] [General]}>/	TRUE	<{[Nature* 's] [first*] [green*]*} {[is gold*]*}>

D3 Phonological Scansion – theory tests

The following table compares the meter assignments for various poems by Phonological Scansion, AnalysePoems, Scandroid, Fabb's Bracketted Grid Theory and Groves' Base Template Theory against an expert assessment of the meter. Cells coloured red show an inaccurate assignment. Phonological Scansion assigns 94% of the meters correctly, including a wider range of meters than any of the other systems. Groves's system achieves a greater accuracy, but only analyses four poems (compared to Phonological Scansion's thirty-six).

Poem	Phonological Scansion	AnalysePoems	Scandroid	Fabb	Groves	Expert
Total Attempted	36	5	14	8	4	36
Total Correct	34	2	10	7	4	36
Percentage Correct	**94%**	**40%**	**71%**	**88%**	**100%**	**100%**
Arnold, *Dover Beach*	mostly iambic	duple (69%)	irregular	iambic		iambic?
Allingham, *Amy Margaret*	Iambic + catalectic trochaic	duple (86%)	irregular			Iambic
Frost, *Out, Out*	Iambic + free verse	duple (73%)	irregular			blank verse, free verse, accentual
Milton, *When I Consider*	Iambic	Duple (79%)	iambic			Iambic
Whitman, *Out of the Cradle*	free verse	Duple (47%)				free verse
Milton, *Paradise Lost* 12.659	Iambic		irregular	iambic		Iambic
Elizabeth Barrett-Browning, *Sonnets from the Portuguese* 21	Iambic		iambic	iambic		Iambic
Fitzgerald, *Rubaiyat*	Iambic		iambic	iambic		Iambic
Browning, *The Lost Leader*	dactylic trimeter		irregular	dactylic trimeter		dactylic trimester
Charlotte Bronte, *Diving*	anapaestic tetrameter		irregular	anapaestic tetrameter		Anapaestic
Christina Rossetti, *Spring Quiet*	trochaic trimeter		irregular	2 stress		trochaic trimeter?
Georgia Douglas Johnson, *The Heart of a Woman*	anapaestic tetrameter		duple		anapaest tetra-meter	Anapaestic
Shakespeare, *Sonnet* 130	iambic pentameter		iambic		iambic penta-meter	Iambic
Wyatt, translation of Petrarch Rime 140	iambic pentameter				iambic penta-meter	Iambic

Larkin *The Explosion*	Iambic + trochaic				Iambic + trochaic	Iambic
Browning, *One More Word* 1	trochaic pentameter		iambic			trochaic pentameter
Swinburne, *Sapphics*	Sapphic					Sapphic
Tennyson, *Milton*	Alcaic					Alcaic
"A little conceit? what a dangerous thing!"	Undecided					unmetrical
Keats, "How many bards gild the lapses of time!"	Undecided					unmetrical
"Ode to the West Wind by Percy Bysshe Shelley"	Undecided					unmetrical
Byron, *The Destruction of Sennacherib*	anapaestic tetrameter					anapaestic tetrameter
Longfellow, *The Song of Hiawatha*	trochaic tetrameter					trochaic tetrameter
Milton, *Samson Agonistes*	iambic pentameter		iambic			Irregular / blank verse
William Browne, *Epitaph On the Countess Dowager of Pembroke*	Trochaic > iambic					catalectic trochaic + headless iambic
Shakespeare, *Sonnet* 116	Iambic					Iambic
Donne, *Holy Sonnet* 14	Iambic					Iambic
Alexander Pope, "Soft is the strain when zephyr gently blows."	Iambic					iambic, general
John Gay, *Polly*	Amphibrach + anapaest couplet					Amphibrach
Byron, *The Bride of Abydos*	3 triple meters					3 triple meters
Tennyson, *The Lady of Shallot*	Mixed			mixed		iambic tetrameter
Swinburne, *Itylus*	trochaic pentameter					anapaest/ iambic tetrameter
Coleridge, *Metrical Feet*	mixed feet					mixed feet
Thomas Hardy, *The Oxen*	accentual, 4 beat					Accentual
Yeats, *Easter 1916*	Accentual					Accentual
Auden, *The Age of Anxiety*	Accentual / trochaic + f. iambic					Accentual

D4 Subject Tests

The following section gives the results obtained from the subject tests described in 3.5.3.

D4.1 Summary

	Written Scansion Test
Subjects	**12**
Average	**34%**
jan2w	30%
jmc54	40%
mwi39	30%
Pcrai	20%
rch22	20%
Rmca	40%
Rspee	40%
scu2c	50%
Smca	10%
Wgouc	60%
gfe24	40%
Mkeal	30%

D4.2 Scansion

D4.2.1 Lines 1-10

12	correct	%	Gone were but the Winter, come were but the Spring	The tide is full, the moon lies fair	The buzz saw snarled and rattled in the yard	No more to laugh, no more to sing	If hairs be wires, black wires grow on her head	A friend, a daisy, and a pearl	Only, from the long line of spray	than in the breath that from my mistress reeks	the Assyrian came down like the wolf on the fold	and yet, by Heaven, I think my love as rare
Phonological Scansion			**1**	**1**	**2**	**2**	**2**	**1**	**4**	**2**	**3**	**2**
Scandroid			2	2	1	1	1	2	3	3	1	2
Experts			1	1	2	2	1	1	3	2	3	2
Fabb/Plamondon/Groves			3		1		1		1	1		1
PS number correct			7	6	1	3	5	5	3	4	2	5
Scandroid			5	6	5	9	5	6	6	4	3	5
Experts			7	6	1	3	5	5	6	4	2	5
Fabb/Plamondon/Groves			0		5		2		2	3	0	4
% PS correct			**58%**	**50%**	**8%**	**25%**	**41%**	**45%**	**25%**	**36.36%**	**18.18%**	**45.45%**
% Scandroid			**41%**	**50%**	**41%**	**75%**	**41%**	**54%**	**50%**	**36.36%**	**27.27%**	**45.45%**
% experts			**58%**	**50%**	**8%**	**25%**	**41%**	**45%**	**50%**	**36.36%**	**18.18%**	**45.45%**
% Fabb/Plamondon/Groves			**0%**		**41%**		**16%**		**16%**	**27.27%**	**0.00%**	**36.36%**
jan2w	3	30	1	2	1	1	1	1	3	1	1	2
jmc54	4	40	2	1	3	2	3	2	4	3	1	2
mwi39	3	30	2	1	3	1	2	2	4			
Pcrai	2	20	2	1	1	1	2	?	2	3	4	1
rch22	2	20	1	2	1	2	1	2	3	3	1	3
Rspee	4	40	2	1	1	1	2	1	3	2	4	1
scu2c	5	50	1	2	1	2	2	2	3	2	4	2
Rmca	4	40	1	2	4	1	1	1	4	2	4	1
gfe24	4	40	1	2	4	1	1	1	3	3	3	2
Smca	1	10	2	2	4	1	2	2	1	1	2	3
Mkeal	3	30	1	1	4	1	3	2	3	2	2	1
Wgouc	6	60	1	1	2	1	1	1	1	1	3	2

D4.2.2 Lines 19-20

12	correct	%	and there him hideth and not appereth	when her lose gowne from her shoulders did fall	The emblem of two people being honest	Unruly Murmurs, or ill-tim'd Applause	By night he fled and at midnight return'd	Ay, some mad message from his mad Grandfather	Doth with his lofty and shrill-sounding throate	And those love-darting eyes must roll no more	Getting and Spending, we lay waste our pow'rs	When to the seasons of sweet silent thought,
Phonological Scansion			**2**	**1**	**2**	**2**	**1**	**2**	**2**	**2**	**2**	**1**
Scandroid					2							3
Experts									1			1
Fabb/Plamondon/Groves			3	2	1	1	2	1	1	1	1	2
PS number correct			3	4	5	4	2	4	1	1	2	0
Scandroid			0	0	5	0	0	0	0	0	0	3
Experts			0	0	0	0	0	0	0	0	0	0
Fabb/Plamondon/Groves			2	2	0	2	1	1	0	5	4	3
% PS correct			**50**	**66**	**100**	**66**	**33**	**66**	**16**	**16**	**33**	**0**
% Scandroid			**0**	**0**	**100**	**0**	**0**	**0**	**0**	**0**	**0**	**50**
% experts			**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
% Fabb/Plamondon/Groves			**33**	**33**	**0**	**33**	**16**	**16**	**0**	**83**	**66**	**50**
jan2w	3	30	3	1	2	2	3	2	4	2	2	2
jmc54	4	40										
mwi39	3	30										
Pcrai	2	20	2	2	2	2	2	1	3	1	1	3
rch22	2	20	3	2	2	2	1	2	2	1	1	2
Rspee	4	40										
scu2c	5	50	2	1	2	1	1	2	4	1	1	2
Rmca	4	40	4	1	2	1	3	3	3	1	1	3
gfe24	4	40										
Smca	1	10	2	1		2	3	2	4	1	2	3
Mkeal	3	30										
Wgouc	6	60										

Appendix E – Application Information

This section gives more information on the Calliope application including design diagrams and program outputs.

E1 Errors

E1.1 Antelope Parser

The Antelope Parser produces some idiosyncratic readings:

Problem	Fix
Phrases joined with ‘and’ have the second phrase governed by ‘and’.	‘and’ is replaced with ‘also’ for the parsing phase
Inverted verb phrases are classified as noun and verb: ‘gone (Noun) were (Verb)’	None
Some nouns are incorrectly classified as proper nouns: ‘I have seen roses (Proper Noun) damask’d’	None
Syncopated Past Participle endings (‘d) are classified as independent Noun Phrases: ‘damask’d’ becomes ‘damask’ (Verb Phrase) ‘‘d’ (Noun Phrase)	Where ‘‘d’ is found, its syntactic value is ignored and it is added to the previous word
‘Cannot’ and ‘can not’ are both classified as two words: ‘can’ and ‘not’ (similarly ‘can’t’ is classified as ‘ca’ and ‘n’t’).	‘Not’ in ‘cannot’ and ‘n’t’ are associated with the previous word

Table 75: Antelope Parser idiosyncrasies

E2 Program Flow

See Figure 4.

E3 Classes

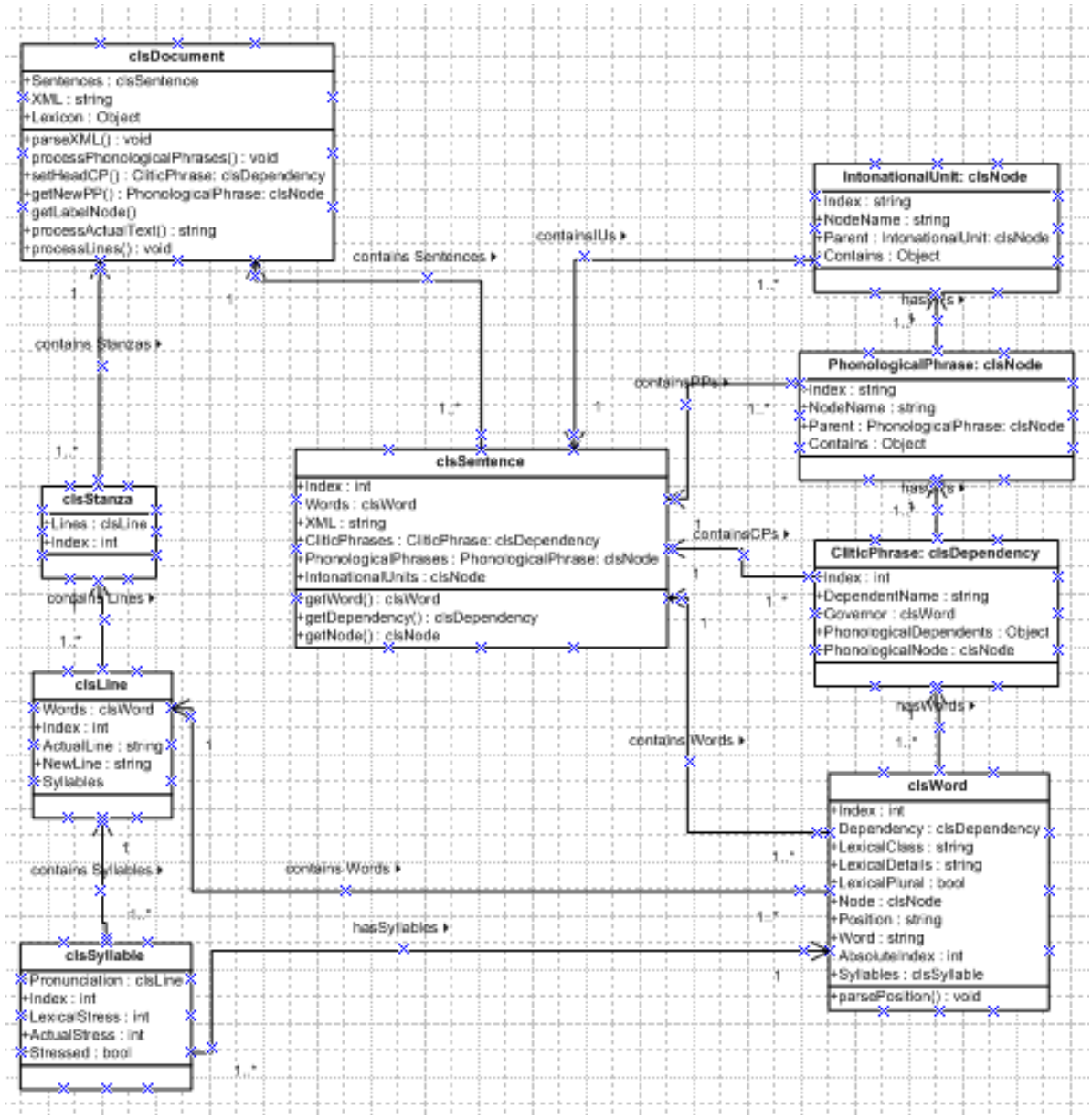

Figure 13: Phonological Classes

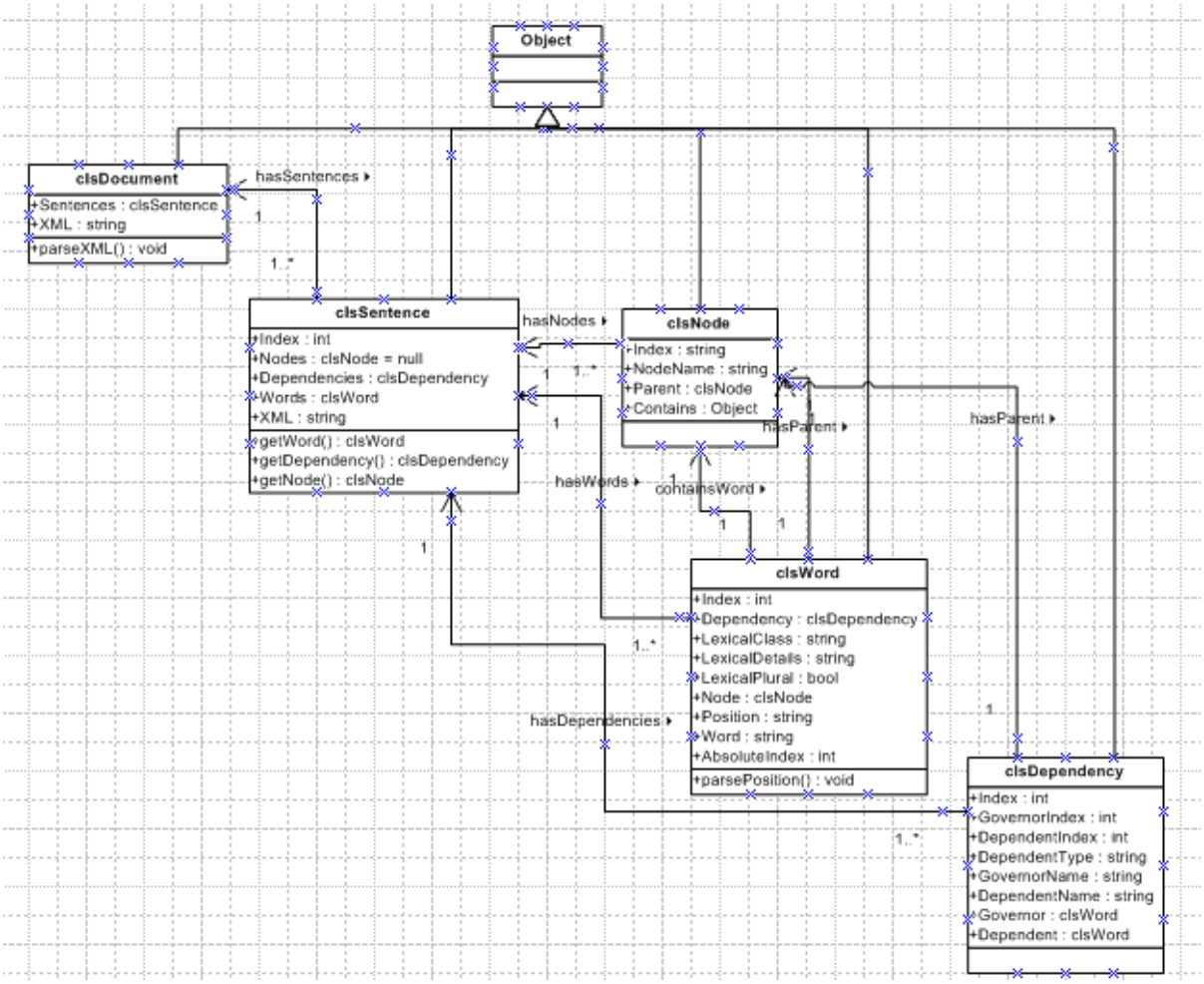

Figure 14: Syntactic Classes

E4 Stress Assignments

The program has three phases for stress assignments:

- Lexical stresses
- Phrase stresses
- Relative stresses

These are discussed in more detail below.

E4.1 Lexical Stresses

Lexical stresses are assigned to a word from its dictionary entry (derived primarily from Ward, 1997). There are three levels of stress – primary (2), secondary (1) and no stress (0). Content words are assigned a primary stress on the relevant syllable, but non-content words are given a secondary stress where a primary stress is marked in the dictionary (non-content words include prepositions and pronouns). This is to indicate the difference in stress levels between content and non-content words.

E4.2 Phrase Stresses

Phrase stresses are assigned by using the Compound Stress and Nuclear Stress Rules developed by Hayes (1984b) and others. The compound stress rule adjusts the stress in compounds which fit particular semantic categories. Stress clashes are then resolved. Finally, increasing phrase stress is assigned to the primary stressed syllables in content words as a phrase progresses.

E4.3 Relative Stresses

Relative stresses are assigned to the syllables based on their phrase stress values. Initially, syllables with no phrase stress (0) are assigned weak stress (w), syllables with secondary stress (1) are assigned low stress (n) and syllables with a phrase stress of 2 or greater are assigned strong stress (s).
These values are modified if there is a sequence of two contiguous relative stresses with the same value. The goal is to avoid contiguous stresses with the same value. However, the system still generates such sequences where there is no relationship between the syllables. The syllables are tested for their syntactic relationship: if one syllable governs the other then it is assigned stronger stress, or the governed syllable is given weaker stress (depending on the value of the syllable immediately before the sequence). If there is no syntactic relationship, then the stresses are left unadjusted. Additional considerations also affect the relative stress resolution of identical contiguous stresses: for example, prepositions will receive greater stress, except where they govern a pronoun which will receive greater stress than the preposition.

Phase	Stresses	Comment
Shall I compare thee to a summer's day?		
Lexical Stress	2102111202	Non-content words "I" and "thee to a" all receive secondary stress (1)
Phrase Stress	2103111405	Content words receive an additional stress to achieve a stress gradient across the phrase
Relative Stress	snwswnwsws	In the sequence "111", "to" is promoted to n stress because it governs "a"
Scandroid	wswswnwsws	To is allocated an ambiguous stress because it is the second weak stress in a series of three (www)
If hairs be wires, black wires grow on her head		
Lexical Stress	1222,222112	
Phrase Stress	1234,567118	
Relative Stress	nsms,nmsnws	In the sequence "222,", "be" is demoted to m because the copula ("wires") of the verb to be is considered to govern the verb. In the sequence ",222", "black" is demoted to m because it is governed by "wires". "wires" is demoted to m because it is governed by "grow". "on" is promoted to n because it governs "her"
AnalysePoems	0202222002 (wsws,ssswws)	
Scandroid	wswssssws	

Figure 15: examples of the stress assignment procedure

E4.4 Discussion

The system generates a relative stress pattern which approximates what I think are the stress values of a phrase, and which produces scansions which are closer to expert opinions than other stress assignment systems (such as AnalysePoems or Scandroid). The system fixes the stress values of the line, and adjusts the metrical pattern round them, rather than adjusting the stress values to fit metrical patterns.

In most cases, this reflects the practice of native speakers: it seems that only where certain types of polysyllabic words appear in compounds are the stresses adjusted to fit the meter.

Moreover, this relative stress assignment allows, for example, the promotion of prepositions to strong metrical positions, which is required in some verses. Plamondon achieves this by adjusting the stress of the line to fit the meter, but the relative stress assignment system uses only linguistic assumptions to produce the same effect.

E5 Output

The following information is produced by the application when it scans a line.

E5.1 Antelope Parser Syntactic output


```
<?xml version="1.0" encoding="utf-8" ?>

- <Document id="1">
- <Sentence id="1">
- <words>
  <word ind="1" pos="VBD">Shall</word>
  <word ind="2" pos="PRP">I</word>
  <word ind="3" pos="VB">compare</word>
  <word ind="4" pos="PRP">you</word>
  <word ind="5" pos="TO">to</word>
  <word ind="6" pos="DT">a</word>
  <word ind="7" pos="NN">summer</word>
  <word ind="8" pos="POS">'s</word>
  <word ind="9" pos="NN">day</word>
  <word ind="10" pos=".">?</word>
    </words>
- <dependencies>
- <dep type="aux">
  <governor idx="3">compare</governor>
  <dependent idx="1">Shall</dependent>
    </dep>
- <dep type="nsubj">
  <governor idx="3">compare</governor>
```

```
  <dependent idx="2">I</dependent>
    </dep>
- <dep type="dobj">
  <governor idx="3">compare</governor>
  <dependent idx="4">you</dependent>
    </dep>
- <dep type="prep">
  <governor idx="3">compare</governor>
  <dependent idx="5">to</dependent>
    </dep>
- <dep type="det">
  <governor idx="7">summer</governor>
  <dependent idx="6">a</dependent>
    </dep>
- <dep type="poss">
  <governor idx="9">day</governor>
  <dependent idx="7">summer</dependent>
    </dep>
- <dep type="possessive">
  <governor idx="7">summer</governor>
  <dependent idx="8">'s</dependent>
    </dep>
- <dep type="pobj">
  <governor idx="5">to</governor>
  <dependent idx="9">day</dependent>
    </dep>
    </dependencies>
- <nodes>
  <node id="1" parent="0" label="SQ" />
  <node id="2" parent="1" word="1" />
  <node id="3" parent="1" label="NP" />
  <node id="4" parent="3" word="2" />
  <node id="5" parent="1" label="VP" />
  <node id="6" parent="5" word="3" />
  <node id="7" parent="5" label="NP" />
  <node id="8" parent="7" word="4" />
  <node id="9" parent="5" label="PP" />
  <node id="10" parent="9" word="5" />
  <node id="11" parent="9" label="NP" />
  <node id="12" parent="11" label="NP" />
  <node id="13" parent="12" word="6" />
  <node id="14" parent="12" word="7" />
  <node id="15" parent="12" word="8" />
  <node id="16" parent="11" word="9" />
  <node id="17" parent="1" word="10" />
    </nodes>
    </Sentence>
    </Document>
```

E5.2 Scansion output

=====TEXT ANALYSIS=======
=====Original Text=====
Shall I compare thee to a summer's day?

=====Revised Text=====
Shall I compare you to a summer's day?

====SYNTACTIC ANALYSIS==========
=====Analysis=====

```
                        ┌──────────────────pobj>──────────────────┐

    ┌──────────>aux──────────┬──────────prep>──────────┤
┌──────────>poss──────────┤

    │       ┌─>nsubj─┼──dobj>─┐       │       ┌─>det─┼possessi┐       │

    │       │       │       │       │       │       │       │       │
```

<a href='word:0:' title="> <a href='word:1:Shall/VBD' title='Shall/VBD'>Shall/VBD</a> <a href='word:2:I/PRP' title='I/PRP'>I/PRP</a> <a href='word:3:compare/VB' title='compare/VB'>compare/VB</a> <a href='word:4:you/PRP' title='you/PRP'>you/PRP</a> <a href='word:5:to/TO' title='to/TO'>to/TO</a> <a href='word:6:a/DT' title='a/DT'>a/DT</a> <a href='word:7:summer/NN' title='summer/NN'>summer/NN</a> <a href='word:8:'s/POS' title="'s/POS'>'s/POS</a> <a href='word:9:day/NN' title='day/NN'>day/NN</a> <a href='word:10:?/.' title='?/.'>?/.</a>

```
  (SQ Shall

    (NP I)

    (VP compare

      (NP you)

      (PP to

        (NP

          (NP a summer 's) day))) ?)
```

=====PHONOLOGICAL ANALYSIS==========

=====Phonological data=====
[day*] [to a summer* 's] [Shall I compare* thee]

{[Shall I compare* thee]*}{[to a summer* 's] [day*]*}

<{[Shall I compare* thee]*} {[to a summer* 's] [day*]*}>

=====Speed of Phonological Phrase Processing=====
speed:31ms

=====Speed of Syllable Assignments=====
speed:485ms

=====STRESS ANALYSIS=====

=====Stress Patterns=====
Shall(0) I(0) compare(0) thee(0) to(0) a(0) summer(1)'s(-1) day(2)?/n/r shall(1) I(1) Com(0)pare(2) thee(1) To(1) A(1) Sum(2)mer(0) 's(-1) Day(2)
nmwsnmnswm

=====Speed of Stress Assignment=====
speed:50ms

=====SCANSION=====

==== relative stresses + phonological data======
<{[nm/ws\n]}mn/sw\]m]}>

=====line=====
Phonological:<{[nm/ws\n]}mn/sw\]m]}>
Stresses:xxwswxxsws

====HARTMAN SCANSION========

Chosen Method: Maximise the Normal

=====Corral the Weird
xxwswxxsws
xx/ws/wx/xs/ws/
RemainingSyllables:2
Start:
End:

=====Maximise Normal
xxwswxxsws
xx/ws/wx/xs/ws/
MaxSyllableCount:8
StartsAt:0
EndsAt:7

=====PHONOLOGICAL SCANSION====
All:<{[nm/ws\n]}mn/sw\]m]}>

KeyStresses:<{[xx/ws\n]}xx/sw\]m]}>
Meter:< iambic pentameter
Summary:(0,3)
 Iamb(R):7, 1IU, 1PP, 1PW Dactyl(R):5, 1IU, 1PW Anapaest(R):2, 1PW
 Iamb(F):6, 1IU, 1CP, 1PW Dactyl(F):5, 1IU, 1PW Anapaest(F):2, 1PW
KeyRevised:<{[xx/ws\n]}xx/sw\]m]}>
Scansion:xx/ws/nx/xs/wm
Certainty:87.5%

=====Speed of Scansion=====
speed:100ms

====SPEECH XML============
SSML:
<?xml version='1.0'?><speak version='1.0' xmlns='http://www.w3.org/2001/10/synthesis' xmlns:xsi='http://www.w3.org/2001/XMLSchema-instance' xsi:schemaLocation='http://www.w3.org/2001/10/synthesis http://www.w3.org/TR/speech-synthesis/synthesis.xsd' xml:lang='en-US'>

<pron sym="shall 1"/>
<pitch absmiddle="5"><pron sym="I 1"/></pitch>
<pron sym="Com"/>
<pitch absmiddle="5"><pron sym="pare 2"/></pitch>
<pron sym="thee 1"/>
<pitch absmiddle="5"><pron sym="To 1"/></pitch>
<pron sym="A 1"/>
<pitch absmiddle="5"><pron sym="Sum 2"/></pitch>
<pron sym="mer’s 1"/>
<pitch absmiddle="5"><pron sym="Day 2"/></pitch>

</speak>

Appendix F – Contents of CD

The CD sent with this dissertation contains a working copy of the Calliope application and associated documentation:

- Calliope1 folder
- Calliope shortcut
- Calliope Manual.doc